Facebook Group: Indian Administrative Service (Raz Kr)

GEOGRAPHY
By
Neetu Singh

2014

Geomorphology

Part 1

Facebook Group : Indian Administrative Serivce (Raz Kr)

UPSC Optional

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Geography

By Neetu Singh

Vol-2

2014

1. Geomorphology
2. climatology
3. Oceanography

Date
17/02/2014

Sustainability : Conservation not preservation

NCERT → Class Notes → Model Answers

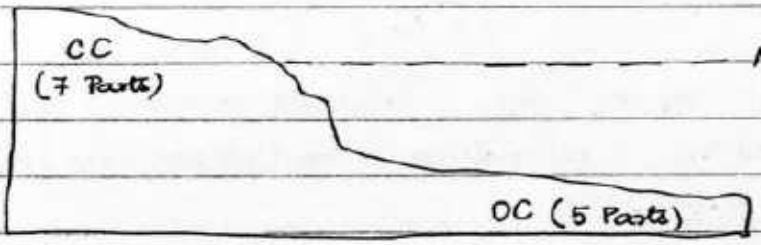
"Physical Vs Human" Part

GREAT INDIAN DESERT is also a PLAIN.

↳ Environment : Lithosphere*, atmosphere*, hydrosphere, biosphere

↳ Pillars of Syllabus : (Total 5 in Block 1) → Geomorphology, Climatology ← Oceanography, Bio & Environment Geography

MACROSCALE RELIEF:
1st Order Relief : Land/Continental Crust & Oceanic Crust



(7+5) parts = Lithosphere

Lithos = Brittle

In geomorphology, we study 7 continents (i.e. CC)
↳ Features of CC
↳ Study of 1st realm of environment

Lithosphere, the outermost layer of the Earth

Causes of features : Endogenic & Exogenic forces

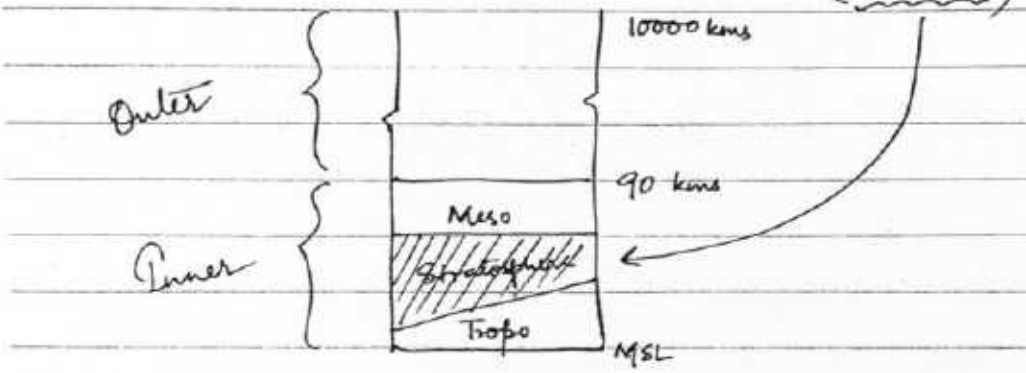
Aravalis are OLD FOLD mountains.
Himalayas → NEW → →

JAN

Processes
⇓

riability developers → Endo
wellers → Exo
} Reason for dynamism for CC.

inatology : 2nd Realm : Atmosphere
:NCCA
ational Carbonaceous Aerosol Programs: Indian Middle Atm. Program (I-MAP)



posphere: Densest; Maxm. atm.; Comprises components responsible for CLIMATE
All weather mechanisms takes place here (WEATHER LAYER)
JET STREAMS

opical Monsoonal Climate : INDIA

Under Climatology

RUSSIAN CLIMATE : Only 2 types — TAIGA, TUNDRA → Russian name for forest; ↳ Largest biome

Climatology → Weather Mechanisms
→ Climatic Prospects

Study oceans and their marginal extensions (in Oceanography)
Caspian Sea = Inland
Mediterrean Sea = Marginal

Marginal extensions have connectivity to oceans.

Ocean Water properties → SALINITY → These are Applications of climatology
→ TEMP. →

Photatrophs & Decomposers ⇒ Both are important

Paper II Physical Geography :

Physical Setting + Resource
Indian Geological Structure
} of India

LOCK (1)

Paper 1 : Geomorphology
Climatology
Oceanography
Biogeography + Envt. Geo.

+

Paper 2:- Physical Setting & Resource
- Indian Geology

×

HUMAN GEOGRAPHY

Demography
→ Population
[Trends & Patterns } P1
Migration with theories
In Indian Context } P2]

Settlement
→ Temporary
→ Permanent
→ Urban Vs Rural

Economic
→ Extractive
→ Manufacturing
→ Facilitative
→ Reproductive (Agriculture)

Other

Today's Social Forestary = "Economic + Green"

NCR & NCT
↳ Political

AREAS outside the political boundaries as well.

Economic Geography : ① Agriculture
② Industry
③ Trade, Transport, Comm.

Kishan Vision Project
CONCOR

heus :
gional
litical – Space Politics
eographical Thought

isa-On-Arrival" → Boost Tourism

nventional Map Marking

"

vient Black Swan → 6th Edition.

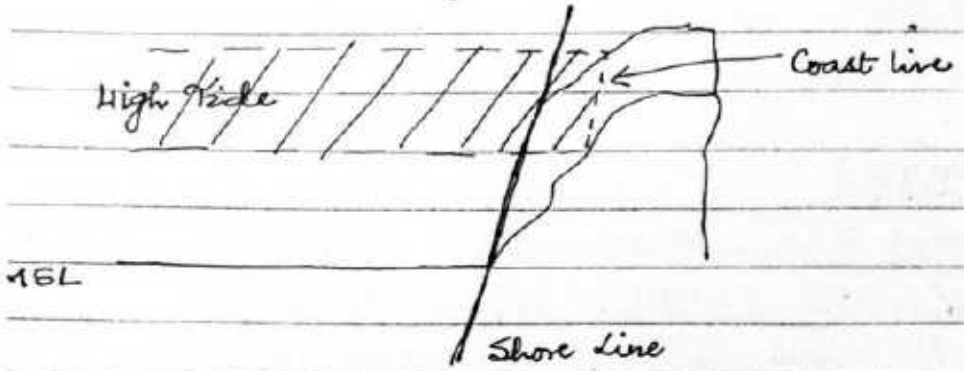
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Geography

Date
18/02/2014

Relief :

1st Order : Macro-divide of Earth (CC & OC)
2nd Order : Mountains (CC) ; Mid-Oceanic Ridges (OC)
3rd Order : Valleys, Deltas (CC) ; Generally missing (in OC)



Endogenetic Forces : Variability generators ⇒ 2nd Order Relief Feature
Exogenetic Forces : Levellers

- Cutting Down (Erosional) — Eg. Valleys
- Depositional — Eg. Delta

RELIEF FEATURES :

Fundamental principles of physical geography takes into account the study of relief features. Technically relief features are distinguished into 3 prominent categories :

1) 1st Order Relief Features .
2) 2nd Order " "
3) 3rd Order " "

The first Order Relief Features :

Represents macro-scale divide of lithospheric features including the CC & the OC. In the context of present map of world,

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st order Relief is depicted as 7 continents & 5 ocean floors. he 1st order relief features largely represent their ORIGIN o be related to cooling & solidification of the Earth's Crust, ome 4000 million years ago. They, however, also represent Subsequent modification regulated by drift of the continent nd the Ocean floor.

2ND ORDER R.FEATURES :

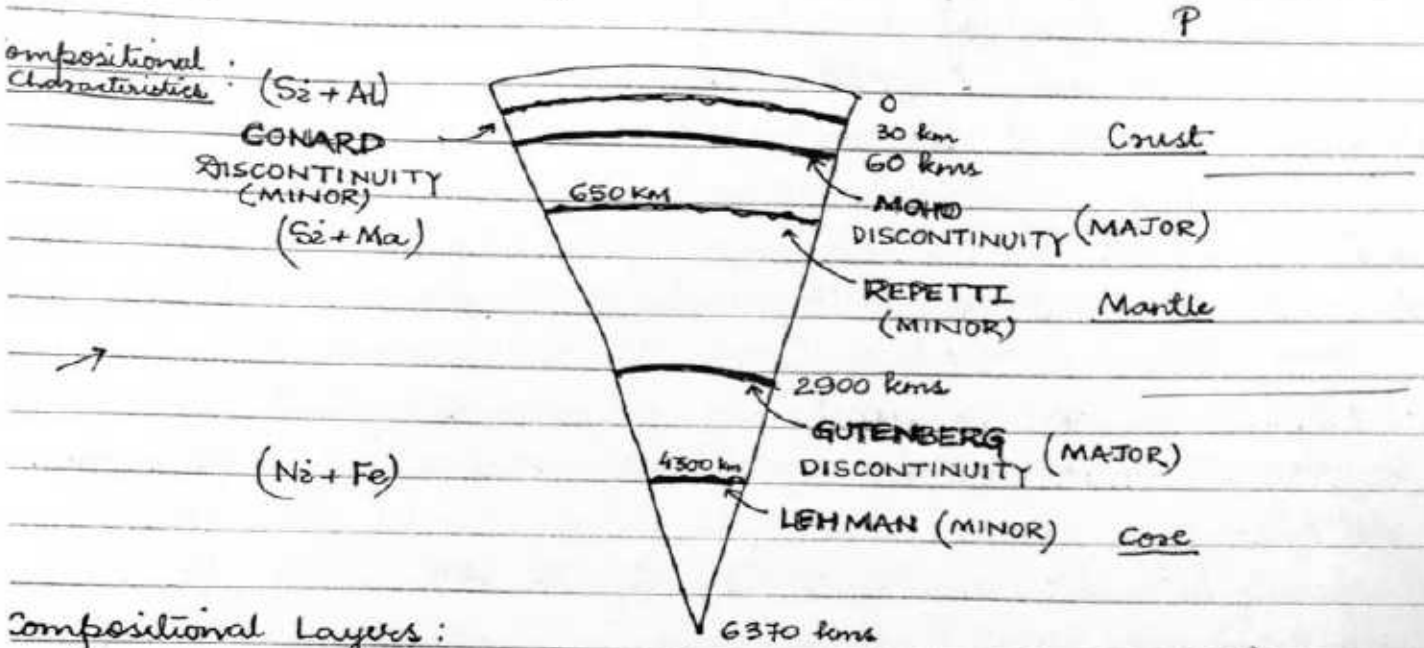
These features in comparision marks their development due to he effectivities of ENDOGENEC FORCES. These forces originate INSIDE the Earth's crust and result into development of VARIATIONS on the earth's surface. The endogenic forces, therefore, are recognised as variability developers. These includes mountain building & vulcanism as important processes. These generate features like mountains & plateaus on the CC, sub-marine ridges & trenches in the OC.

3RD ORDER R.FEATURES

These include the effectivities of exogenic forces. These forces are defined to be originating on the surface of the earth & work as LEVELLERS. For exogenic forces, 3 fold task therefore is commonly applied. That are: eroding, transporting & depositing. n the effectivities of these activities, carried on by exogenic orces, they developed. 3rd O.R.F includes erosional features like alleys & depositional features like deltas. The exogenic forces ncludes rivers, wind, glaciers, sea waves and underground water. s all these levellers are effective only on the CC, there is eneral absence of 3rd ORF in oceanic crust.

Continental Shelf

General absence of 3rd O.R.F. in the oceanic crust is attributed to the fact that MEAN SEA LEVEL FORMS BASE LEVEL OF EROSION. It is however that the change of climate results in fluctuations in the base level of erosion generating the possibilities of development of 3rd order relief features along CONTINENT MARGINS. (meaning continent shelfs).



Compositional Layers:
↳ Crust, Mantle, Core

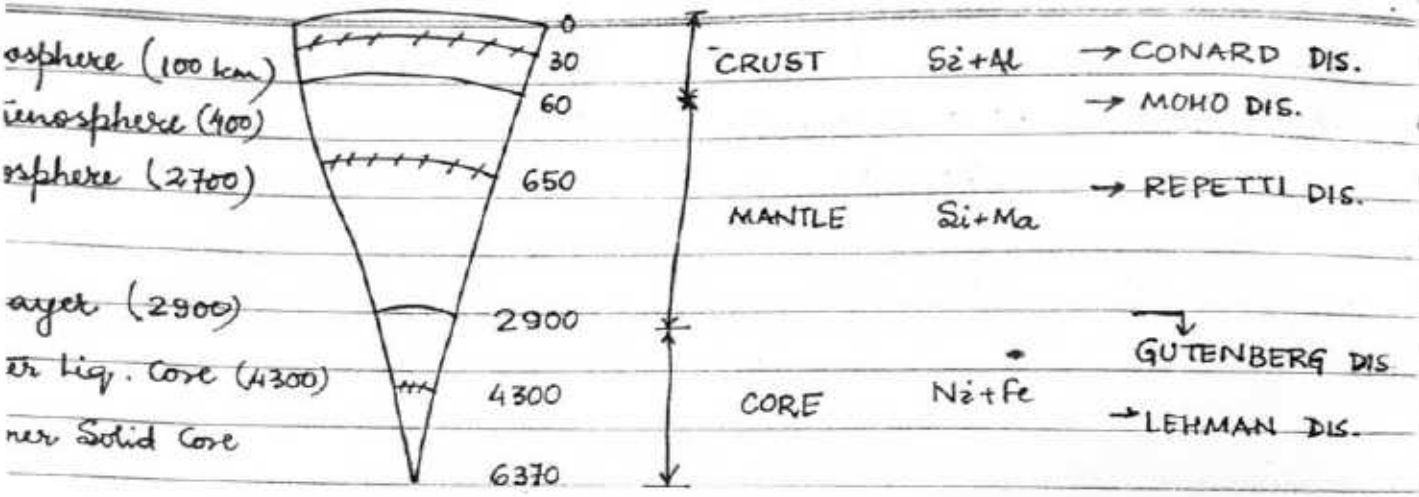
Mechanical Layers:
↳ Lithosphere (brittle): 100 km ✓
* ↳ Asthenosphere (Semi-liquid Plastic): Upto 400 km Much molten than Mesosphere
↳ Mesosphere (Brittle): 2700 km
* ↳ D' Layer (Plastic): 2900 km
↳ Outer Liquid Core: 4300 km (Lehman Discontinuity)
↳ Inner Solid Core:

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rth's Interior

① Lighter →

② Plastic →



lastic

RTH'S INTERIOR:

. terrestrial planet Earth in its genesis process experience idual cooling, solidification which was combined with FASSING & SORTING effects. In the combination of these processes, ucture of Earth's interior involved the development of 3 CONCENTRIC ers:

THE CRUST, THE MANTLE & THE CORE.

THE CRUST is the outermost layer of Earth's interior that extends upto the depth of 60 kms. At this depth, compositional discontinuity alled MOHO Discontinuity is identified. It is in accordance that the utermost layer of the Earth is compositionally lightest density material ade SiAl (Silica and alumina). Within SiAl at approximate depth f 30 kms, CONARD DISCONTINUITY is demarcated distinguishing outer, lighter SiAl from denser inner SiAl. Mechanically, entire crust forms part of lithosphere that is outer brittle layer of the Earth.

(Lithosphere = Crust + 40 kms of Upper Mantle)

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THE MANTLE: It is the second layer of the earth's interior which accounts for approx. 80% of the Earth's volume extensive from Moho discontinuity (60 kms) to Gutenberg Discontinuity (2900 kms). Mantle represents transitional compositional layer called SiMa (Silica + Magnesium). Compositionally, this layer is divided into 2 unequal parts as outer lighter SiMa & inner denser SiMa demarcated at a depth of approx 650 kms called REPETTI Discontinuity. Mechanically, mantle involves 4 out of 6 layers of the Earth's interior: the brittle lithosphere accounts for upper 40 kms of mantle largely comprising oceanic crusts. In between 100 kms to 400 kms of depth, partially molten SUBSTRATUM asthenosphere is demarcated. This mechanical layer represents its development due to temperature increase caused by frictional drag of lithosphere floating over asthenosphere. It is this floatage that forms the cause of genesis of endogenic forces. Beneath asthenosphere, the mechanical layer called mesosphere extends upto the depths of 2700 kms. Mesosphere forms mechanically brittle layer representing the effect of termination of frictional drag and increased pressure of overlaying layers. The lower 200 km (2700 - 2900 kms) forms the 5th mechanical layer in mantle called D' layer. In the effect of radioactive core, substansive rise of temperature makes this layer partially molten.

THE CORE: It is the innermost layer of the Earth that compositionally forms the densest layer, comprised of Nickel & iron thus called, NiFe. Mechanically, this innermost layer representing the effect of radioactive disintegration and thus actual temperature increase involves the only liquid layer of Earth's interior called outer core i.e. lighter NiFe also. It is this layer that forms the cause of earth's magnetic properties called GEOMAGNETISM. at approx. depth of 4300 kms i.e. Lehman's Discontinuity. Outer liquid core

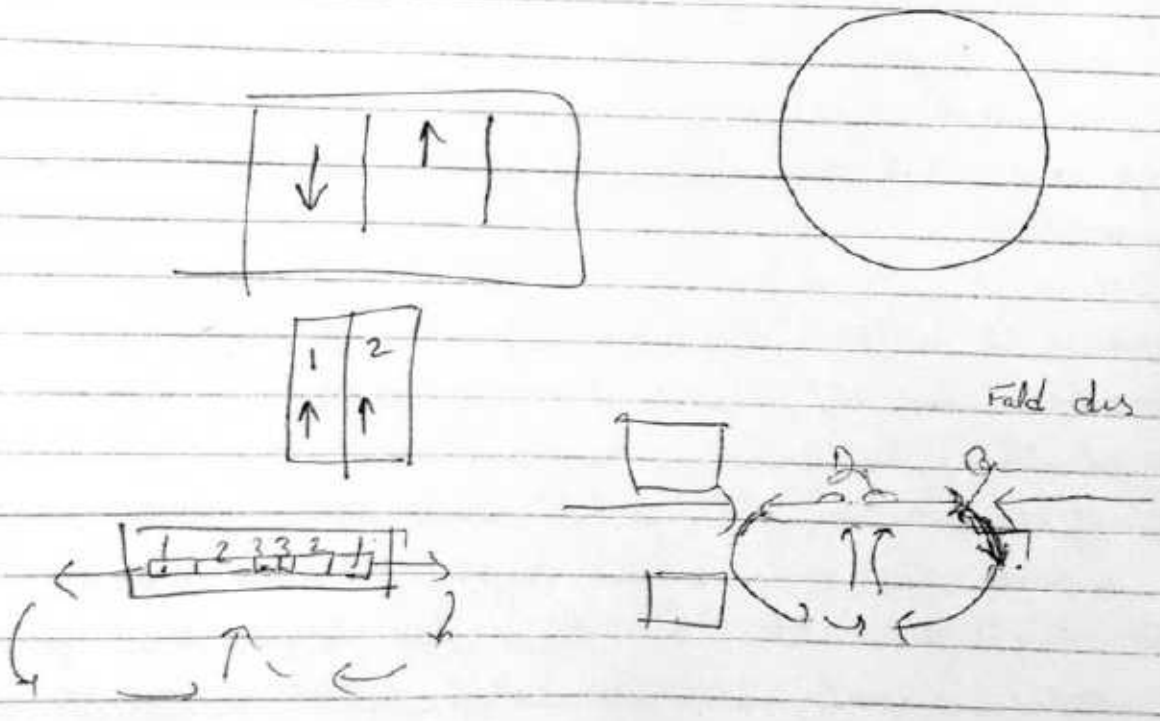
ves way to inner solid core which acts as SOLID layer
e to the pressure of over lying layers. The centre of the earth
demarcated at the depth of 6370 kms.

The interior structure of the earth is proves consistent increase in density with increase in pth. However, the temperature increase reveals variation, cifically the rate of increase of temperature decreases with creasing depth though the temperature increases thro'out the earth's interior.

EARTHQUAKES*

udden, endogenic force that do not generate 2nd order lief feature but avails the windows to the study of earth's terior.

NCERT 11th Standard : Read Earthquakes



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Date 3/02/2014

* Endogenic forces are 3 but 2nd order relief features developers are two

- Levellers + exogenic forces → generated on earth's surface → cause erosion, transfer, depo
- Sub-stratum – asthenosphere
- Plastic – 2
- Solid layer – 3
- Liquid layer – 1



Sudden endogenic forces

Plutonic → Somewhere inside the Earth

Earthquake is defined to be the sudden tremors experienced on the surface of the Earth due to the release of accumulated energy in the deformed rocks. These tremors correlates to the focus and the epicentre. The focus is defined to be that plutonic spot where the energy gets released propagating via the adjacent rocks. These energy waves called seismic waves generate epicentre vertically over the focus on the surface of the earth where maximum impact is felt. The earthquake study involves the analysis of characteristics of seismic waves. It is this analysis that helps in outline :

) Study of earth's interior
) Earthquake as geomorphic agent.

Based on the propagation, location and behaviour, Seismic waves are classified into 2 categories :

Seismic Waves

Body Waves — Surface Waves

ompressional — Distortional — Rayleigh — Love

'p'

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eomorphic agent → Write only about surface waves.

ODY WAVES

hese represent those energy waves which propagates via rth's interior to be recorded on the seismograph in diff rts of the world. These energy waves are classified in bsolute accordance to their nature of propagation. The marcated categories of body waves includes: compressional aves and distortional waves.

The compressional waves form those body aves which propagates via the earths interior by compressing e media. In the propagation of these waves rocks gets ompressed only to expand beyond their original volume as he energy waves travels via it. In this sequence, propagation f these energy waves generates alternating sequence of ompression & expansion influencing the volume of the medium. hese energy waves are also called primary or 'p' waves. s they are the fastest moving seismic waves moving at a elocity of 7 kmps via the crustal rocks and therefore are first be recorded on the seismograph.

In comparision, the distortional waves form he body waves which propagates by inducing shearing stress. t is because these waves propagate by pulling rock particles p & down ⊥r to the direction of wave propagation. In this ffect, it generates distortional effects i.e. influencing shape f the rocks. It is therefore that these body waves disappears n the liquid media. These energy waves are also called secondary waves or 'S' waves as they represent 3.5 kmps f propagation velocity, travelling via the crustal rocks.

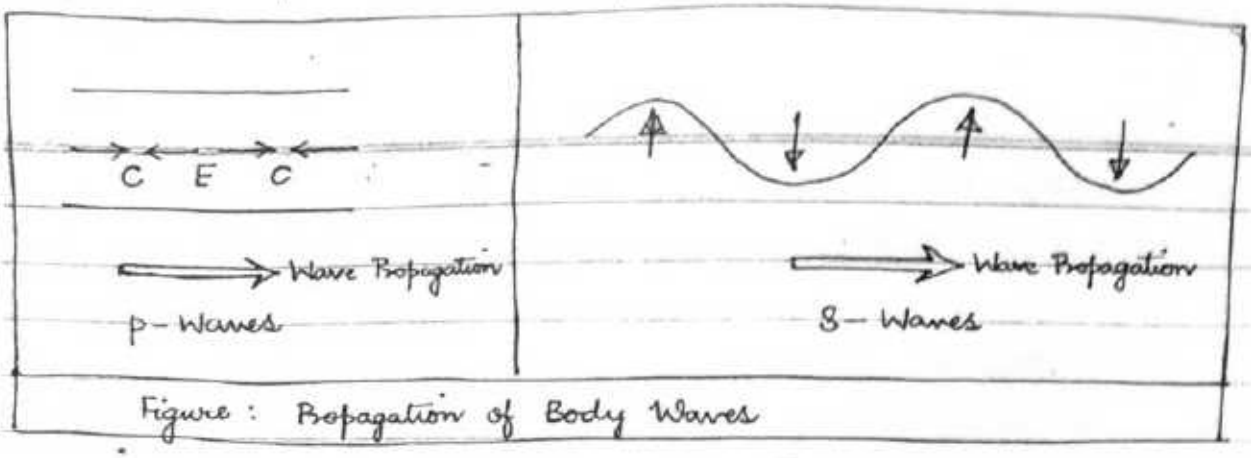


Figure: Propagation of Body Waves

BEHAVIOUR OF BODY WAVES & EARTH'S INTERIOR

The primary waves in its compressional propagation is capable of propagating via all the likely mediums in the earth's interior. However, in the change of the medium it marks the modification in its velocity as well as in the direction of its flow. As the inelasticity of the liquid makes p-waves significantly reduce its velocity, this medium justifies disappearance of secondary waves as liquid cannot bear shearing stress. In the beginning of 20th century, analysis of seismographic records make the scholars conclude shadow zone which are defined to be that opposite part of the earth's surface in content of earthquakes focus where no energy wave is recorded on the seismograph. Enlargement of the study of shadow zones proved that shadow zone for S-waves form half the circumference of the earth on the opposite side of the focus. as these energy waves do not propagate beyond Gutenberg Discontinuity (2900 kms) proving outer core to be liquid. In comparision, refracted course of the primary waves causes the development of divided shadow zones along either side of the focus b/w 103° and 143°, that is not more than quarter of the circumference of the Earth. It is in regards to the analysis of seismic wave behaviour that both the plastic medium (asthenosphere & D' layer) are referred to be LOW VELOCITY ZONES. Similarly, Gutenberg Discontinuity & Lehman Discontinuity are highlighted to be regulators of Seismic wave behaviour. It is in accordance to the minute analysis of the

mograph records that the detailed analysis of the Earth's interior ucture have been facilitated.

unamis of marginal water bodies

:ARTHQUAKE AS GEOMORPHIC AGENT / SURFACE WAVES

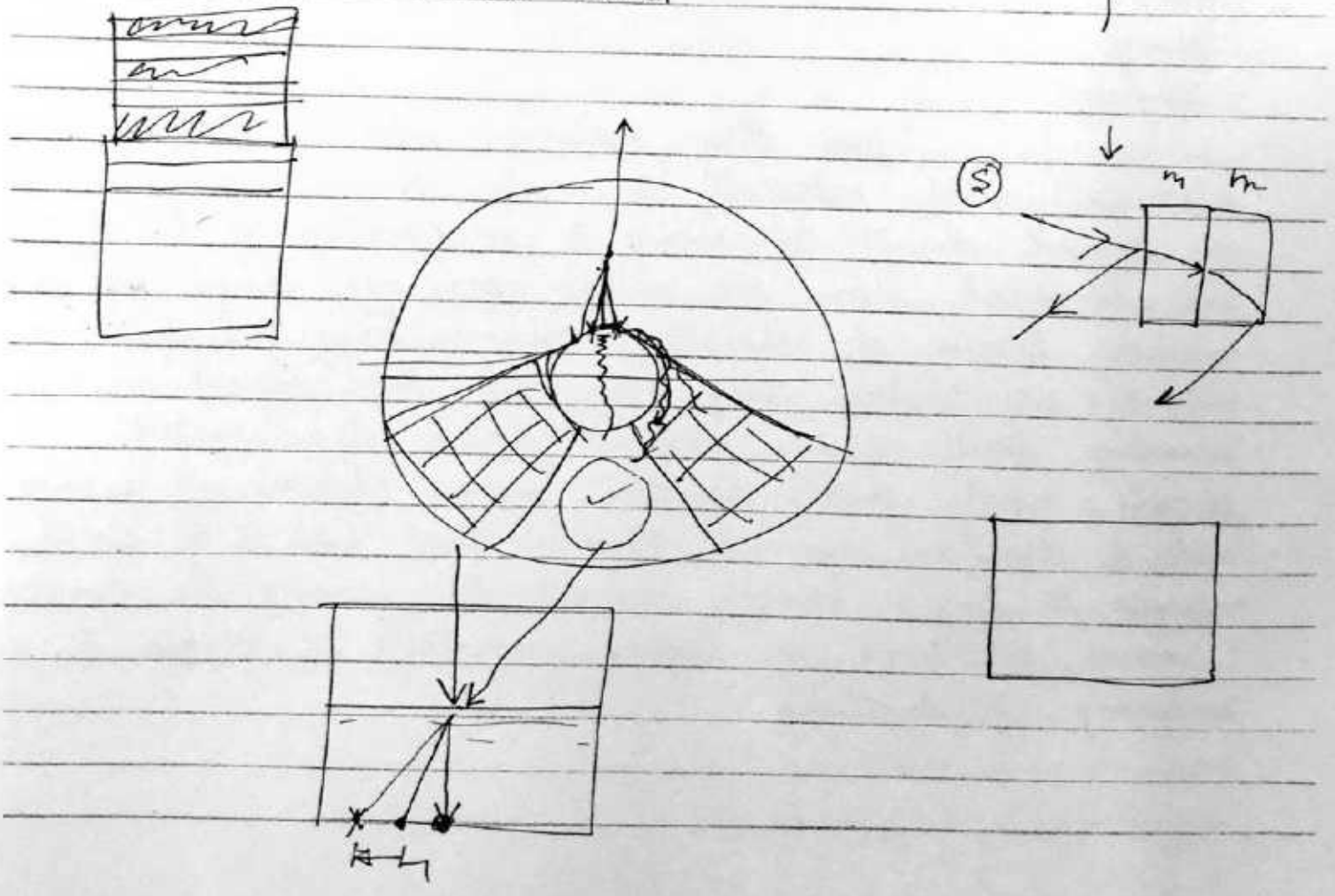
he energy waves released at the focus when readily escapes § the surface generating epicentre, there is the development of <u>rface seismic waves</u> which propagates only on the earth's rface It is these surface waves that forms the cause of mors experienced making earthquake a natural hazard. mmonly the surface wave are the slowest moving energy aves with velocity not more than 2.5 kmps. These energy aves generates tremors are capable of inducing secondary zards as landslides, avalanches or liquefaction. primarily, wever, they represent the possibilities of ground shift as the ost recognised geomorphic effect. The possible nature of ground hift includes horizontal ground shifts & vertical ground shift. he <u>horizontal ground</u> shift is caused by love waves that volves propagation called <u>weeping motion</u>. In comparision, rtical ground shift is caused by Rayleigh waves that registers eswell motion. These primary effectivities on the continental rust represent restricted geomorphic influence compared to e <u>oceanic crust</u>. High intensity submarine quake when results nto development of Rayleigh waves, there is the genesis of evastating secondary hazards called tsunamis or SEICHES.

<u>Tsunamis</u> are defined to be seismic sea aves which unlike ordinary seawaves involve the movement of tire water mass. These sea waves in the <u>open water</u> involves ypical characteristics of being long waves (wavelengths in the

range of 150-160 km), short heighted (wave ht. not more than 3 m) and fast moving waves with velocities upto 800 kmph. Toward the shore line, however, these waves completely transform into slow moving gigantic waves with wave hts. upto 30 m. It is these gigantic waves that poses as secondary hazards caused by earthquakes. In accordance to its surfing effect as well as delayed backwash, tsunamis in marginal water bodies are called seiches which are more specific in creating secondary hazards as have proximate shoreline.

1935 → Richter Scale

Ahaggar Mts. → Ancient / Part (conical mountains) on shields } in ALGERIA in AFRICA
→ Special categories of mountains
→ Conical shield



02/2014

Lappland → European Shield
Central Russian Upland : Ancient plateau
Ural Mountains — Old fold mountains (due to compressional stress)
Alps → Young fold mountains
Pamir Plateau :
(Roof of world

CHRONOLOGICAL DEVELOPMENT OF RELIEF :

Movement of lithosphere over asthenosphere in the entire history of the planet earth have been because of development of different types of relief features. The analysis of these relief features in the chronological sequence is attempted as:

Ancient relief
Old "
Young "
Youngest "

first three categories represent 2nd ORF and ancient relief represents the shields of the world which re formed due to the cooling & solidification of the Earth's ust. Developed some 4000 million years ago, shields represent mation process of lithosphere. However, being subjected to fectivities of levellers, they have lost their original size & ht. presenting part of the continent rather than complete nuents, mostly shields represent ancient plateaus as Canadian ields & Brazilian Highlands. However, some parts of the shields ntinues to project conical characteristics forming the examples ancient mountains as Aggthar Mountains of Algeria & kamersley of Australia.

The old relief dating back to 600 million years ago, represents 2 endogenic effectivities with old fold mountains & old block mountains distinguished compressional stress resulting in the formation of old fold mountains includes Appalachians, Urals and Great Dividing Range (of Australia) as excellent examples of. In the effect of tensional stress, the old block mountains involves the examples of Mantequiera (Brazil), Drakensberg (S. Africa). In the old relief, formation of table land is also spef specifically highlighted as lava plateau (plateau of Borborema in Brazil, Piedmont plateau, Patagonia in Argentina and Intermontane Plateau as Tarim basin in China).

The young relief represents active boundaries & it is therefore that maxm. of the focus of geographical studies remains confined to young relief. It is the most well deciphered development mechanism of these relief features that is applied to interpret the formation of older counterparts. Among the young fold mountains Himalayan Ranges of Asia and Western Mountains of North America forms significant examples with range of Intermontane plateaus as Tibetan plateau, Iranian plateau, Columbia plateau & Colorado plateau. The young block mountains largely relates to the east African Rift Valley which also corresponds to young lava plateaus as Ethiopian Highlands & Kordofan Upland (Sudan). Th

The youngest relief depicting the 3rd OR are largely lowlands developed by running water (rivers). Among the important examples: Central Lowlands in N. America, Amazon basin in S. America, North European Plains in Europe are included.

Human Geo -

LTURAL REGIONS

ndamentals of human geography correlates to the cultural ntities of the population. These identities involves all the pects of human living including the religious beliefs, the nguage spoken, fundamentalist or cosmopolitan ideologies, ture of economic activities, food & clothing culture as ell as demographic attributes. To analyse these diverse d dynamic elements of culture, it is the classification opounded by historians Boak & Webb that is taken into nsideration. These historians have classified global cultural gions on the basis of all the cultural parameters rring the exception of language. In accordance to ir classification, the world is divided into 7 cultural gions:

Boak & Webb

Major

i) Christianity
ii) Islamic
iii) Indic
iv) Sinitic (SEIMITIC)

Minor

(i) Oceanian
(ii) Meso African
(iii) SE Asian

Oceanian → Australia & New Zealand

→ { Melanesia, Micronesia, Polynesia

CHRISTIANITY REALM:

This realm as the major cultural realm is identified with largest geographical extension incorporating entire Europe, and NEW WORLD. This cultural region is analysed geographically in the content of religious sects that includes protestant Christianity, Roman Catholic and Eastern Orthodox Christianity. The Protestant Christianity correlates to cosmopolitan cultural identities where religion has significant restricted role in dictating day-to-day living. It is western Europe that forms the SOURCE region & anglo-America along with Australia, New Zealand forming the DESTINATION. The source region with small geographic size, larger population load depicts the example of INTENSIVE CULTURE. Demographically the source regions is experiencing the threat of DEPOPULATION as maxm. population of these countries is confined in old age COHORT group. Compared to it, the destination with larger geographic size & smaller population forms the example of EXTENSIVE CULTURE. Moreover, demographically this represents progressive to stable population profile with vast resource ~~waste~~ babe and sound commercial potentiality. The destination like source regions represent the similarities in the Caucasoid Race (White People). The Roman Catholic Christianity Regions confined in Southern Europe as its source area forms Fundamentalist christianity regions. Vatican City makes the node of this cultural identity with religion dictating day-to-day activities., commercial & economic growth is significantly restricted with South European countries representing the examples of economies in transition.

mographically, however, southern European countries represent
milar characteristics to that of Western European countries that
the threat of depopulation. It is the colonial legacy
at makes Roman Catholic Christianity develop its destination
. Latin America. This destination with moderate population
ad involves extensive culture with developing economic
ities combined with NEGROIDS i.e. black race. The Eastern
thodox Christianity like Roman Catholics correlates to the
ndamentalist religious ideologies. Geographically it includes
e the European break-away countries of former USSR
nd countries of Balkan Peninsula.

Portugal, Spain, Italy, Vatican city: Roman Catholics Source
Estonia, Latvia
Russia, Belarus, Ukraine, Moldova
Balkan Peninsula → Break away countries of USSR } → Eastern Orthodox

Scandenivia
Low Countries
British Isles
France, Germany, Poland, Finland, + Alpine Countries
} → Protestant (Source).

Peninsula b/w Black Sea & Adriatic Sea

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Lecture # 5

Date : 21|02|2014

Islamic Cultural Realm : This major cultural realm is extensive in tropical western margins of the continents forming the example of desert or dry cultural region. It involves mere continuous geographical expanse from Northern Africa to South West Asia. Moreover, the disintegrated Central Asia, breakaway countries of former USSR also represents this cultural realm. Overall the complete region represent the example of fundamentalist religious ideologies and dominated caucasoid race. They mark distinction from each other in terms of nature of economic activities.

The entire range of Gulf countries depicts the examples of "Rich countries" which represent crude oil as the cause of their economic status as well as their opening to the global community. Distinguished from them are the entire range of Northern African countries, Central Asian countries, as well as caucasus states which represents developing economic states involving the combination of agricultural as well as manufacturing industries. The extreme range of cultural and economic characteristics of this region is depicted in the reference to Turkey, the only secular state of the region, proving to be culturally most advanced along with Afghanistan and Yemen, which represents poorest demographic and cultural identities with backward economic status. This entire cultural region incorporates the similarity of involving strong dictate of religion in the day-to-day living.

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THE INDIC CULTURAL REGION:

s region represents S. Asia which involves multilingual, ulti-racial, multi-religious cultural identities. The only mmon element identified with this cultural region is 'RAL AGRARIAN economic setup. Starting from its genesis its present profile, S. Asian culture represents to be docentric. For the countries with approximately 70% of the al population living in rural areas, agricultural economic ip is justified.

On the religious front, India, the secular te depicts distinction from the Islamic states of istan, Bangladesh and Maldives; Hindu state Nepal, tribal ligion Bhutan and Buddhist dominated Sri Lanka. In the nilar category all the constituents of Indic Cultural gion represents ORIENTAL CULTURAL IDENTITIES. These ntities depicts joint family norms, population as cial capital, societal and religious values with minating community living.

EIMITIC CULTURAL REGION :

e oriental culture overlaps with this region. Geographically, his cultural region belongs to East Asia where the ominating similarity is only Mongoloid race, i.e. yellow ople. In this cultural region, religious and economic iversity is of prominence where Japan represents SHINTOISM, hina - Confucianism; North Korea, South Korea and Taiwan - uddhism and Mangolia - Animism. Economically, the leveloped economy Japan and Mini Japan Taiwan depicts listinction from developing China and South Korea; whereas . Korea and Mongolia continues to dominate agrarian nd less developed profile.

It is with Japan, the oriental-most country where the unique combination of RICE-FISH CULTURE in the OCCIDENTAL traits are identified. This developed country represents urban industrial characteristics combined with cultural societal bondage.

7) SOUTH EAST ASIAN REALM: This part of Asia is geographically divided into mainland and islands which forms the cause of general absence in the development of coherent cultural characteristics for the mainland of SE Asia. Buddhist religious ideologies dominates. However, Malaysia - Islamic, Laos - Animism and Singapore - Cosmopolitan depicts the diversities. Fragmented islands also further adds to the cultural diversity with Phillipines forming the example of orthodox Christianity and Indonesia as Islamic. The entire cultural region involves dominating Mongoloid race. However, in the island region, there is substantive share of Negroid population. It is the lack of coherent characteristics that justifies SE Asia to be a minor cultural realm.

MESO AFRICAN REALM: This minor realm represents 4th world communities. It is called meso-African as maximum of African countries involves such primitive population where every tribal community has its own strong cultural bondage but with complete absence of cultural links with other tribal community.

Within Africa, Bedowins - the Camel Herders of Sahara Deserts, Masais - the Cattle Herders of Savannah, Pygmies - the hunting-gathering population of Congo Basin, San Bushmen of Kalahari Desert - very well depicts this cultural realm.

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* Lapland – Only plateau that extends in 4 countries
* S. Africa - Burundi - Global Hunger Index

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side Africa, Meso-African realm correlated to with — kimos of Canada, Aborigines of Australia, Monavis of w Zealand, Mongols of Mongolia, Lapps of Finland, noyeds of Russia — as all these represent primitive bal cultural identities.

THE OCEANIAN REALM :

e Oceanian realm excluding Australia and New Zealand left out with fragmented smaller islands that are argely uninhabited. It is in accordance that they represent inor, Melanasian, Micronesian and Polynesian cultural egions.

EOGRAPHY OF LANGUAGE

ultural identities involves language as important element nd thus is analysed geographically.
incorporates demarcation of isoglosses that are imaginary es within which same language is spoken.
ith more than 5000 spoken languages in world, isoglosses re demarcated on the basis of linguistic families.
the reference of linguistic families, demarcated cultural egions includes :

Indo-European – This linguistic family includes English, German, French, Spanish, Portuguese, Slavick, Russian, Celtic nd Indo-iranian languages (Pakhtu, Iranian or Persian, indi)

Ural – Altaic : Includes Finnic, Kazak, Uzbek languages.
Sino-Tibetan : includes Mandarin, Chinese, Tibetan, Thai
Japanese-Korean : includes Japanese, Korean
Austro-Asiatic : includes Khmer, Laotion, Burmese, Vietnamese
Malaya-Polynesian : includes Dayak, Phillipino

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(g) Dravidian :- includes Tamil, Telegu (S. India and Sri Lanka)
(h) Afro - Asiatic :- includes Arabic

Saharan
Sudanic
Niger - Congo
Khoisan

In Africa

"GEOMORPHOLOGY"

1) Geological time scale : Relative & Absolute time
2) Forces affecting surface of Earth : Endogenic , exogenic
3) Endogenic theories :
   - Continent Drift Theory
   - Sea floor Spreading
   - Geomagnetism*
   - Plate Tectonic Theory
4) Mountain building , Vulcanism
5) Exogenic forces
6) Geomorphic Cycles - Davisian
7) Isostasy

Strata : Rock layers

EOLOGICAL TIME SCALE : → Rel. → Abs.

eomorphological studies incorporates chronological account as important requirement. It is because each eveloped landform feature depicts its characteristics in ccordance to the time of its formation. The chronological tudies are divided into two well defined categories. alled : Relative Time and Absolute Time.

he Relative Time : Forms the older approach of hronological studies which involves the attempts of dentifying the age of the rocks in relation to the adjacent rock stratas. This chronological approach is credited to NICHOLAS, who identified relative time to be referred to as chronology of succession. He based the relative time into 4 well-defined principles including:

) Principle of superimposition (Lithostratigraphy)
) " " Faunal succession (Biostratigraphy)
) " " Cross-bedding relation.
) " " Fragmental inclusion.

The principle of superimposition states "that the low ying stratas are always older than the overlying stratas representing earlier episode of deposition". This rinciple is further ~~focalated~~ facilitated by the principle of Faunal succession which identifies that low lying older stratas ill always involve fossil remains of primitive organisms compared to the overlying stratas. The principle of cross bedding ~~Strata~~ relation depicts relative time in context of magma intrusions. It states that "intruded and solidified magma within the existing lithospheric stratas will always be

younger than the host rocks." This principle is prominently applied in the study of INTRUSIVE VULCANISM. The principle of fragmental inclusion recognizing the effectivity of exogenic forces states that "rock fragments will always be of older rock strata as such rock stratas have been subjected to erosional effectivities after their formation."

All these principles of relative time continues to be applicable in the geomorphic studies However, with the practical ~~implication~~ implementation of these principles in the land-form studies, prominent limitations called UNCONFORMITIES were realised. Unconformities are defined to be gaps existing between the stratas as they are subjected to dynamic equilibrium of endogenic and exogenic forces. In order to minimise the generalised studies, the requirement of absolute time scale was felt.

(1.2) The Absolute Time Scale:

By mid 19th Century, well defined Absolute Time Scales were developed by different geographic schools from among which, the EUROPEAN GEOLOGICAL TIME SCALE is primarily utilised in geomorphology.

Supereons → Eons → Eras → Periods → Epoch → Age
(4 in total) (10 in total)

)PEREON = PRECAMBRIAN" 4.6 by to 600 my

NS :

ISCOAN

CHAEAN

)TEROZOIC

Pre Cambrian

1. EO ARCHAEN
2. PALEOARCH
3. MESOARCH
4. NEOARCH

Eras of Archaen Eon

5. PALEO PROTEROZOIC
6. MESO -
7. NEO

Eras of Proterozoic Eon

NEROZOIC

8. PALEOZOIC ERA

Cambrian
Ordovician
Silurian
Devonian
Carboniferous
Permian

Fish

Periods

APMCN

4.6my - 600my ① Azoic
600my - 225my ② Paleozoic
225my - 70my ③ Mesozoic
70my - 1my ④ Cenozoic
1my - present ⑤ Neogoic

9. MESOZOIC ERA

Triassic
Jurassic
Cretaceous

Periods

10. CENOZOIC ERA

ALEOGENE PERIOD

Paleocene
Eocene
Oligocene

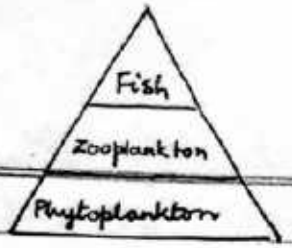
Epoch

NEOGENE

→ Miocene
→ Pliocene

QUATERNARY

→ Pleistocene
→ Holocene



Azoic → No visible life, 4600 my – 600 my

Paleozoic → 600 my – 225 my, old life, aquatic life (Primary Period)

Mesozoic → Mid-life, 225 – 70 my; aquatic, terrestrial, aerial developm
Secondary Period

Epochs → Triassic, Jurassic, Cretaceous

) Cenozoic → 70 my to 1 my ago, New Life, tertiary period

Epochs: Paleocene, Eocene, Oligocene, Miocene, Pliocene

Neozoic (Quaternary Period) → 1 my – till date, recent life,

Epochs → Pleistocene (1 my – 10,000 yrs)
Holocene (10000 yrs – till date)

4) Azoic Era:

The oldest geological era that is extensive between 4000 M years to 600 M years ago. It incorporates only one epoch called pre-cambrian developed due to original cooling and solidification of the Earth's crust. Azoic constructs called shields or cratons have lost their original extension as well as height. It is, however, that they represent fundamental complex or basement rock around which subsequent developments of landform features have been registered as relief feature. Completely all shields qualify to be plateaus as Canadian Shield. However, some of the resistant parts of shield continues to project as ancient mountains as Ahaggar Mountains Economically, Azoic constructs corresponds to METALIPHOROUS MINERAL RESOURCES as in the effectivity of degasing, swift cooling, restricted sorting effect, solidifying even heavy metals in the developed shield. Richest iron ore reserves of PILBARA (Hamersley, Australia) and CORAJAS (Brazilian Highlands) forms exellent examples.

LEOZOIC ERA :

presenting old life, paleozoic era or primary period arked its extension between 600 – 225 M years ago. This eological era is well demarcated in the identified ochs. The lower paleozoic epochs sustained the development f aquatic life forms paving way to DEVONIAN EPOCH .e. the fish epoch. In this epoch, well developed aquatic cosystem is traced back. to Following Devonian epoch, prominent changes in the climate and resultant destruction of the habitat resulted in mass death and burial of aquatic organisms including carbon secreting organisms. Economically, therefore, paleozoic era is known for CARBON i.e. coal bearing stratas which belongs to CARBONIFEROUS and PERMIAN EPOCHS. The geological constructs of paleozoic era includes broken rock materials of Azoic era as well. It is, therefore, that metallic mineral resources are also associated with this era. Geomorphologically, this era depicts its importance in the formation of OLD MOUNTAINS (fold as well as block). Among the prominent examples are the Appalachians (USA), which correlates to world's richest anthracite coal seams (layers)

MESOZOIC ERA : MID LIFE

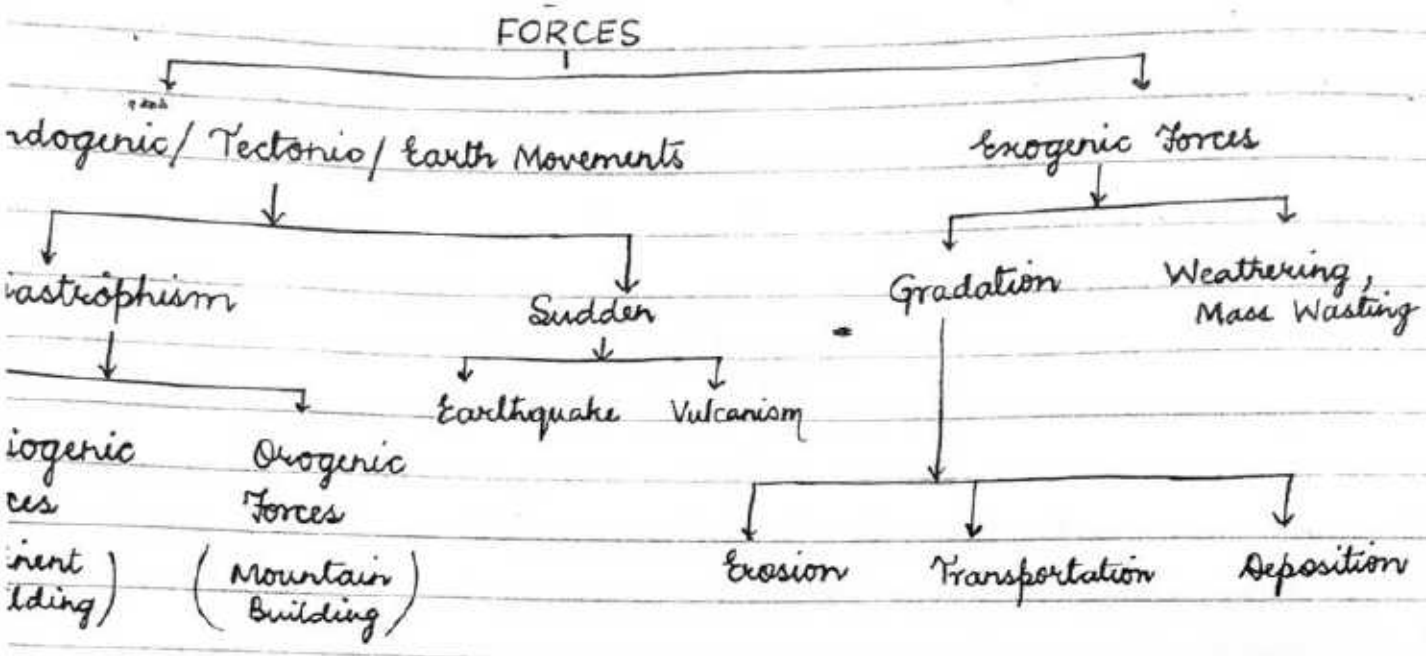
It is recognised to be extensive between 225 M years – 70 M years ago, called secondary period. It qualifies to be the QUIET TECTONIC ERA as it depicted TRANSITIONAL DRIFT OF LITHOSPHERIC PARTS IN ISOLATION. This era is primarily known for well developed life forms in all the habitats: aquatic, terrestrial and aerial. These fully developed life-forms are traced to CRETACEOUS EPOCHS.

CENOZOIC ERA — NEW LIFE :

It is tertiary period extensive between 70 M years — 1 M years ago. This era corresponds to all the young relief. It is par excellent active tectonic era and is also known for evolution of new life. All the tertiary reliefs are as young fold mountains or young block mountains dominates physiographic map of world because of their elaborate heights. It is cenozoic relief that are directly deciphered in geomorphological studies. These reliefs largely depicts to be deprived of resources primarily because difficult physiography restricts the fundamental exploration of the resource relief. Cenozoic era with its well defined 5 epochs paved way to the present NEOZOIC ERA, denoting recent life and quaternary period. Neozoic era began 1 million years ago and is extensive till date. Tectonically it correlates to continuation of Cenozoic mountain building. However, have active earthquakes and vulcanism. This era is divided into Pleistocene and Holocene epochs with each of the epoch denoting the example of important developments. The Pleistocene epoch forms important present day map of the world evolved. Moreover, it represents formation of all the youngest reliefs i.e. river plains of the continents. In terms of resource, the Pleistocene epoch correlates to the crude oil and natural gas related sedimentary stratas. The present epoch called Holocene that began some 10,000 years ago before present is known ~~as~~ for Homo Sapiens.

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2. FACTORS, FORCES & PROCESSES THAT INFLUENCES SURFACE OF THE EARTH



ynamic characteristics are the topographical features developed on the surface of the Earth and is primarily related to the dynamic equilibrium of endogenic and exogenic forces. The endogenic forces are also called the tectonic forces which originates inside the surface of the Earth and registers influence on the Earth's surface as the development of these forces are directly related to the movement of lithosphere over aesthenosphere. These processes are also called Earth's movement. In the context of time taken by these forces, they are categorised into :

Diastrophism

Sudden Forces

Diastrophism : It forms slow processes that take million of years to generate variations on the surface of the Earth. In accordance to their nature of effectivity, they are classified as epeiriogenic and orogenic forces. The epeiriogenic forces are continent building forces that are

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vertical in nature. These forces correlates to the vertical upliftment or subsidence of large mass of land. Practically epeiriogenic forces are not identified in the formation of continents as all the continents of the world have marked their genesis around the shield or craton. It is these fundamental complex that supported the subsequent developments of endogenic and exogenic processes evolving present shape and size of the continents. In these developments, attachment of "Terranes" are specified as important development. Terranes are defined to be "small fragmented shields which eventually buckles up with main shield enlarging the size of the continent. Among the prominent examples are included : CHILOE ISLAND in S. America and VANCOUVER ISLAND in North America.

Orogenic Forces :

These are the horizontal forces that results in the formation of mountains. These forces are analysed as DEFORMATION or STRAIN. In accordance to the dimensions of orogenic forces, the deformation is classified into folding and faulting.

Folding as deformation is the outcome of compressional stress with the rocks acting in ductile manner. In the effect of compressional stress, original rock strata generates out Anticline and Syncline. The Anticlines are defined to be upthrown part of the fold. Syncline is, in comparision, the down-thrown part of the fold. It is in accordance that anticlines make the mountain peaks and synclines, the valley. The geometry of the fold is applied to interpret the nature of compressional stress applied to the rocks in the

development of the fold. It includes HINGE, AXIAL PLANE, and LIMBS. The limbs are defined to be the parts of the Anticline or the syncline. The intersection point of limbs is called the HINGE and the plane developed by combining the HINGE in both horizontal and vertical direction is called AXIAL PLANE. It is on the basis of the angles that limbs make with the axial plane that the types of the fold are classified.

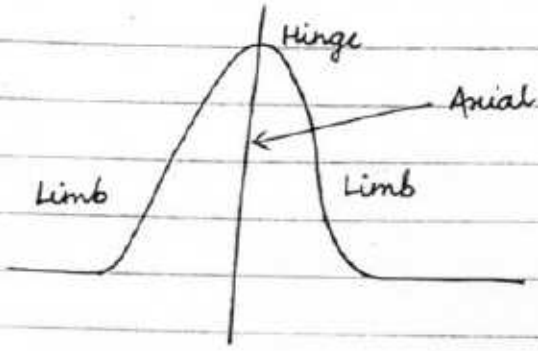


Figure: Geometry of fold

- Fold terminology in 2-D:

HINGE & LIMB →

* The hinge point is the point of maximum radius of curvature for a fold.

* The limbs are the flanks of the fold and the hinge is where the flanks join together.

* The axial surface is the surface defined by connecting all the hinge lines of stacked folding surfaces. Can be planar.

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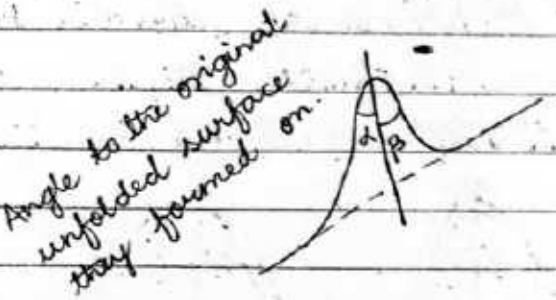
LECTURE # 7

Date : 23/02/2014

DIFFERENT TYPES OF FOLDS :

The nature of the compressional stress experienced by the ductile rocks is interpreted by the angle that the limbs makes with the axial plane. Primarily, therefore, folds are classified into 5 different categories:

i) Symmetrical Fold
ii) Asymmetrical Fold
iii) Isoclinal Fold
iv) Recumbent Fold
v) Nappe



1) Symmetrical Fold :

It is identified to be the fold type where limbs make similar angle with the axial plane. This represents similarity of compressional stress experienced from either sides. This type of folds are identified with Western Cordilleras of N.Am. and Andean Cordilleras of S.America.

2) Asymmetrical fold :

In this type of fold, the limbs make different angles with the axial plane. It, therefore, represents variation in the compressional stress from either sides. Both the Himalayan Cordilleras and Alpine Cordilleras are examples of asymmetrical folds

3) Isoclinal fold :

It is commonly designated as special type of symmetric fold with a tilted axial plane. The Middle Himalayas throughout its expanse are of this fold type.

i) Recumbent fold :

When the tilt of the axial plane becomes stronger making it horizontal or near horizontal, the fold developed is recumbent fold. It is also called sleeping fold and do not represent complete mountain range. However, are correlated to all the four young fold cordilleras.

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Nappe or Broken fold

It is also a major fold wherein the induced compressional stress ∅ surpasses the ductile characteristics of the rocks. This type of broken fold corresponds to either " terminus of greater Himalayas which corresponds to the SYNTAXICAL BENDS along River Indus and River Tsangpo (Dhang in Tibet)

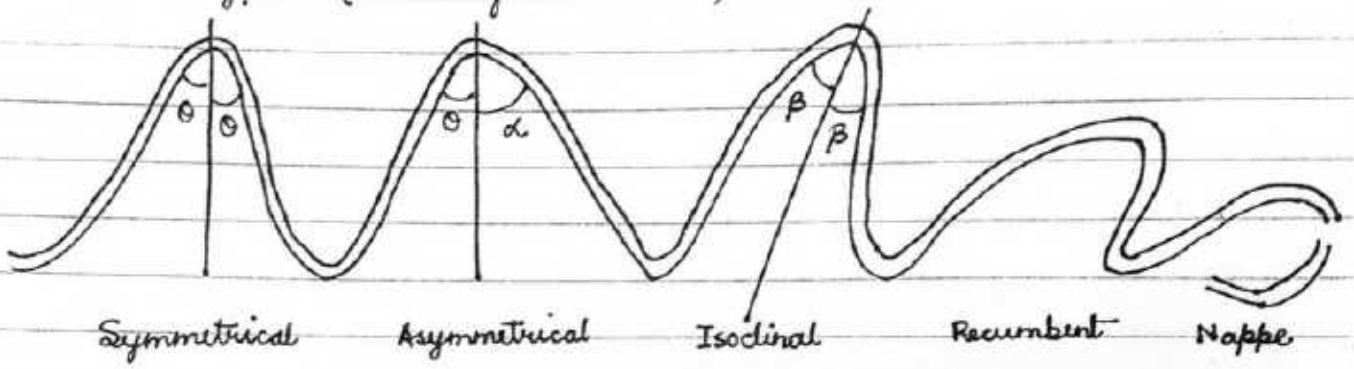


FIGURE : DIFFERENT TYPES OF FOLDS

MINOR FOLD TYPES :

These folds incorporate variable combination of geological construct and compressional stress. These are largely deprived of global representation. Prominent examples include fan fold, chevron fold & monoclinal fold.

Fan fold – Represents the category where multi-directional compressional stress have facilitated genesis of IMMATURE anticline and syncline called ANTICLINORIUMS & SYNCLINORIUMS respectively.

Chevron fold : This fold has unique structure generating geometrically correct triangular folds. It is variations in the compressional stress that open & closed folds are also categorised to be minor type of fold.

Monoclinal Fold :

It is a step-like fold in rock strata consisting of a zone of steeper dip within an otherwise horizontal or gentle-dipping sequence. It has only one limb depicting the condition where part of rock have ductile characteristic and another part brittle. If the complete rock is brittle in nature, compressional stress generates reverse faults.

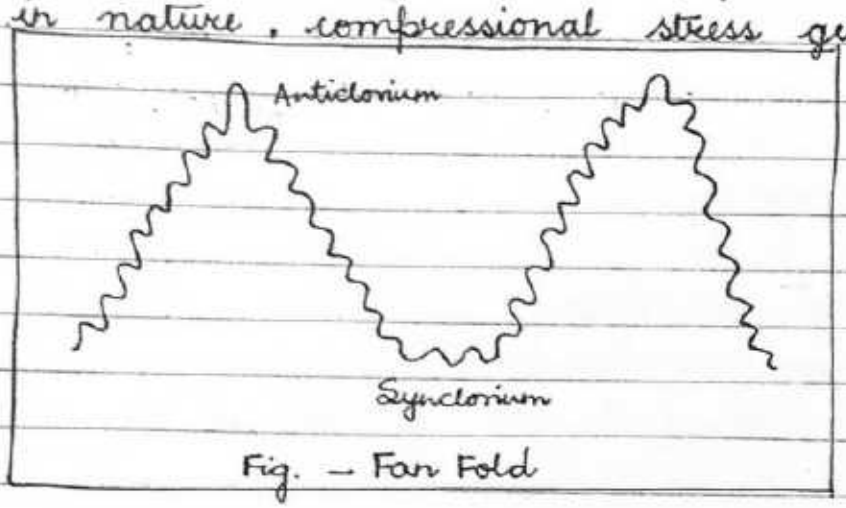


Fig. – Fan Fold

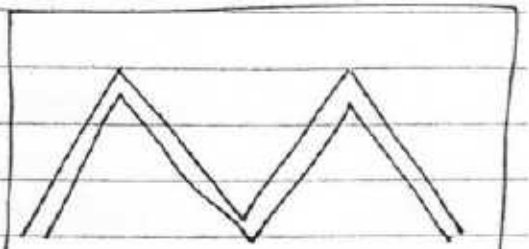
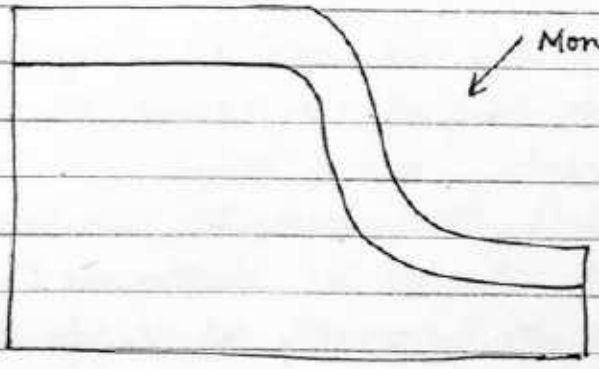
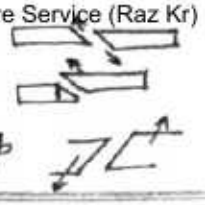


Fig. – Chevron Fold



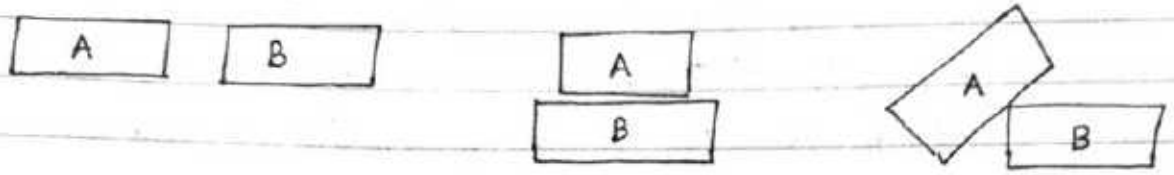
Reverse Fault : These are of 2 types namely THRUST FAULT & OVER THRUST FAULT. The overthrust fault is that type of reverse fault where there is absolute horizontal override of one slab on the other. However, elevation increase due to the ride of one slab over the other with

Fault — Normal, Reverse, Strike Slip



Hanging Wall
Footwall

re angle of 45° or more denotes the THRUST FAULT. It is this category that makes reverse fault applied to mountain building process with all the global examples confined along mightiest young fold mountains Himalayas.



Overthrust Fault — Thrust Fault

Faulting (Tensional Stress)

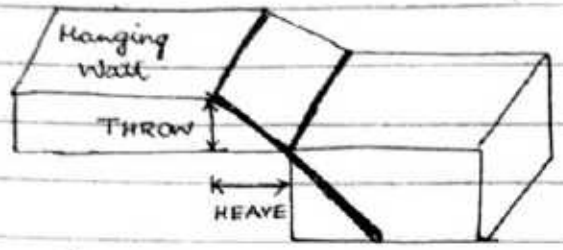
Faulting is defined to be plastic deformation leading to brittle failure or breakage. This strain or deformation is commonly correlated to the tensional stress which in turn is classified as NORMAL & OR TRANSVERSE tensional stress.

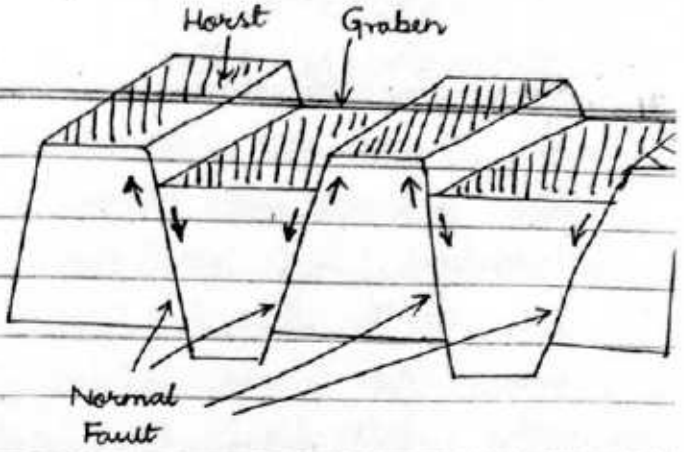
The normal tensional stress is correlated to the pulling apart effect which on brittle failure leads to the development of NORMAL FAULTS.

The transverse stress correlates to slide pasting effect, which when results in brittle failure, causes the development of transverse fault.

In both the cases, the development of fault stand synonym to horizontal displacement. This horizontal displacement in the geometry of the fault is called HEAVE. In the case of development of only heave, the developed fault is called STRIKE SLIP FAULT. For the mountain building process, simultaneous requirement of THROW i.e. vertical displacement is mandatory. The developed topography is called HORST & GRABEN topography or BLOCK MOUNTAIN WITH RIFT VALLEY, depicting DIP SLIP FAULT.

Pangea in Greek means "entire Earth"





Horst & Graben Topography

MINOR FAULTS : Transform Faults

1) These represent the development of brittle failure due to the differential rate of movement of the crustal slabs in the SAME DIRECTION.
2) These are, thus, examples of Strike Slip Faults
3) These faults led leads to the formation of lava plateaus on the continental crust and development of fracture zones with ancillary ridges in the submarine floor.
(4) These do not correlate to dip Slip fault.

THEORIES ON "ENDOGENIC FORCES"

Chronological development of endogenic forces with Azoic, Palezoic and Cenozoic formations are explained in the light of plate tectonic theory. This theory in turn marks its development in the gradual evolution process which involves

1) Continent Drift Theory
2) Sea-floor spreading

Antonio Pellegrini (1858)
F. B. Taylor (1910)

CONTINENTAL DRIFT THEORY

Its genesis is traced back to early 19th century when the pillar of modern geography, German scholar Humboll (1801) outlined striking similarity of the shoreline of South Atlantic Ocean.

His work was followed on by Antonio Pellegrini (1858) who attempted the first map of the region highlighting the similarity of shape of the shoreline.

The complete theory on the drift of the continent of the specified region was, however, attempted by F.B. Taylor (1910).

In the global perspective, the study of continent drift is credited to the German Climatologist – Alfred Wegener (1915) as the compilation of his conclusions of the analysis of PALEOCLIMATOLOGY.

In his attempt of identifying the nature of the climatic conditions experienced by different locations in different geological time, he concluded prevailing of contrasting climatic conditions compared to present latitudinal locations.

To prove these contrasts, he concluded "drifting of the continents from their "original" latitudinal locations/ positions to the present position.

In order to specify the cause of the drift he concluded "Free float of SIAL over SIMA in the effect of "Gravitational Force" & "Tidal Force"

Based on these factors, the propounded continental drift theory identifies the existence of the super-continent PANGEA to be existing till CARBONIFEROUS EPOCH surrounded by super ocean called PANTHALASA.

The beginning of disintegration of Pangea traced

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Angaraland Gondwanaland Tethys Sea	Equator-ward movement — Indian, Arctic, Southern Ocean west-ward movement — Atlantic & Pacific

from Carboniferous epoch, was simultaneous to the division of Panthalasa. Initially, Pangea got disintegrated into unequal parts called Angaraland and Gondwanaland.

Angaraland/Laurasia, formed the northern part of the Pangea, comprised of North America and Eurasia. The Gondwanaland, the southern part of Pangea incorporated S. America, Africa, India, Australia and Antarctica as major constituents. In between Angaraland and Gondwanaland, the intruded water of Panthalasa formed Tethys Sea. Continuation of disintegration due to gravitational force leading to equator-ward movement generated buckling up of Eurasia with Africa and India like landmasses along with the formation of Indian Ocean, Arctic Ocean and Southern Ocean. The disintegration due to the tidal force leading to the westward movement facilitated formation of Atlantic Ocean and substantive contraction of Panthalasa making the largest ocean of the world with the oldest oceanic rocks, the Pacific Ocean.

Drawbacks:

1) SiAl does not float over SiMa
2) Lunar & solar tidals cannot contribute to continent shift

ARGUMENTS IN FAVOUR OF THE CONCEPT:= Wegener's Concept includes the convincing arguments, contributed also by him, in favour of his ideology. These arguments include:

(1) Jig Saw Fit Evidences
(2) Paleoclimatic Evidences
(3) Paleomagnetic Evidences

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Isostatic change = Results in alteration to the global sea levels due to changes in either the volume of water in the oceans or net changes in the volume of the ocean basins.

The JIG-SAW FIT evidence : It has 3 different dimensions :

Jig = saw fit of shape (Wegener) (Sir Edward Bullard)
" " " " process
" " " " structure

1) FIT OF SHAPE :- The jigsaw fit of shape was emphasized by Wegener in highlighting near complete fit of shape of the S. Atlantic shoreline. Sir Edward Bullard (1960) further elaborated Wegener's conclusion through computer fit and concluded that the best fit is obtained if the coastlines are matched at the depth of 1000 m below current sea level, barring the exception of minor unconformities. The unconformities (gaps and overlaps) are explained by :

(a) Coastal erosion since continental separation
(b) " deposition " " "
(c) Rises in sea level (eustatic change) since sepn
(d) Changes in land level (isostatic change)

* Examples include erosion as well as deltaic deposits on either shoreline as that of River Niger and River Sao Francisco.

FIT OF PROCESS: The fit of process includes generation of normal tensional stress leading to the development of Dip-Slip fault generating Horst-Graben topography which includes Serra da Mantiqueira and Admawa highlands as the old block mountains with proto Atlantic floor proven to be the example of Rift Valley.

FIT OF STRUCTURE : The fit of structure primarily credited to Wegener involves the mention of absolute

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* Wegener could not explain how the plates move.

fit of the super imposed stratas compositionally on either sides of the Atlantic Ocean, which also includes embedded fossil remains of specialised land organism as "Mesasauraus". Moreover, economically important reserves specifically of Gold have been emphasized by the scholar as the evidence of fit where "Lode-deposits (metalliferous) of Guinea Coast is projected as placer deposits (mineral) of Belem (Brazil). The combination of these evidences proves the drift of South America away from Africa, thus, Continent Drift Theory.

2) PALEOCLIMATIC EVIDENCES :

These evidences are credited to A. Wegener. Based on his analysis, Wegener concluded that Gondwanaland constituents which are lower latitudinal locations in present perspective and far off from each other experienced ice-age together. This was simultaneous to warm-tropical climate being experienced by Angaraland constituents. Wegener identified that fitting Gondwanaland towards south polar location helps in explaining glaciation experienced by these landmasses and as none of these landmasses are in combined position today nor are in colder latitude, there has been the drift of the continent. For Angaraland, flowrishing tropical vegetation of the geological past due to the drift of the continents in comparative ~~lower~~ colder climate have paved way to the formation of rich coal belts availing added evidence of continent drift. The paleoclimatic evidences are recognised to be the most compelling argument in favour of continent drift

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theory presented by Wegener.

However, as his approach was significantly "different" from the solid ideologies of geomorphological studies, his work was completely sidelined as the work of imagination. The arguments presented against the concept included:

Inappropriate mechanism

Treatment of oceanic crust as passive crust

Failing to explain 2nd order relief.

The primary reason to discard Continent Drift Theory was that the concept was technically wrong. SiAl cannot float over SiMa as Moho discontinuity falls well within lithosphere.

The description of equator-ward and westward drift proves to be faulty as most of the Gondwanaland constituents have actually crossed equator.

The tidal force is proven to be "too weak" to cause the drift of continents.

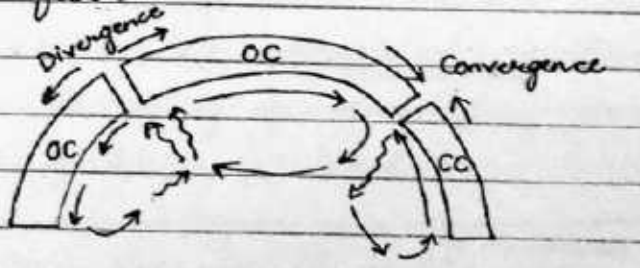
The entire concept is oriented towards the movements of continents and presumed the oceanic crust to be absolutely passive.

Only availed the description of relative positioning of the first order relief and failed to justify geomorphic requirements in analysing the formations of 2nd order relief.

2/ SEA-FLOOR SPREADING CONCEPT : HARRY HESS

1) Based on the study of submarine topography specially specifically the Challenger Ridge of southern Atlantic Ocean and Albatros Plateau of S.E. Pacific, American scholar Harry Hess proposed the concept of sea floor spread.
2) He based his concept on the "convective magma currents approach", propounded by Arthur Holmes in accordance to which movement of magma currents in Aesthenosphere forms the cause of movement of lithosphere, creating tectonic forces.
3) In accordance to convective magma current concept, magma cells generate converging, diverging as well as transverse movement of lithosphere over aesthenosphere.
4) Harry Hess applied diverging and converging movements to develop his sea-floor spread concept. He emphasized his concept to be recognising drift of continent as outlined by A. Wegener along with spreading of the sea-floor.

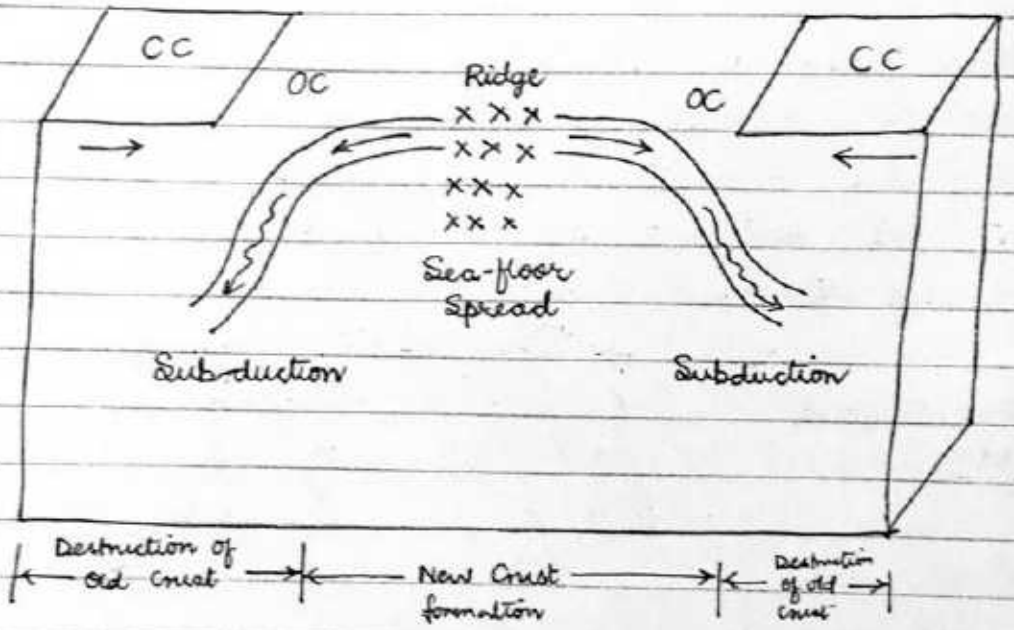
Fig. = Convective Magma Cells



5) The concept of sea floor spreading emphasizes on the construction of new oceanic crust at the spreading boundary and destruction of old crust at converging boundary. Taking empirical evidences along the spreading boundary, Harry Hess identified existence of paired rocks along either sides of the submarine ridge. He identified paired rocks to be similar in terms of their formation

is well as their composition. He concluded that with increase in the distance from the central ridge, age of paired rocks goes on increasing, justifying the seafloor spread. The destruction of the old crust at the converging boundary proves ACTIVE sea-floor and so also explains the fact that the oldest oceanic crustal rocks are not older than Paleozoic era whereas oldest continental rocks are as old as Azoic era.

The concept of sea-floor spread proves that Atlantic crust forms the example of spreading boundary whereas the Pacific crust represents destructive boundary.



GEOMAGNETISM

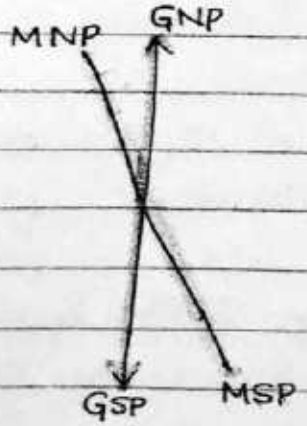
The magnetic property of planet earth generated due to iron rich interiors is called geomagnetism. This field of study in geomorphology is credited to William Gilbert, American scholar.

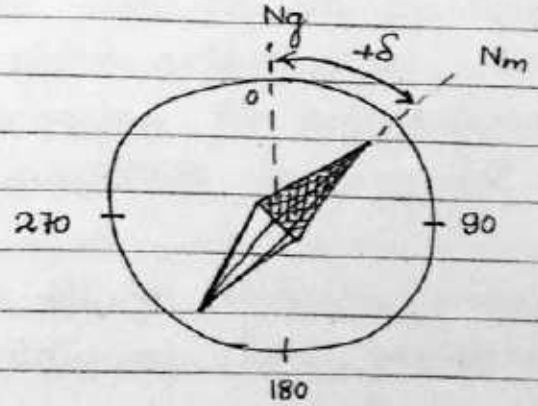
He identified that the Earth's magnetic field is like a giant bar magnet which is caused due to iron rich core. Due to prevailing temperature, molten core registers movement, creating dynamic characteristics of the magnetic field. The analysis of geomagnetism for geomorphological purposes involves 3 interrelated components :

① Magnetic Declination
② Magnetic Inclination
③ Magnetic Equator

i) Magnetic Declination :- (Also called horizontal component)

- It is the angle on the horizontal plane between magnetic north and geographic (true) north. By convention, the declination is +ve when magnetic north is east of true north, and negative when it is to the west.
- In the present perspective, magnetic declination reveals north magnetic pole to be slightly west of GNP and thus, MSP slightly east of GSP.
- The molten mobile iron-rich core, which spins like a dynamo forms the cause of gradual westward drift of geomagnetic field, resulting in changing magnetic declination in entire geological past.





"POLAR WANDERING - NCERT"

Magnetic Inclination / Dip Angle / Magnetic Dip :=

This component of geomagnetism is identified to be the vertical component. It is defined to be the angular inclination between freely suspended magnetic needle & the horizontal plane of the earth's surface.

Magnetic dip results from the tendency of a magnet to align itself with lines of force. With increase in the sign of latitude, the dip angle increases, making it 90° at the magnetic poles and 0° at geographic equator.

Magnetic Equator :

The locus of the points having zero dip is called the magnetic equator (or aclinic line).

PALEOMAGNETISM :

It is the study of the record of "direction and intensity of the magnetic field", locked in rocks formed in different geological time period. These records are classified as :

Thermoremanent

Depositional remanent

These categories depict maintenance of record, not just in the igneous rocks but also in sedimentary and metamorphic rocks.

The major contributers to paleomagnetic studies include Runcorn, Fred Vine and Matthews.

Runcorn

His paleomagnetic studies in the early 1950's provided compelling evidences that facilitated "Revival of Continental Drift Theory.

He recognised that in the past 500 million years,

North magnetic pole have registered wandering from near ~~Hawiian~~ Hawaiian Islands via Eastern Siberia to its present position at Bathurst Island, Canada.

ii) In this apparent north magnetic polar wandering, he facilitated the understanding of combined existence of Eurasia and N. America.

iv) Moreover, it helped in the conclusion that the Appalachians, Kjollen, Urals, Verkhoyansk and Chukotsky forms the part of the same old fold orogenesis. It is in accordance that their present respective position readily justifies continent drift.

Fred Vine & Matthews in Paleomagnetism

i) In 1963, Fred Vine and Matthews added to the studies of paleomagnetism by identifying Zebra Strips.

i) Analysis the magnetic records of the oceanic basins and the ridges, they concluded that older existing basins show polarity different than Earth's magnetic field, referred to as reverse (weak) polarity. On the other hand, the younger ridges represent normal polarity.

ii) This paleomagnetic conclusion is recognised to be evidence of sea-floor spreading as the older diverging basins generate condition for new ridge to get formed.

v) Moreover, with increasing distance from the central ridge, with increase in the age of the paired rocks, there is also gradual increase in magnetic anamolies.

vi) This Zebra strip concept facilitated consolidation of sea-floor spreading propounded by Harry Hess.

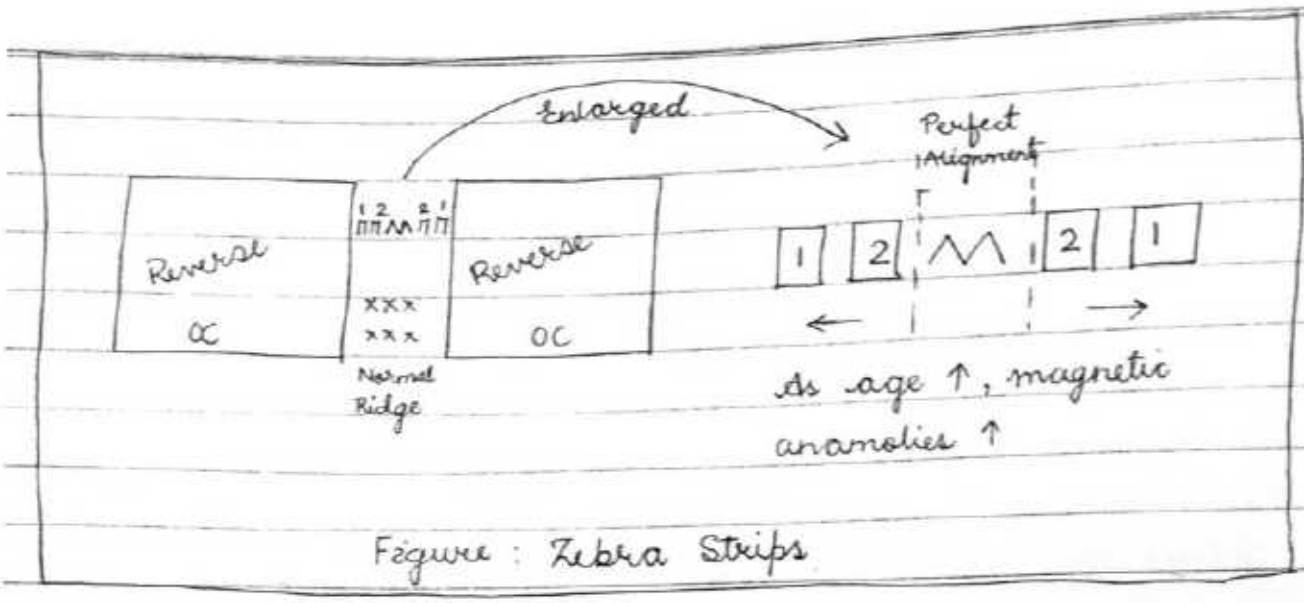


Figure : Zebra Strips

PLATE TECTONIC THEORY

In late 1960's consolidation of different work of different geomorphologist towards analysis of locational characteristics of tectonic forces led to the plate tectonic theory. Among the major contributors – Parker, Harry Hess, Woodridge, Morgan, Tuzo Wilson® are included. The tectonic theory is based on the recognition that under the effect of convective magma currents, lithosphere floats over aesthenosphere.

The different parts of lithosphere are being referred to as PLATES.

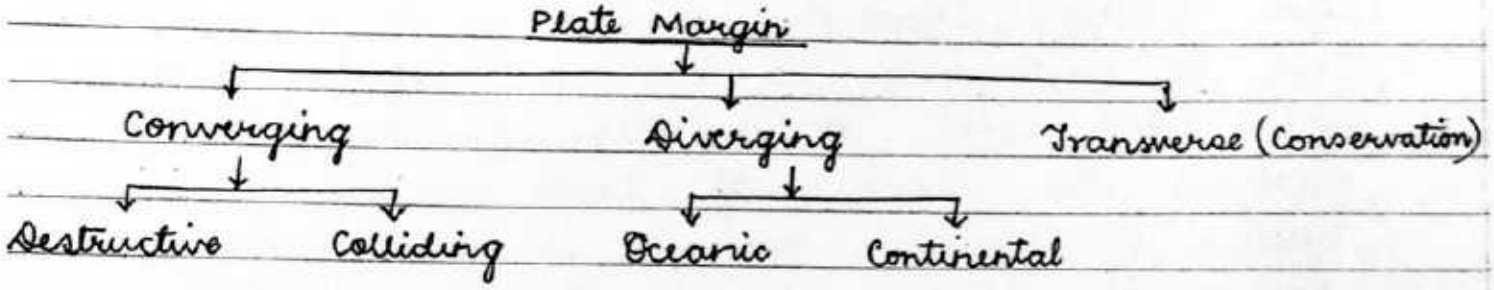
With their active tectonic characteristics, movement of these 7 tectonic plates in relation to each other forms the cause of development of tectonic boundaries or plate margins.

At present there are 7 major tectonic plates namely

i) Pacific Plate
ii) N. American Plate
iii) S. American Plate
iv) Eurasian Plate
v) African Plate
vi) Indo-Australian Plate
vii) Antarctic Plate.

Each of these major plates involves different number of minor plates with their respective tectonic boundaries.

As all the endogenic forces takes place at tectonic boundaries or plate margins, the study of tectonic theory is primarily confined to the study of types of boundaries. The major categories of boundary include :



CONVERGING BOUNDARY :

The converging plate boundaries are recognised to be geomorphologically most important boundary as it correlates to the continental crust. Along this type of boundary, generated compressional stress results in fold orogenesis. Deformation of rocks causes earthquake. However for volcanic activity, destruction forms a mandatory clause. It is in accordance that converging boundary is classified into :

(a) Converging Destructive & (b) Converging Colliding.

a) Converging Destructive :

The converging destructive boundary on the map of world relates to Circum-Pacific Belt and Mid-Continental Belt. The circum-pacific belt corresponds to the entire shoreline of Pacific Ocean involving convergence of Pacific plate with the

TRANSVERSE BOUNDARY :

This ' boundary was outlined by TUZO WILSON who called this - ' boundary transform boundary. Along this boundary slide pasting effect of the continental crust paving way to faulting, vulcanism & earthquake – have been recognised. The only location of this plate boundary is SW N. America where California 'Minor plate' is slide pasting towards south / south-east whereas rest of N. America is moving towards ~~South~~ / ~~South~~-North/ north-west. It is this boundary that justifies the presence of block mountains in the Western Cordilleras. Moreover, it also forms the cause of absence of development of continuous "North American Trench."

APPRAISAL OF PLATE TECTONICS :-

Since the publication of Plate Tectonic Theory in late 1960's, complete orientation of the studies of endogenic forces were channelised, as the theory not just availed the understanding of tectonic boundaries developed in TERTIARY PERIOD but also provided the recognition to the dynamic characteristics of tectonic boundaries, in reference to which, changing locations of tectonic boundaries in the geological past have been taken up to be the cause of different location of older reliefs. In accordance to the theory formation process of older mountains are explained in the light of the fact that :
Law of nature do not change with time.

(ii) Tectonic forces remain applicable only to the plate boundaries.

This full-proof theory have been eventually added with two modifications, recognised as:

(1) HOTSPOTS

(2) ACCRETION WEDGES

Hawaiian I. located in the north central part of Pacific Ocean forms the excellent example of hotspot developed due to INTRAPLATE OCEANIC DIVERGENCE, which is not given recognition in the original plate tectonic theory. The hotspots are defined to be localised spot of lithospheric opening from where there is continuous ejection of molten rock material. Long-lived hotspots are identified with the age of upto 100 M years.
In the beginning of the Cenozoic Era, converging boundary of Northern Circum-Pacific generated the hotspot where continuous ejection of magma, in accordance to the plate drift, facilitated formation of island chain called Hawaiian Islands. The oldest island of this chain is called EMPEROR SEA MOUNT and the youngest island in formation is LOHII. This active intra-divergence forms added evidence of plate movements and tectonic activities. At present, the term hotspot is only applied to almost all vulcanic islands.

The ACCRETION WEDGES or converging boundaries developed due to attachment of terrains with the mainland, have been added to the plate tectonic theory as additional example of tectonic boundaries.

SUDDEN ENDOGENIC FORCES

VULCANISM :

Solidified → ↑ Extrusive → Surface → Vulcanic
Solidified → ↓ Intrusive → Plutonic

Magma Components :

→ Silica ↑ ⇒ Mobility ↓ ⇒ Viscosity ↑
→ Iron ↑ ⇒ Density ↑

Ejected Matter
- MAGMA
  - sic
  - 40% Si
  - ateaus
  - altic
  - re Fe Content
  - nser
  - Acidic
    - → > 70% Si
    - → Cones
    - → Granitic
    - → Less Fe Content
    - → Viscous
- GASES
  - → H_2O (g) [~70%, most prominent]
  - → CO_2
  - → SiO_2
- PYROCLASTS
  - → Solid particles ejected

The sudden endogenic force that is capable of developing 2nd order relief feature as lava plateaus and vulcanic cones is defined to be all set of processes that leads to movement of molten rock material towards the earth's surface. This molten rock material called MAGMA, in its composition and trapped gases reveals various intensities of ejection. These intensities based categories of vulcanism is best described with the reference to the plate tectonic theory. The distinguished categories in increasing order of intensity of ejection includes:

* Converging
Destructive

Hawaiian	↕	Quiet
Strombolian		Transverse
Vesuvian	↕	Strong
Pelean		

i) Hawaiian Type of Activity :-
This type of vulcanic activity is the quietest which is correlated with complete absence of trapped gaseous constituents and involves quiet ejection of mobile magma in well-defined individual flow of lava streams, technically called as PELE'S HAIR. This type of activity is confined along oceanic divergence and derived its name from the "smoking mountains" of Hawaiian Is. as Mt. Mauna Loa.

ii) The Strombolian Type :
This vulcanic type represents the category which is slightly stronger than the Hawaiian type. It involves some trapped gases but is not associated with precipitation after vulcanic ejection. This type of activity attains its name from * Mt. Stromboli of Italy. is designated to be the "LIGHTHOUSE OF MEDITERRANEAN". Tectonically this type of activity is correlated to continent divergence as Mt. Kilimanjaro and Transverse boundary as Mt. Whittney, CA, USA.

ii) The Vesuvian Type:
It reveals stronger activity and is also identified to be the commest type of vulcanic activity confined along converging destructive boundaries. This activity therefore represents all circum - Pacific belt and mid-continental belt with

* Converging
Destructive

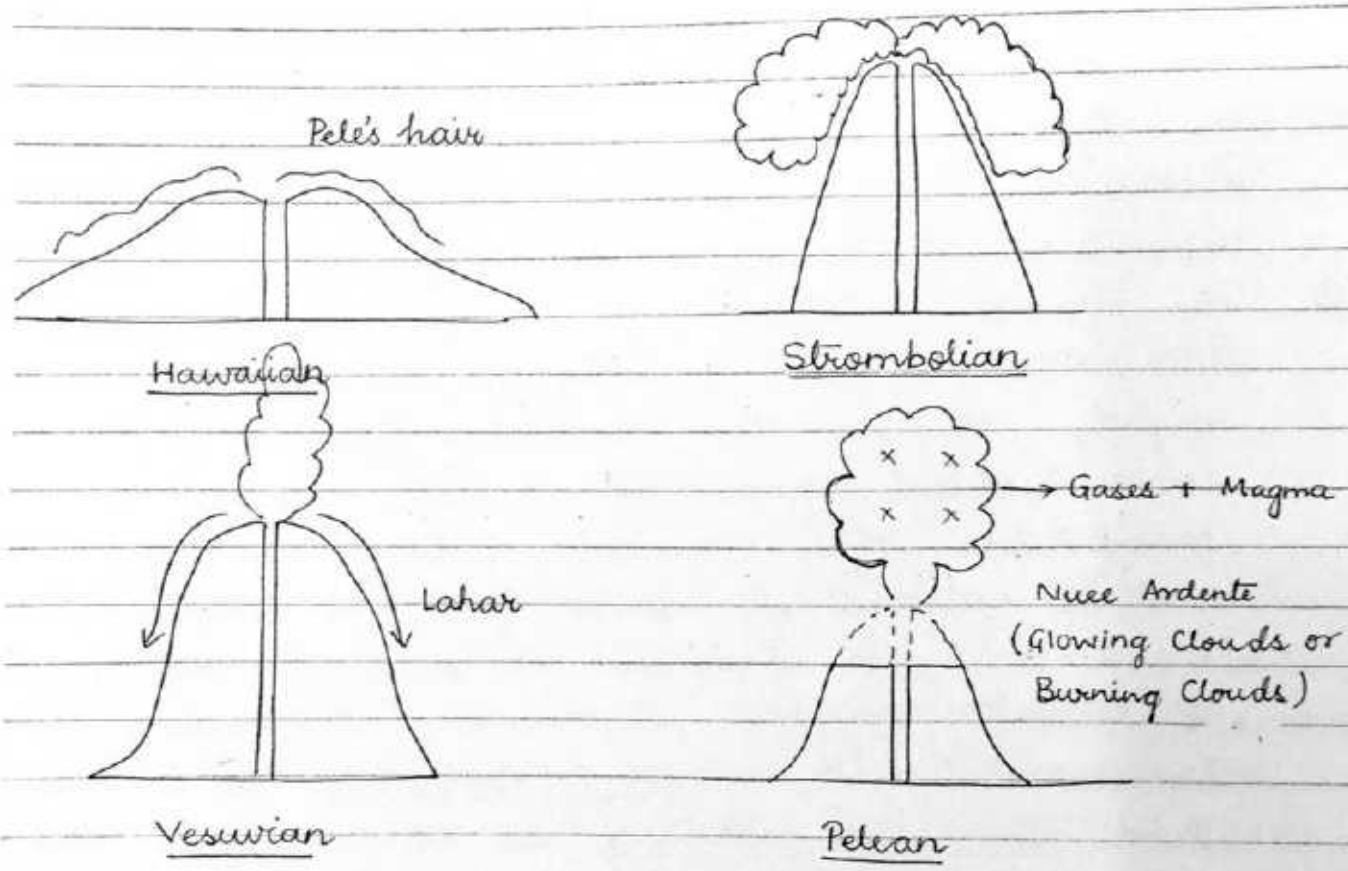
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Mt. Vesuvius as the most studied volcanic cone, providing the source of the name. The intensity of ejection in this activity is such great that trapped gases are thrown to greater heights always resulting in condensation & precipitation. This type of activity thus is also always correlated to LAHAR, i.e. down-slope movement of water saturated pyroclast generating Piedmont deposits of volcanic soil.



Pelean Type :

When Vesuvian type of activity becomes further intensified, it is called pelean type of activity. In this activity existing cone gets blown off generating largest amount of pyroclasts. However,

with complete absence of 'Lahar'. Another characteristic is the 'Burning or glowing clouds' that are the thrown magma patches upto the height of the clouds. This type of activity has been experienced only in Krakatoa in Greater Sundae Is. (East Indies) and Martinique Island in Lesser Antilles (West Indies).

Original plate tectonic drawbacks :
(i) Hotspots
(ii) Terranes

Write note on Volcanoes ⇒ Extensive Vulcanoes, lava plateau

Critically examine the continental drift theory of Alfred Wegener (250 words)

Based on the composition of ejected magma, volcanoes are classified into three well demarcated categories. These include :

Basic Cone

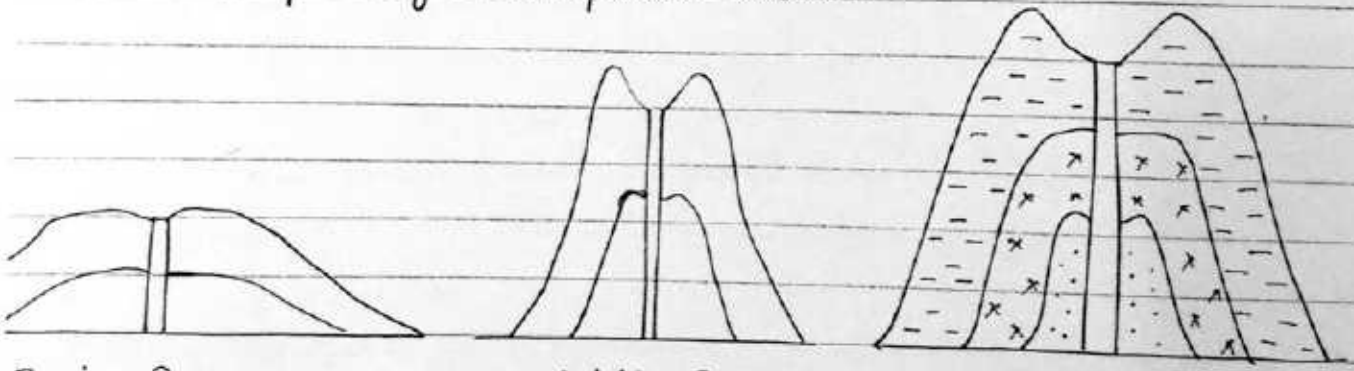
Acidic Cone

Composite Cone

BASIC CONE : It represents the volcano developed by mobile type of magma with broad base, gently sloping sides and shorter height. Mt. Mona Loa makes excellent example of such cones.

ACIDIC CONE : The acidic cones incorporate more viscous magma, have narrow base, steeply sloping sides and taller height. Mt. Vesuvius forms excellent example.

COMPOSITE CONE : As every central type of ejection correspond to well-developed stratas developing different episodes of ejection ; compositionally, each strata involves elementary difference making every vulcanic cone example of composite cone.



Basic Cone Acidic Cone Composite Cone

TYPES OF VOLCANIC CONES :

The Central type of activity also correlates to the development of distinctive 2nd order features. These includes :=

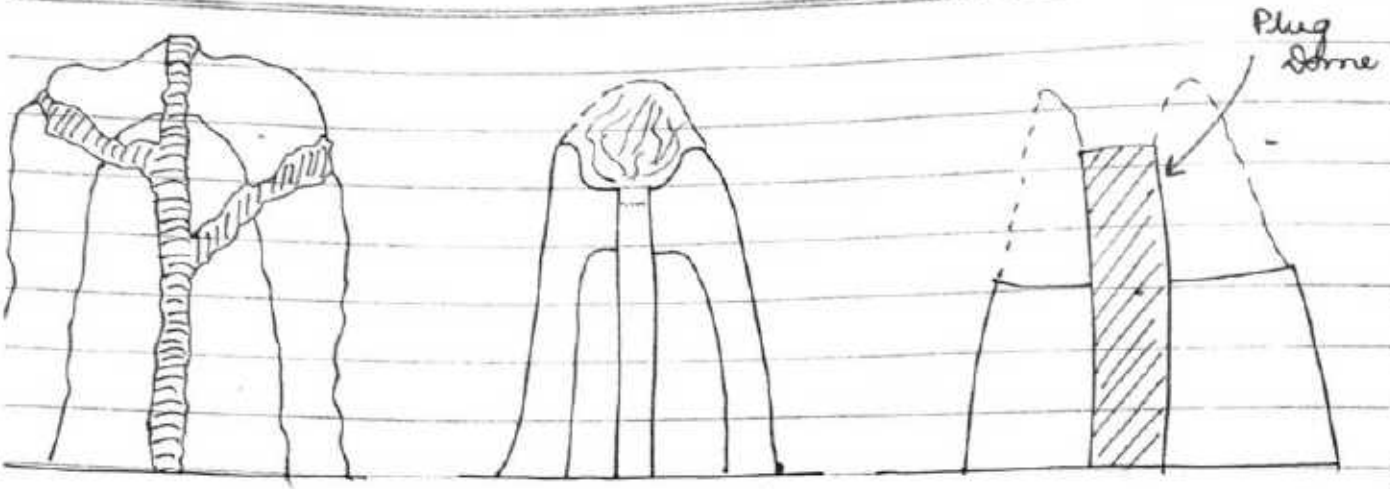
(i) Parasitic Cone
(ii). Cinder Cone
(iii) Caldera Dome
(iv) Plug Dome (Special type of dyke")

The parasitic cone develops when the pipe of the volcano have more than one vent which results in the development of more than one crater in the volcanic cone. Mt. Shasta (USA) and Mt. Fujishan (Japan) makes the examples.

The Cinder Cone in comparision represents that type of volcano cone which involves geometrically correct conical shape evolved due to deposition of pyroclasts In the crater Mt. Mayon (Phillippines) makes excellent example. The feature called Caldera is defined to be enlarged crater. This enlargement can be correlated to exploding of the existing cone or collapse of crater. Enlarged crater called caldera do incorporate possibilities of "exposed pipe" i.e referred to be plug dome.

Both crater (of dormant volcano) and caldera can incorporate accumulation of rain water developing lakes. It is only with caldera lakes, however, that the presence of central islands is projected.

Prominent examples of crater lakes of the world includes L. Toba (Indonesia) , L. Taupo (New Zealand)



Parasitic Cone Cinder Caldera & Plug Dome

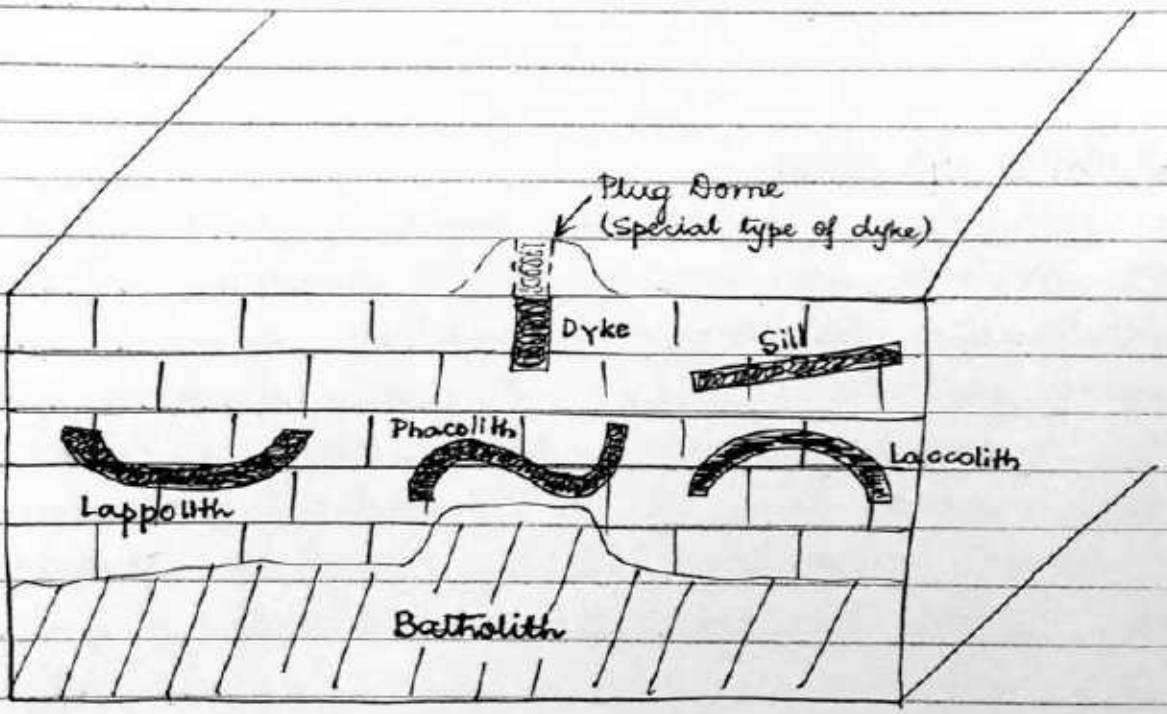
<u>ntrusive Vulcanism</u> : → Massive
→ Tabular → Concordant
→ Discordant

lovement of molten rock material towards the surface of the earth wher makes it solidified wilhin the existing lithospheric stratas, intrusive vulcanism is identified. Magma in its passage towards the surface partially melts the host rock and occupies the created location with variable shape and sizes. The intrusive vulcanic features therefore are classified in absolute accordance to their shape and size.

Based on size, massive plutons are distinguished from tubular plutons. The <u>massive plutons</u> are the gigantic size of magma deposits within the lithospheric stratas where demarcation of alignment of magma deposits is unlikely. Such huge masses are largely developed in most plutonic locations where the intruded magma is having the capacity of melting maximum of the host rocks. Being most plutonic, such type of deposits are largely not exposed on the surface of the Earth. Such massive plutons are called <u>batholiths</u> and smaller extensions are called <u>stock</u>.

Batholiths are always associated with xenoliths that are remains of host rock. The Dartmoor Range in Great Britain (U.K.) is believed to be the example of exposed batholith. The tabular plutons represent those magma intrusions which are not just smaller in size but are also well specified with alignment of magma deposits. This category is distinguished as concordant and discordant features.

Concordant : Horizontal layers
Discordant : Vertical layers
Acidic magma : Cools swiftly, forms irregular shapes.



as Concordant Tabular : The concordant tabular features are horizontal or near horizontal i.e. in accordance to the alignment of original beddings of lithosphere Sill makes par excellent concordant feature. It

represents solidification of more mobile magma in horizontal sheets. Excellent exposed example of sill is PALISADE SILL along Appalachians near Hudson River. The variance of concordant features involving slight acidic magma includes laccolith, lappolith and phacolith. The laccolith are dome-shaped tabular magma deposits which makes one of the commonest concordant plutons. Most recognised example of exposed laccolith is GHAAP PLATEAU (South Africa). The lappolith are the saucer-shaped tabular magma deposits. They represent their exposed example as BUSH VELD (South Africa). Phacolith makes the wave-shaped concordant pluton significantly smaller in size.

Discordant Tabular :

This category includes the tabular plutons that have vertical or near vertical magma deposits perpendicular to original bedding of rock. This category includes DYKES. The only identified feature with a variant called plug-dome exposed dykes makes few of the important mountain of the world including UKMES RANGE in Zimbabwe, and SUGAR LOAF MOUNTAINS in Brazil.

Vulcanic Landform features
- (i) Lava Plateaus
- (ii) Cones & related features
- (iii) Intrusive Volcanoes

MOUNTAIN BUILDING WITH GEOSYNCLINE & THEORIES

Mountains are defined to be the most projected 2nd order relief features with conical profile, steep sloping sides, tall peaks and deep valleys. This feature, therefore, involves the combination of both endogenic upliftment and exogenic down-cutting. Geomorphologically, mountains are classified both on the basis of process of formation and time of formation. In the process of formation - fold mountains, block mountains and relict mountains are distinguished.

i) The relict mountains represent the category that are effectively eroded by exogenic forces. Relict mountains therefore, corresponds to Ancient and old mountains classified on the basis of time of formation. The ancient mountains that are conical remains of ancient shields belongs to Azoic era. Old mountains representing paleozoic era includes Caledonian i.e. Devonian and Hercynian i.e. Permian categories of mountains. The Caledonian categories include Appalachians, Eurasian mountains, and Sierra De Mantiquera as major examples. On the other hand, the Hercynian category includes great Dividing Range, Drakensberg as important examples.

ii) The young mountains belonging to Cenozoic era with their original heights represent to be fold or block categories and not relict.

Relict ⇒ Not growing anymore Getting levelled by levellers. Azoic, Paleozoic	Cenozoic ⇒ Fold or Block

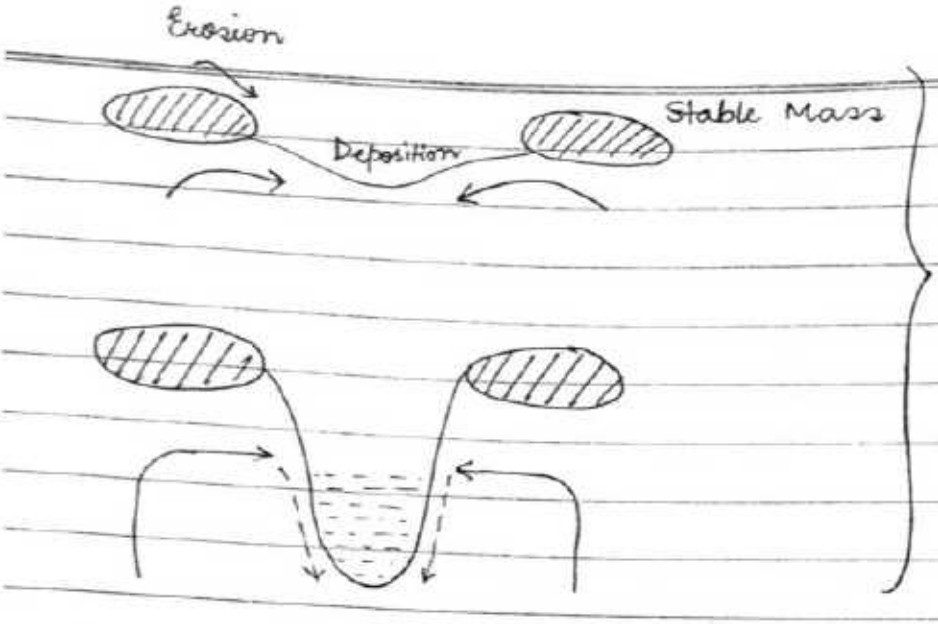
metamorphic rocks as quartzite, shale and slate. In comparision, the eugeosyncline relating to oceanic margins apart from involving sedimentary and metamorphic stratas also includes igneous i.e. vulcanic stratas.

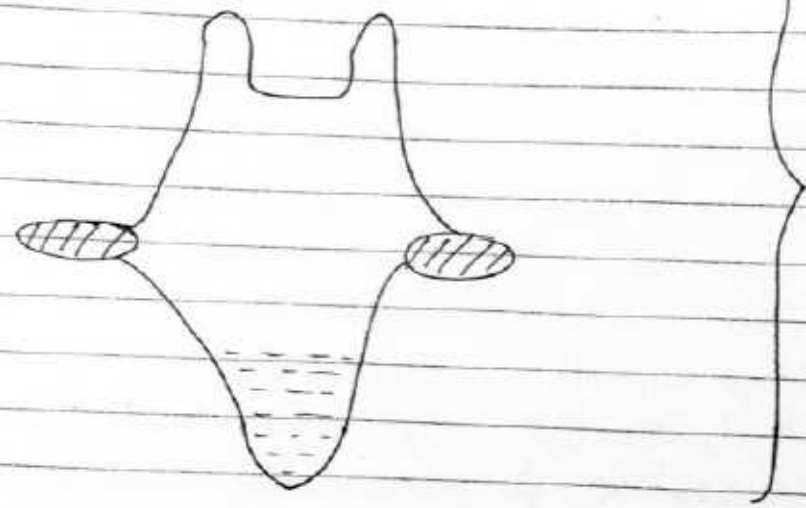
The contribution made by Schuchert in the development of concept of geosyncline is classifying geosynclines on the basis of processes of sedimentation and orogenesis. He identified 2 types of geosynclines called monogeosyncline and polygeosyncline. The monogeosyncline correlates to single major episode of sedimentation followed on by single major episode of folding. Appalachians forms the excellent example of such geosynclines. In comparision, the polygeosyncline involves many episodes of sedimentation and many episodes of folding, resulting in parallel running mountain ranges. Himalayas make excellent example of polygeosyncline. Applying to this category of classification, geomorphologically almost all old fold mountains are referred to be representing monogeosyncline and almost all the young fold mountains are referred to be representing poly-geosyncline.

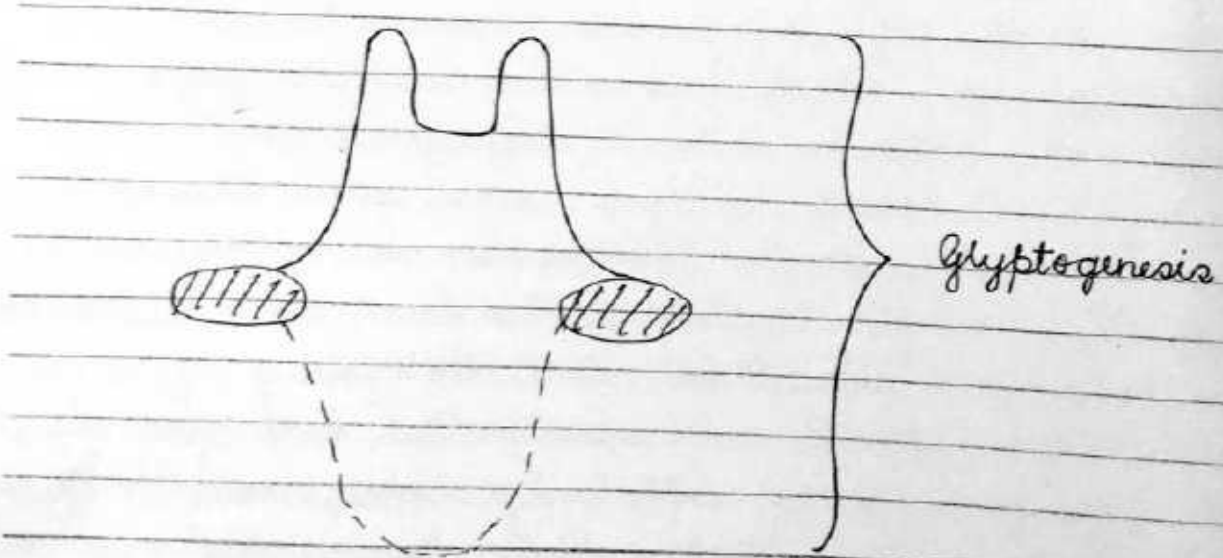
The established concept of geosyncline was applied for mountain building process by Kober. His mountain building approach incorporate 3 well-defined stages — Lithogenesis, Orogenesis and Epeirogenesis. His descriptive approach was made rationally-oriented by Arthur Surther Holmes who integrated

convective magma currents as the cause of deepening of geosyncline and generating compressional stress.

- The lithogenesis stage involving shallow and broad geosyncline bounded by stable masses on the either side is believed to be experiencing continuous sedimentation due to the erosion of the stable masses. By the end of the stage of lithogenesis, the geosyncline registers its transformation from broad shallow to significantly deep and narrow.
- Beginning of compressional stress sets in the beginning of the folding of these sediments i.e. the stage of orogenesis. It is in this stage that along with the development of parallel mountain ranges there is also the development of medial mass. These are comparatively less elevated regions depicting weaker compressional stress with increasing distance from the stable mass. All the inter-montane plains and plateaus are medial masses Eg. := Tibetan plateau, Hungarian plains etc.
- The stage of orogenesis concludes at Glyptogenesis which represent complete folding of sediments with or without closing down of geosyncline. Kober's analysis was given the scientific correction by Arthur Holmes, increasing the validity of the concept. Practically, however, this approach explains fold mountains, that too young fold mountains along mid-continental belt proving to be insufficient in explaining orogenesis. It is only plate tectonic theory that explains orogenesis in complete perspective.


Erosion
Stable Mass
Deposition
Lithogenesis


Orogenesis


Glyptogenesis

Map: Europe –
Dinaric Mountains
Hungarian Plains
Carpation Mts.

N. America –
Columbia
Colorado (intermontane)

Asia –
Kunlun Shan
Tibetan Plateau (Medial mass)
Himalayas

PLATE TECTONICS AND OROGENESIS:

(i) Tectonic theory recognises mountain building to be effective at plate boundary.
(ii) Dynamic locations of boundaries are applied to explain older mountains.
(iii) Explains young mountains in all its categories and location as :=
(a) Mountains at colliding boundary
(b) Mountains at destructive boundary
(c) Mountains at diverging boundary
(d) Mountains at transverse boundary.

A) Colliding boundary mountains as Himalayas depicts Miogeosyncline with no vulcanism.
B) Destructive boundary mountains are also fold like Western Cordilleras representing Eugeosyncline with active vulcanic cones. It can be divided as:
(i) Andean Type (Oceanic - Continent Convergence)
(ii) Mariana Type (Oceanic & Oceanic - continent convergence)
C) Diverging boundary mountains are block mountains

on continent crust as Mitumba, Muchinga along great African Rift Valley and includes extensive submarine ridges that are oceanic volcanic mountains as mid-Atlantic ridge.

) Transverse boundary mountains are again block mountains with stippage movement as Sierra Nevada and ~~Wastach~~/ Wasatch (Utah, USA).

EXOGENIC FORCES

Exogenic forces are defined to be the forces that originates on the surface of the Earth and induces effect on it. These are defined to be the levellers by the father of geomorphology, Davis.
The exogenic forces includes :

(i) Weathering
(ii) Mass Wastage
(iii) Gradation

Weathering is defined to be "static / in-situ" disintegration or decomposition of the rocks. It is, when combined with mass wastage i.e. gravity transfer of the broken rock, acts as levellers. Compared to it, gradation is the continuous levelling process which involves three simultaneous activities called :

- Degradation (erosion)
- Transportation
- Agradation (deposition)

These are carried on by the agents of gradation which individually develops different topography and/or cycle in their role as levellers. These agents include :

(i) Running Water (Fluvial topography / cycle)
(ii) Glacier (Glacial Cycle)
(iii) Wind (Eolian Cycle)
(iv) Sea Waves (Coastal Cycle)
(v) Underground water (Karst topography / cycle)

Among these 5 agents running water or fluvial action is recognised to be <u>universal</u> in its

nature as it is effective in all the geographical locations. This universal agent involves the categories of erosional capacity depending on the volume of water. It is in accordance that splash erosion, sheet erosion, gully erosion and stream erosion are distinguished. The splash erosion is carried on by individual rain drop with their falling force in the moistureless sandy surface of deserts. In such locations absence of vegetative barrier also facilitates development of flash floods following sudden torrential rain. Such flash flood water creates sheet erosion i.e washing away of entire top layer of sandy surface. More permanent fluvial erosion includes gully erosion carried on by streamlets that are smaller streams with more seasonal characteristics. The true fluvial action corresponds to stream erosion i.e. river erosion. In this category, fluctuations in the volume of water in the annual seasonal cycle regulates the erosional capacity i.e. That is called river regime. Prominently the river regime is classified into 3 categories :=

(i) Simple regime
(ii) Complex regime
(iii) Compound regime

Simple regime is represented to be water volume fluctuations with one high level followed on by low levels. Such type of regime is applied to the river systems which have same season of

glacial melt and precipitation combined with less dense network. R. Volga (Russia) and the lengthiest river of China, Chang Jiang, represents the example

Complex Regime:

The complex regime is developed when annual water volume fluctuations develop 2 maximas. This type of regime is typical to Mediterranean climate with winter concentrated rains as it is in lengthiest river in Europe, River Danube and it is also identified in equatorial climate where during equinoxes there is always rain maximas. R. Congo forms excellent example.

Compound Regime:

This regime incorporates lesser fluctuations in river regime as per the effectivity of dense drainage network. All the big river systems as Gangetic system, Mississippi - Missouri river system, Paraguay - Parana river system forms excellent examples.

Map: S. America: River Amazon (Complex)
R. Paraguay - R. Parana (South) (Argentina, Brazil)
Rio Deo Plata (mouth)

Africa: R. Nile (L. Victoria - Source)
Simple regime
R. Congo (Dem. Republic of Congo)

Asia: Chang Jiang (China)

Europe: Southern Germany → Black Sea (Danube River)
Mediterranean Climate

Facebook Group: Indian Administrative Service (Raz Kr)

LECTURE (13)

I March 2014

Long-profile of river :

Young	Mature	Old
Source		Destination / Mouth
Erosion	Transportation	Deposition

NORMAL CYCLE OF EROSION / DAVISIAN CONCEPT / FLUVIAL CYCLE OF EROSION

American geomorphologist William Morris Davis propounded normal cycle of erosion way back in 1889, to identify the nature of influence induced by universal leveller – running water or stream. He based his concept on the empirical studies carried on in different physiographic regions. He outlined that depending on the nature of prevailing climate, distinctive type of levellers will register their effectivity in the development of landscape. He emphasized that exogenic downcutting begins only after conclusion of endogenic upliftment. Moreover, the exogenic downcutting has well-defined stages of erosion and that end-product of erosional cycle is called, erosion surface, will be featureless lowland. To develop his concept, he utilised universal agent in the hypothetical location after the completion of the upliftment process. His entire concept is based on the fact that landscape is function of structure, process and stages, i.e.,

$$L = f(\text{structure}, \text{process}, \text{stage})$$

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STRUCTURE :

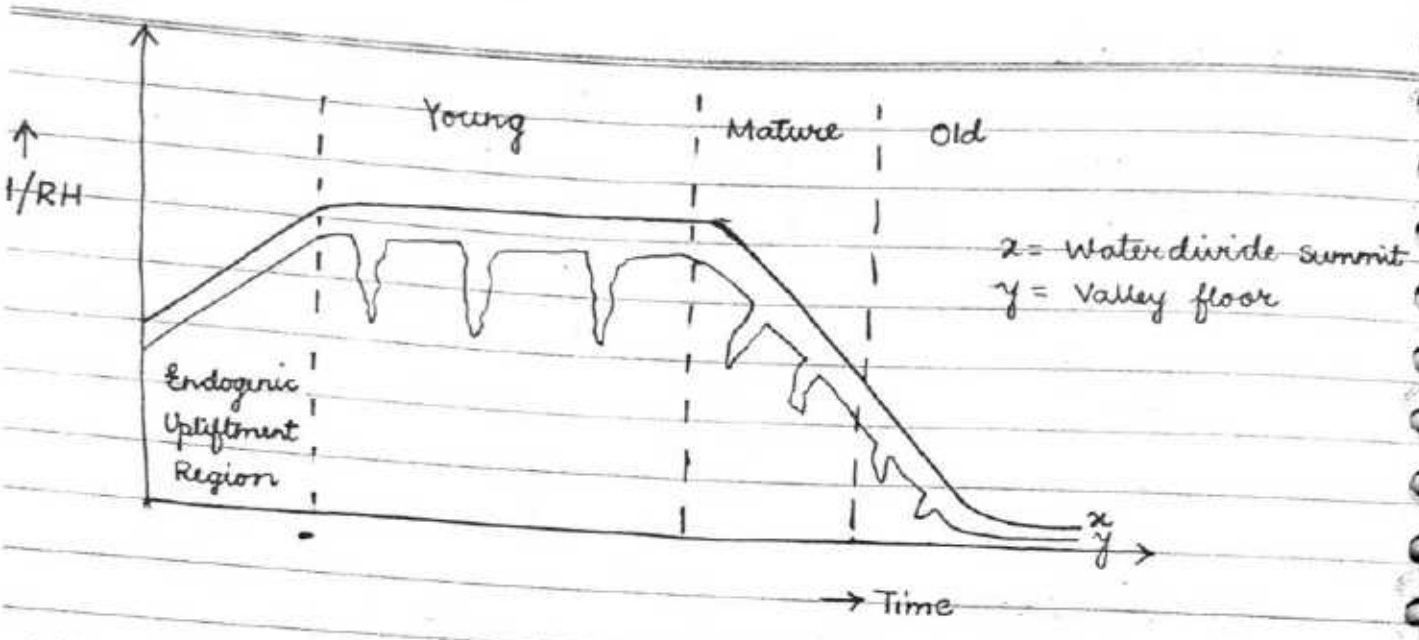
In the term structure, Davis emphasized on compositional constituents of the relief features. He emphasized that the relief features made up of less resistant construct will fast forward the cycle of erosion. However, if relief is comprised of resist material, it will delay the completion of cycle. In the expression structure, he also included minor structural elements like presence of or absence of joints.

According to Davis, it is variation in the structure of relief that the differential erosion generates bigger range of third order relief features.

PROCESS :

In the term process, he included both endogenic and exogenic processes and emphasized that endogenic processes are variability developers and exogenic processes are levellers. Elaborating exogenic category he identified that 5 different levellers reveal similarity in attempting to work to wipe off the generated variations. Moreover, they all are engaged in three-fold task : eroding, transporting and depositing and all of them increases relative height in order to decrease the absolute height. These agents, however, reveals variations in their effectivity in terms of different areas of functioning, different methods of functioning and development of different set of relief features. His entire concept, however, is built up in the analysis of stages.

STAGE :



Young stage : The young stage of the cycle of erosion begins with the end of endogenic upliftment. In this stage, developed active channels generate their well defined ungraded long-profiles moving from the source to their respective mouths. Each channel in this stage reveals 3 fold task : deepening, widening and deposition. In the reference of steeper gradient, this stage however marks prominence of valley deepening or vertical abrasion. It is therefore that the topographical features like V- or I - shaped valleys, rapids, cascades, waterfall and structural ~~beach~~ benches dominates the landscape. With the occasional cutting of the banks of the river in terms of the increased water volume, there is the gradual "shift" of the water divide summit. This shift may or may not relate to river water piracy. When the water-divide summit between the two ungraded channels is completely eroded, landscape enters mature stage. This transition is also referred to be the development of graded long profile among the

main channels. The 'swiftness' of erosion of water divide summit apart from being dependent on structure of the relief is also dependent on the factors like drainage density.

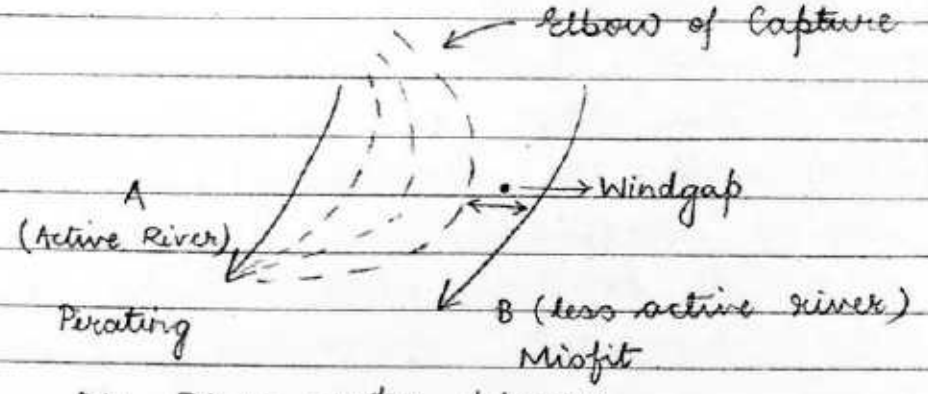


Fig. River water piracy

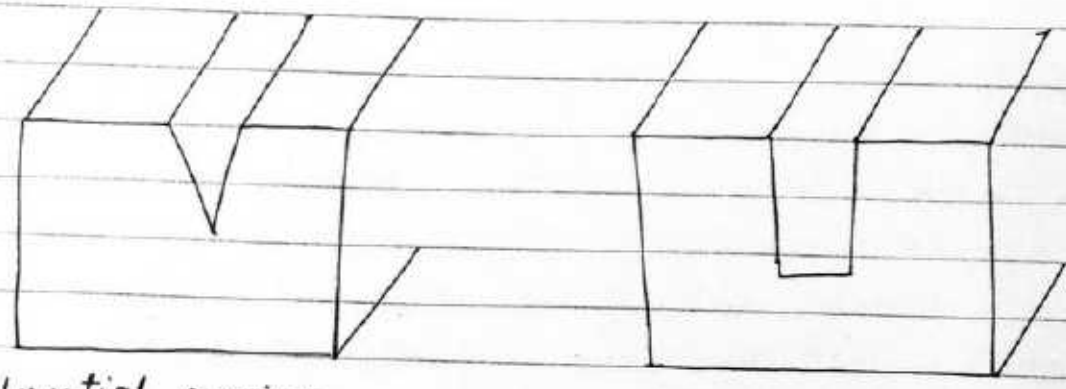
(ii) Mature Stage :=

The mature stage of normal cycle of erosion as begins with decrease in the absolute height and thus related gradient is prominently correlated to valley-widening activity. In this stage of landscape, dominating features thus includes flood plains with natural levees and bluffs along with wide meanders with river cliff and slip-off slope. This stage of cycle of erosion, however, marks continuation of valley deepening and deposition as well.

(iii) The Old Stage :=

When the water divide summits ~~of~~ between the tributaries (small channels) gets eroded it generates graded long profile among the tributaries making the landscape enter the old stage. The delay in attainment of graded long profile by the tributaries

- related to lesser volume of water. The beginning of ld stage, therefore, largely corresponds to near omplete termination absolute height which makes the ctive channels dominantly engaged in deposition. t is with this stage, therefore, that deltaic deposits are associated along with intense meanders with ox-bow lakes. This stage of cycle of erosion marks its completion with the development of featureless lowland called peneplains that are end-product of normal cycle of erosion, with dotting presence of "resistant thus remnant" monadnocks (isolated rock hill or mountain).

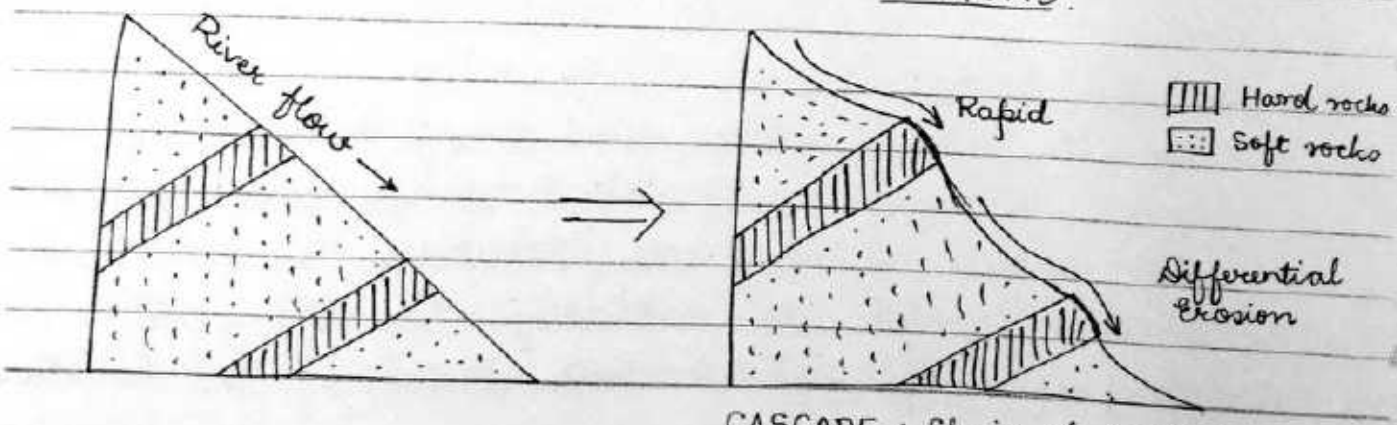


Differential erosion
+
More volm of water

V-SHAPED VALLEY

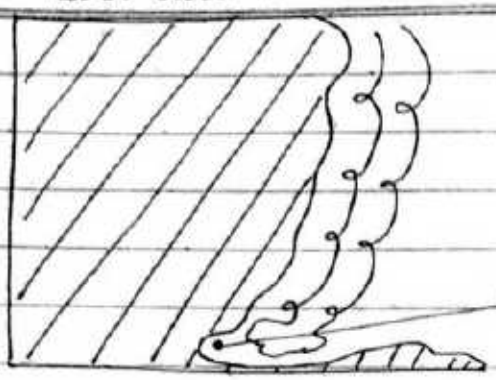
Homogeneous Structure
+
Lesser volume of water

I-SHAPED VALLEYS / GORGES / CANYONS



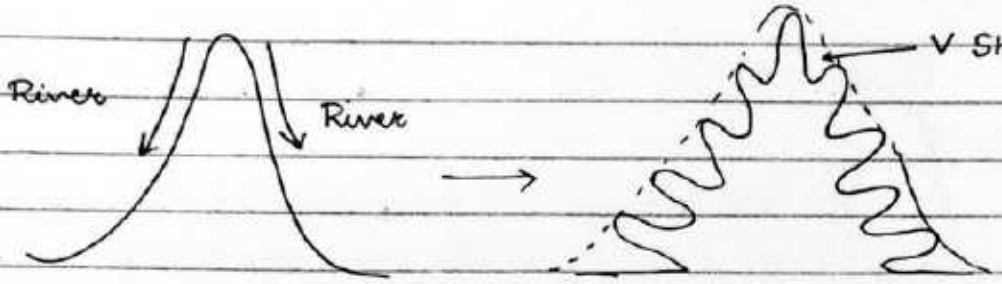
CASCADE : Chain of rapids

River Flow →



⇒ Waterfall
(enlarged rapid)

Plunge pool
(Abrasional hollow)



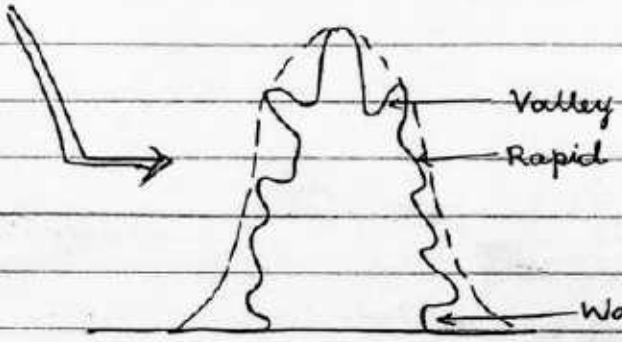
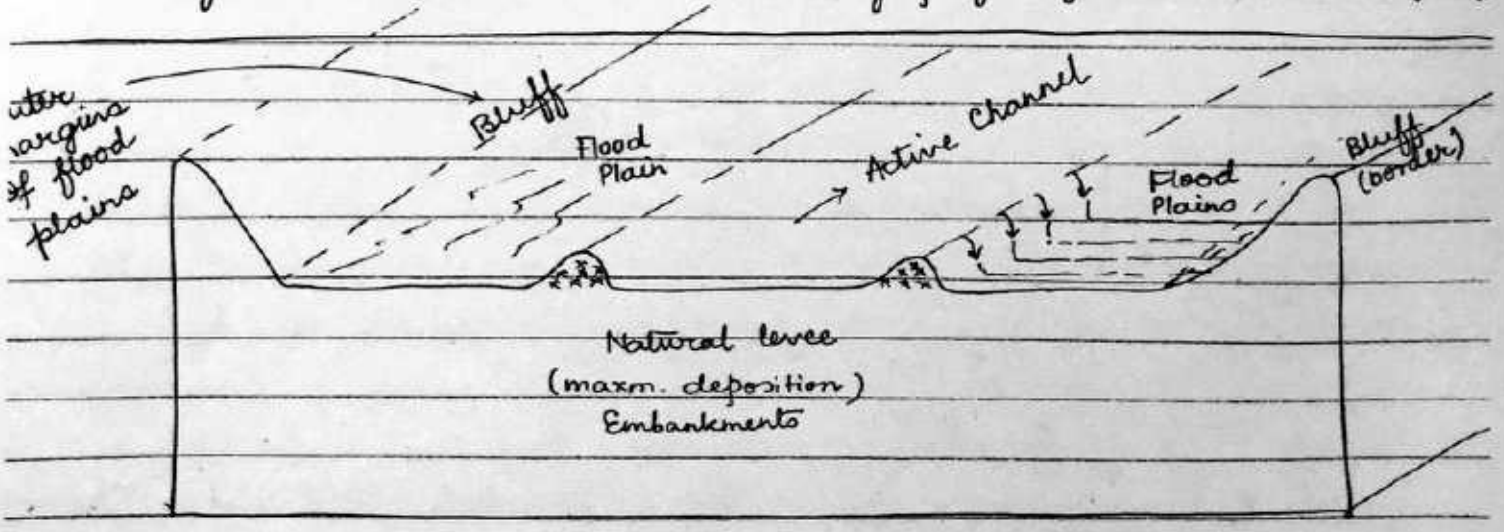
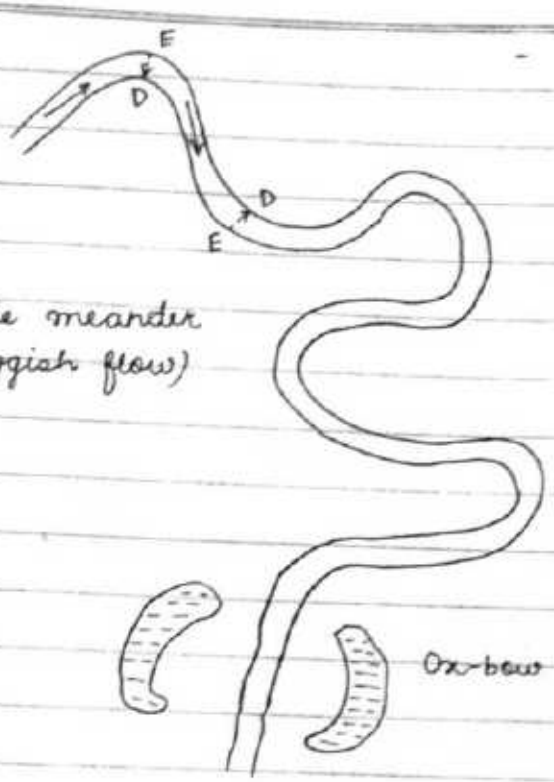


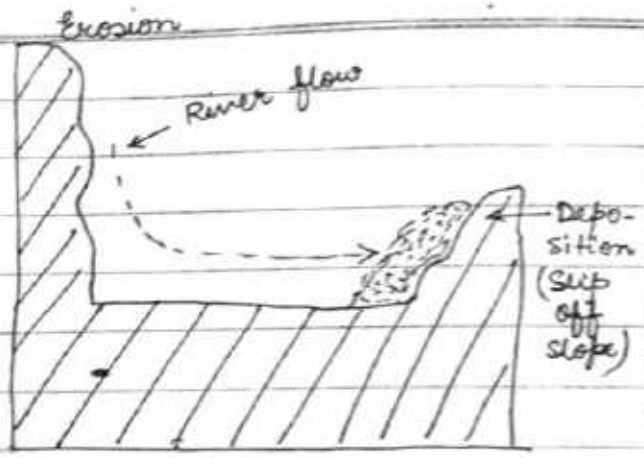
Figure: Structural Benches (Long profile of mountain course of river



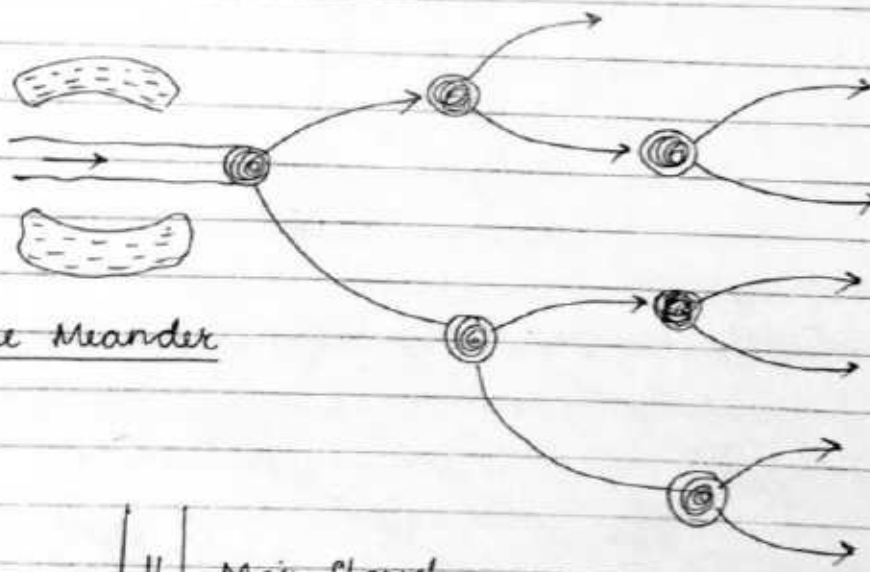
Mature Stage: Flood Plains & associated features



Wide meander (Sluggish flow)



Ox-bow lake : Cut off loops of intense meander, formed during dry season, temporary.



Intense Meander

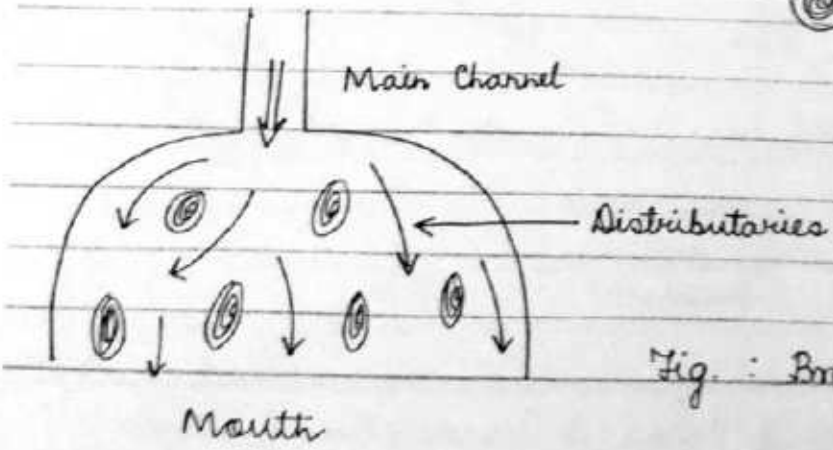


Fig. : Braided (Obstructed) Channel

Types of Delta:

(i) Lobate / Arcuate – Triangular shape eg. R.Nile, R.Niger

(ii) CuSpate / Toothshaped – ⊔⊔⊔ eg. R.Padma, Jamuna

(iii) Digitate / Bird's foot –  eg. R.Mississippi - R.Missouri

(iv) Estuarine Delta –



eg. R. Amazon

Date : 02/03/2014 LECTURE : (14)

Appraisal of Davisian Concept :=

Davisian Concept was originally published in 1889, oriented towards explaining the nature of functioning of exogenic forces. His approach continued to dominate geomorphological studies till the plate tectonic theory in 1960s. In between this timeframe, German geomorphologist Penck attempted strong criticism to Davisian ideology in terms of treating endogenic and exogenic forces to be following each other rather than being complimented However this scholar due to his untimely death could not produce the appropriate published idea. Recognised as the most mis-interpreted geomorphologist Penck could not create the desired counter balancing school. It was therefore that only in the light of plate tectonic theory that the approach was put to criticism. The tectonic theory clearly identifies that upliftment is diastrophism and for such long time downcutting cannot remain passive. As for example, Cenozoic orogenesis is yet not over. However, all the mountain of this era correlates to wide range of third order

reliefs making endogenic upliftment and exogenic downcutting being prevalent simultaneously. In the light of the theory it was also highlighted that the azoic constructs continues to project table-land characteristics i.e. with well recognised absolute height. This justifies that downcutting process is also long time-taking and for such long period, endogenic forces cannot remain positive. This theory therefore prominently criticised use of term CYCLE in the Davisian Concept. It is this criticism that made the approach of William Gilbert, called as the Approach of Dynamic Equilibrium come into lime light. Gilbert was senior American scholar who propounded that the landscape is in dynamic equilibrium to the tendencies towards variability and tendencies towards uniformity i.e. the endogenic and exogenic forces compliment each other in development of relief features. Technically correct Gilbert approach, however, is utilised only towards the study of REJUVENATION and Davisian Concept due to its simplicity is taken as the prominent support system to the study of exogenic forces.

REJUVENATION / POLYCYCLIC LANDFORM / 2nd CYCLE OF EROSION :

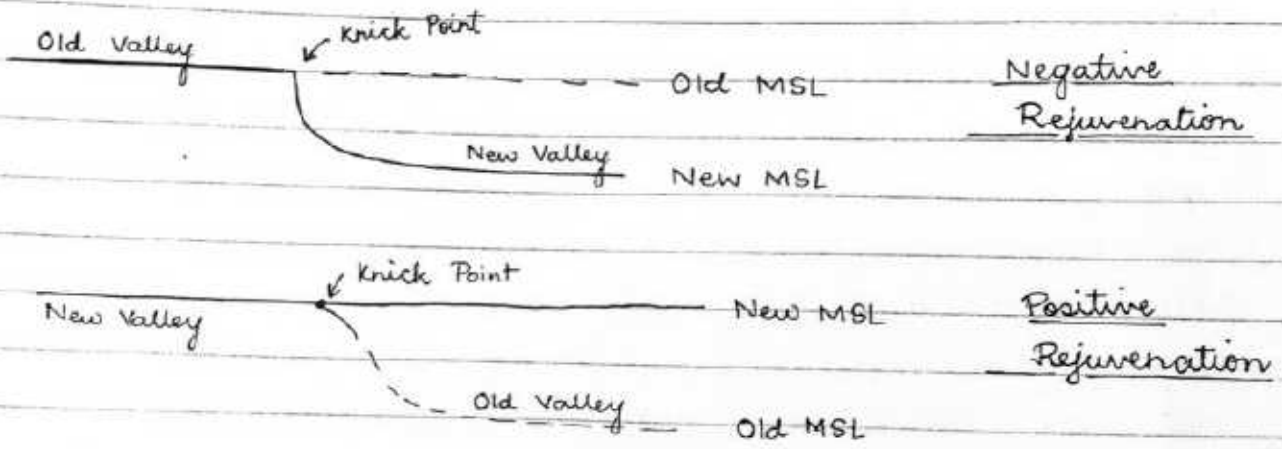
Rejuvenation is defined to be modifications in the work of levellers induced either by climatological agents or geomorphic agents. These modifications are boldly classified into :=

1) Positive rejuvenation
2) Negative rejuvenation

The positive rejuvenation represents those modifications which facilitates levelling. Such type of rejuvenation is commonly associated with the change in the mean sea level after the termination of ice-age. The Pleistocene ice age termination marked the change in the MSL of upto 200 m enundating the coastal areas and facilitating levelling tasks at regional to local levels. Positive rejuvenation can be correlated with subsidence of land and rise of sea-floor.

The negative rejuvenation in comparision generates polycyclic landforms or 2nd cycle of erosion. It is defined to be those modifications induced by climatic factors or geomorphic agents that interrupts levelling. In negative rejuvenation the expression of re-attainment of youth is thus practically applied. Important recognised examples of negative rejuvenation or polycyclic landform are valley in valley topography, entrenched meanders and knick point. The valley in valley topography marks its development when the shallow bed of the river involves the development of deeper valley within due to increased gradient induced by endogenic upliftment. This valley in valley topography is prominently identified in Yenisey River basin of Siberia Russia. The river is prominently known to have flattest course among the big rivers of the world. The entrenched meanders are those wide meanders where there is strong effectivity of valley deepening. It identifies the case of re-attainment of youth of the topography that have reached its mature stage.

Colorado River of USA correlates with most recognised example of entrenched meanders. Knick point is defined to be transitional point between old and new valleys. It is largely identified along with the base level of erosion, representing the effect of climate change in generating rejuvenation. Theoretically, knick point is utilised to represent even the example of positive rejuvenation.



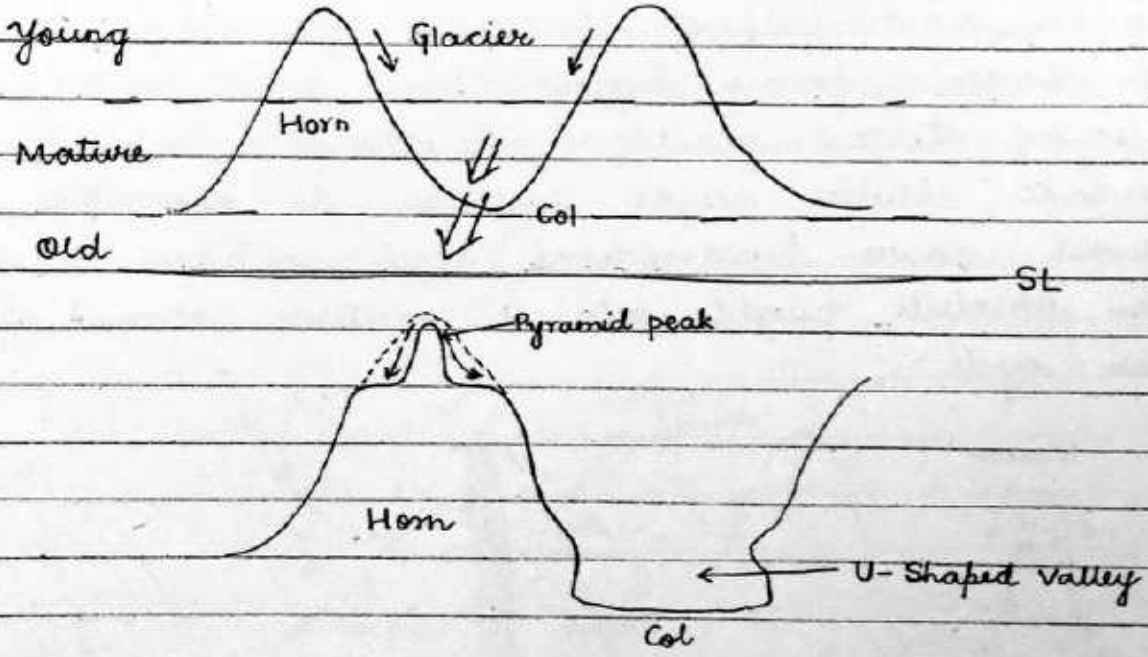
GLACIAL CYCLE OF EROSION

Moving mass of snow and ice is called glacier, which are locationally classified as continental glacier and Alpine glacier. The continental glacier belonging to higher latitude as in Greenland and Antarctica are bigger masses of snow and ice. However, effective in already levelled continental mass, this type of glacier is not referred in the glacial cycle of erosion. The Alpine glacier representing mountain glacier, though have lesser glacial mass, is considered to be more effective leveller and is therefore taken into consideration to the development of the glacial cycle of erosion.

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Q. Explain glacier as active agent of gradation.
↳ Do not write about continent glacier

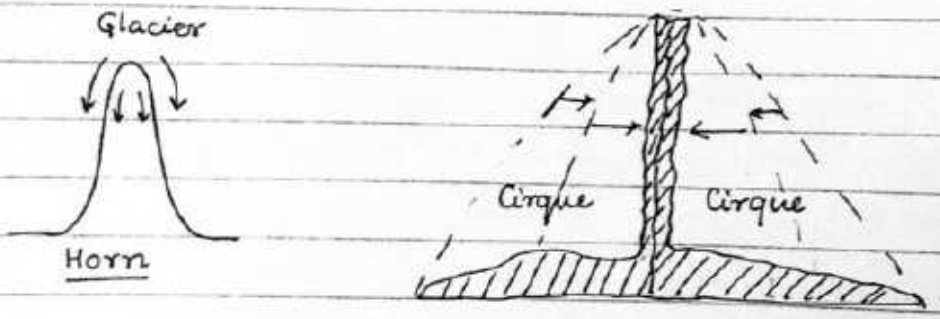
It was Davis who propounded this cycle of erosion in 1900 identifying it to be more theoritical counterpart of fluvial cycle. It is because in the glacial cycle, baselevel of erosion is unclear. Glacier as an active agent of gradation involves plucking effect or scooping effect as the most important method of erosion which generates valley deepening and widening i.e. ablation as additional effectivity. After endogenic upliftment getting over and favourable alpine climate generating snow field, there is the beginning of glacial cycle of erosion with well-defined stages of young, mature and old.



The Young Stage :-
This stage marks its beginning with the beginning of movement of alpine glacier from the snow field via horn to col upto the ablation zone. In the process, alpine glacier generates horn topography,

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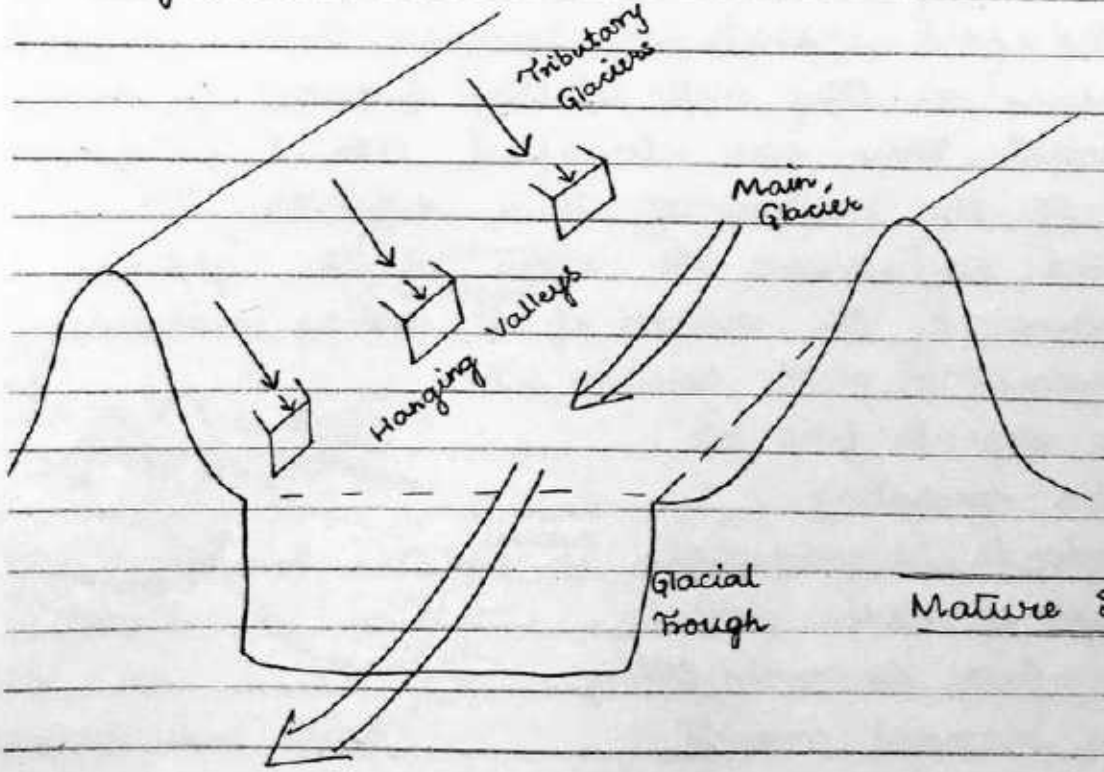
ol topography as well as the depositional features. ; is however that in this stage development of orn topography dominates the landscape. This topography narks its development in the combined effect of vertical xbrasion i.e. valley deepening and basal sapping i.e. barallel retreat of mountain wall. In the effectivity of these two processes, the developed erosional feature is called CIRQUE i.e. the deep alpine valley with steep standing backwall. Development of cirques along more than one slope of original horn simultaneously leads to formation of Arete that are knife edged remains of original horn between the two cirques. In the genesis of more than one arete, narrowed down apex of horn with significant rough profile forms pyramidical peak. Throughout the young stage, glacial cycle, therefore, marks the dominance of relative height increase. As soon as the narrowed down pyramidical peak collapses, decrease in the absolute height sets in mature stage of glacier cycle.



Young Stage

(ii) The mature stage of glacial cycle :

In the mature stage glacial cycle is dominated with the development of col topography. Compared to the horn, col correlates with less steep gradients and thus much fuller accumulation of glacial mass. It is in accordance that the col topography represents to be typical vertical abrasional feature called GLACIAL TROUGH or U-SHAPED VALLEY. This U-Shaped valley is classified into main trough and hanging valley. The main trough represents the vertical abrasional valley developed by the main glacial mass, whereas, the hanging valleys forms the small glacial troughs developed by tributary glaciers with lesser erosion capacity. This typical col topography continues to represent increase in relative height carried on by alpine glacier.



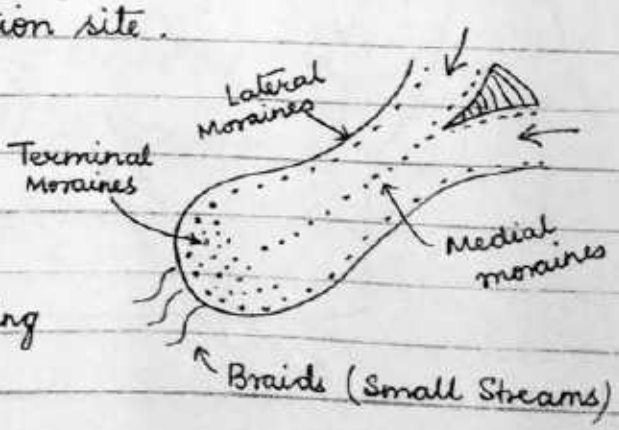
In this stage, developed erosional depressions of both horn and col, develops well-defined GLACIAL STEPS. These steps becomes evident only after termination of glacial mass, largely during summer season. It is therefore that glacial steps always corresponds to fresh-water lakes called TARN LAKES of Cirques & Paternoster Lakes in rest of the depressions.

The Old Stage :

Continuation of bassal sapping in horn when makes the backwall of cirques completely removed, termination of horn with sufficiently deepened col, sets in the old stage of glacial cycle. In this stage denuded height generates unfavourable climate with the beginning of termination of glacial mass making deposition as the only activity, the alpine glaciers are engaged in. The developed feature is called MORAINIC DEPOSITS. Moraines are defined to be the unconsolidated rock deposits of variable texture deposited by the glaciers as they melt in their process of down-slope movement. These are classified into 3 categories according to the location of their deposition.

Lateral moraine i.e. along the sides of the glaciers
Medial moraine i.e. the merger of 2 lateral moraines.
Terminal moraines i.e. at the ablation site.

The morainic deposits fills all the depressions generating featureless lowlands i.e. end-product of glacial cycle of erosion called CRYNOPLAINS which do marks dotting resistant thus remnant mounts called NUNATAKS.



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Geography

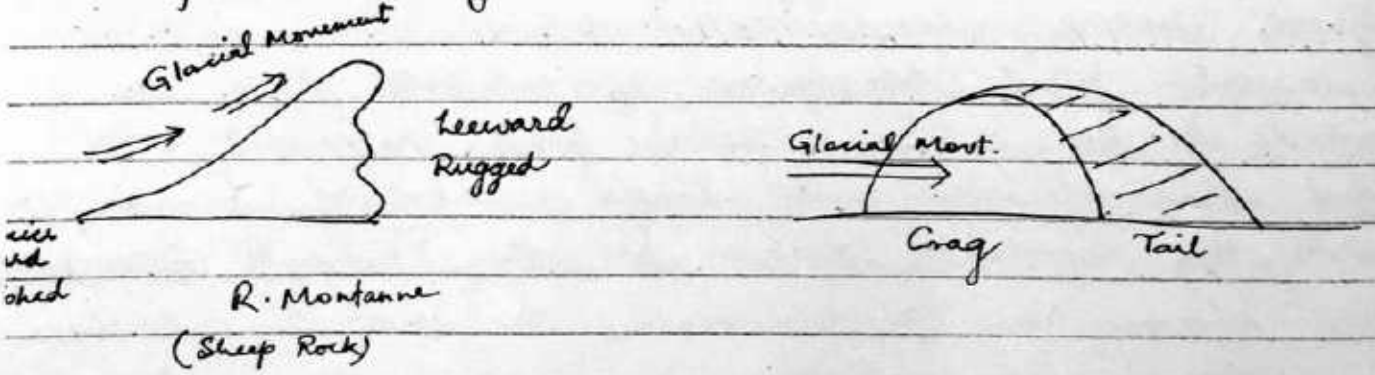
Lecture (15)

Date 3rd March 2019

CONTINENT GLACIER AS AN ACTIVE AGENT OF GRADATION :

In the higher latitudinal areas, continent glacier form effective agent of gradation moving in already levelled continental surface. These glacial masses are not involved in stages of cycle of erosion. They, however, incorporates erosional capacity in polishing distinctive size pebbles in accordance to their movements. The identified minimal erosional features developed by continental glaciers includes Rochee Montanne & Crag and Tail topography.

Rochee Montanne represents the rock that experiences glacial erosion along both its slope with the glacial-ward side being polished & smooth, & leeward side plucked, thus, rugged. This feature because of its shape is also called Sheep rock. Compared to it, Crag & Tail topography involves one polished side of the rock due to the movement of continent glacier. The extension of such rock beyond the reach of continent glacier maintains its original profile called tail. The Crag & Tail topography, therefore, are typical to marginal areas of continent glacier.



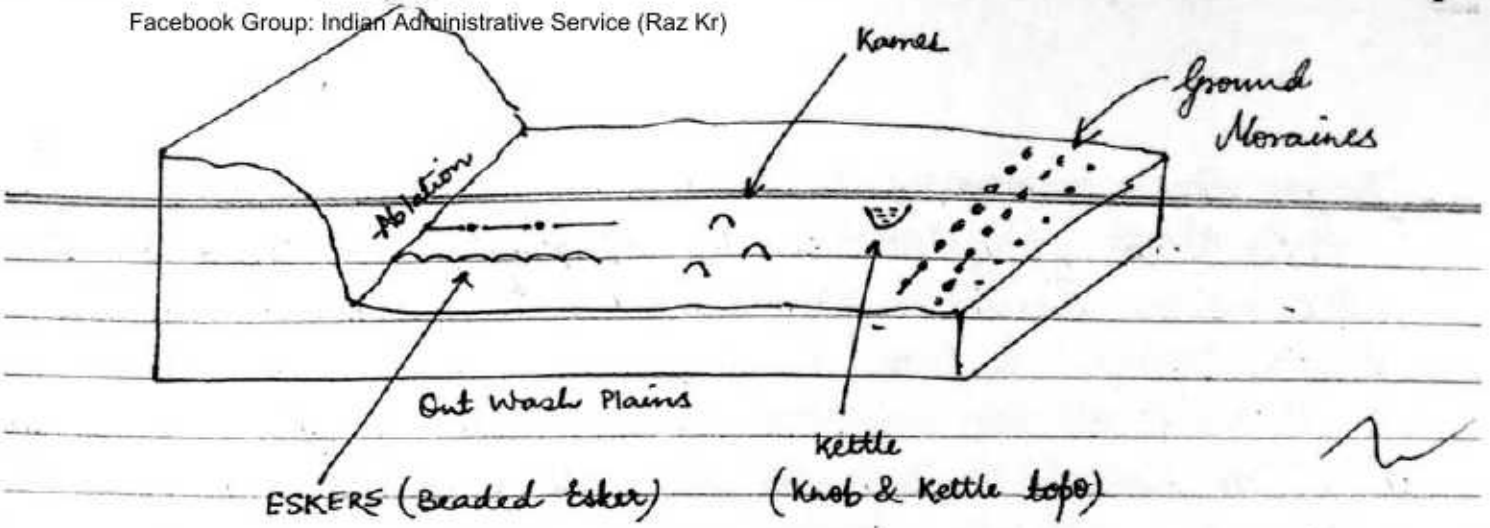
The seasonal ablation of continent glacier results in the deposition of distinctive textured materials in diff. shapes. All these deposits are collectively called TILL DEPOSITS.

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his includes both glacial & fluvio-glacial deposits. The
<u>lacial category</u> includes the only, though, most recognised
'acial depositional features called drumlins. These are the
otter Boulder Clay deposits representing fine texture but
rge sized deposits known for absolute homogenous sequence
deposits making these deposits nick-named as BASKET OF
gs topography. typical to the landmasses along Arctic &
ulhern Ocean, it is in UNGAVA peninsula (Canada) that
rld's largest <u>Drumlins</u> are identified.

The fluvio-glacial category corelates
the generated BRAIDS due to the melting of
tinental glacier. These deposits are typical to the
<u>TWASH PLAINS</u>. It is these plains that represents glacial
osited soil with significant fine textured deposits.
- distinctive deposits of Outwash plain are demarcated
accordance to their shapes of deposits. Among the
ost reg recognised features of outwash plain, <u>ESKERS</u>
at are continuous ridges of deposits, KAMES that
= individual mounds of deposits, <u>GROUND MORAINES</u> that
finest texture unconsolidated deposits and <u>KETTLE</u>
t is water filled depressions of outwash plain are
luded. As the entire outwash plain represent water
gged characteristics with large number of braids, it
vails the excellent location for the study of <u>GEOHYDROLOGY</u>

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{ Only if Glacial Cycle of Erosion :: Avoid Continent Glacier }
↳ Davision Approach

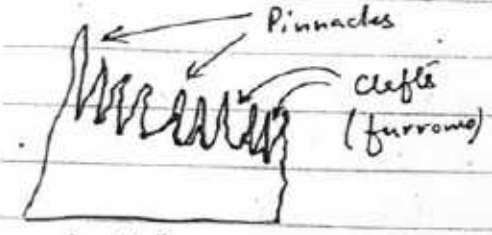
KARST TOPOGRAPHY : (Map of Europe: Eastern Shoreline of Adriatic Sea ↳ Croatia & Bos.–Herz.)

↳ KARST CYCLE :

The Karst geomorphic cycle is identified to be unique in its characteristics as it involves the specific requirements of thick bed of limestone exposed on the surface with humid climatic conditions. This cycle therefore is highly localised with its name derived from Karst Province of erstwhile Yugoslavia along Adriatic Sea Shoreline. This cycle is credited to Beede (1911) and Cvijic (1918), the disciples of David. They identified in their respective study that Karst Cycle forms the simplest geomorphic cycle as it involves mono-process i.e. solution or corrosion. The cycle however involves difficulties as the baselevel of erosion is unclear. Moreover, the movement of underground water cannot be standardised. Like all the other geomorphic cycles this cycle is also correlated with young, mature & old stages. The young stage of Karst cycle marks its beginning with the beginning of surface solution by both rainwater and running water.

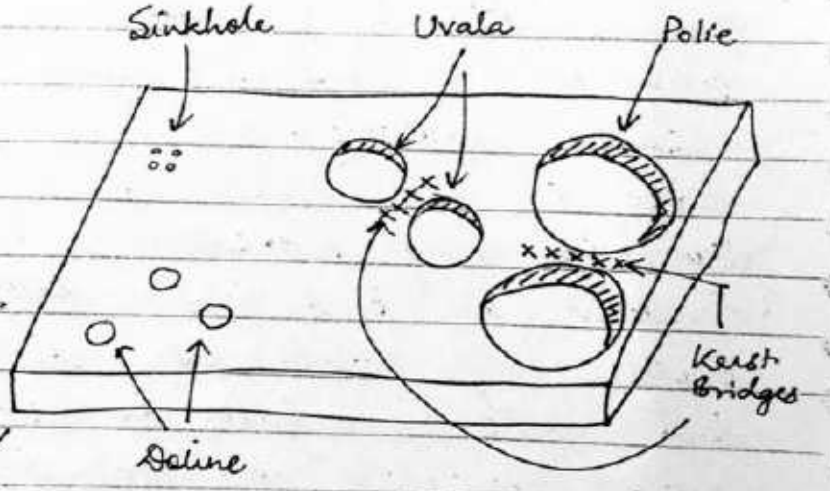
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this stage, the solutional topography developed on the rface are not just distinguished in accordance to the fective agent but also to the nature of limestone beds. e Rainwater solution effective in this stage of Karst cycle, it is rrelated to vertical beds of limestone. The developed solutional atures are called Lappies. These are typically associated ith pinnacles that are the leftout limestones seperated by issolved furrows called CLEFTS. Commonly, however limestone ds are horizontally aligned. In such type of beds, distinctive zed surface solutional craters are generated. In the increasing der of size, it includes Sinkholes, doline, Uvalas and olie. The larger size surface solutional craters as 'alas & polies, also corresponds to KARST WINDOWS & RST BRIDGES.



Lappies
(Uncommon)
→ Bedding Vertical } : Rainwater }

Bedding horizontal } → Rain Water }



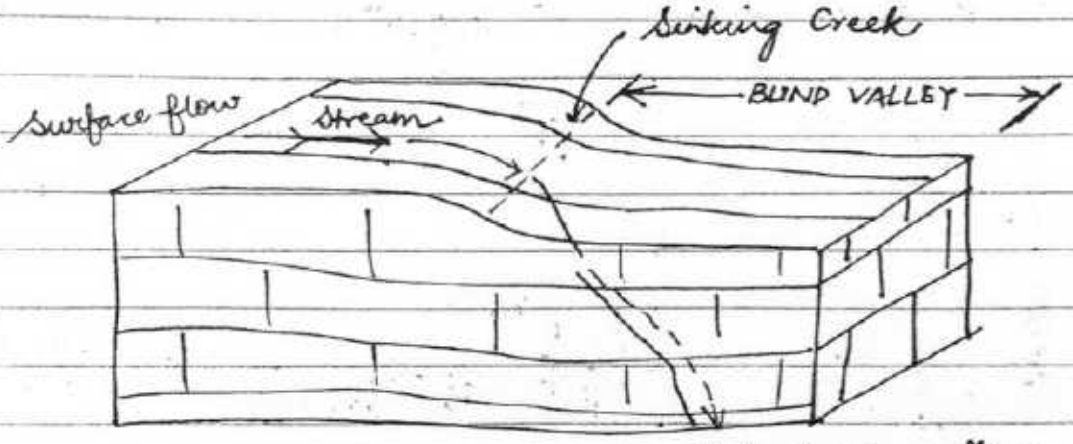
Karst Windows → Depressions

Surface Solutional Craters

Young Stage → Rainwater + Running Water

Running Water surface/topographical features

The surface solution carried on by running water in the Karst region generates additional topographical features which includes blind valley i.e. river valley permanently deprived of active channel and sinking creek i.e. the point where surface stream converts as sub-surface flow. When the Karst topography is dominated by development of numerous sinking creeks, surface runoff disappears with sub-surface solution becoming prominent. That is the beginning of mature stage of Karst Cycle.

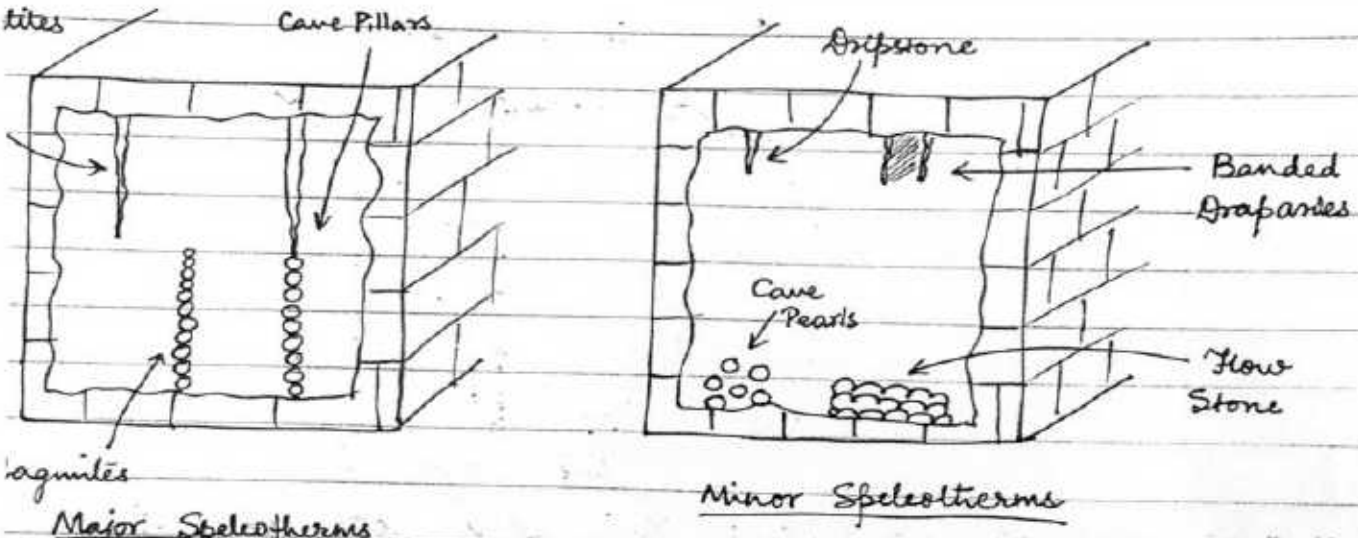


Sub-Surface Flow

Mature Stage of Karst Cycle: This stage correlates to underground water as an active agent of gradation. The source of this underground water includes both penetrated rainwater & sinked-in running water. The only erosional feature that underground water as an active agent of gradation develops is called Karst Caves. These are sub-surface solutional hollows which are known for their extreme humid climate & involves an entire range of depositional features developed by underground water as an active agent of gradation. All these cave deposits are called speleotherms which are categorised as major speleotherms & minor speleotherms.

major category includes stalactites, & stalagmites and
e pillars whereas minor speleotherms representing immature
iants of the major speleotherms includes Dripstone,
ded Draperies, Flow stone and Cave pearls.

leotherms. All cave deposits



Topography → Surface & Sub-surface soils
cycle → (, ,) + (Young / Mature / Old)
ngroundwater → Exclude surface

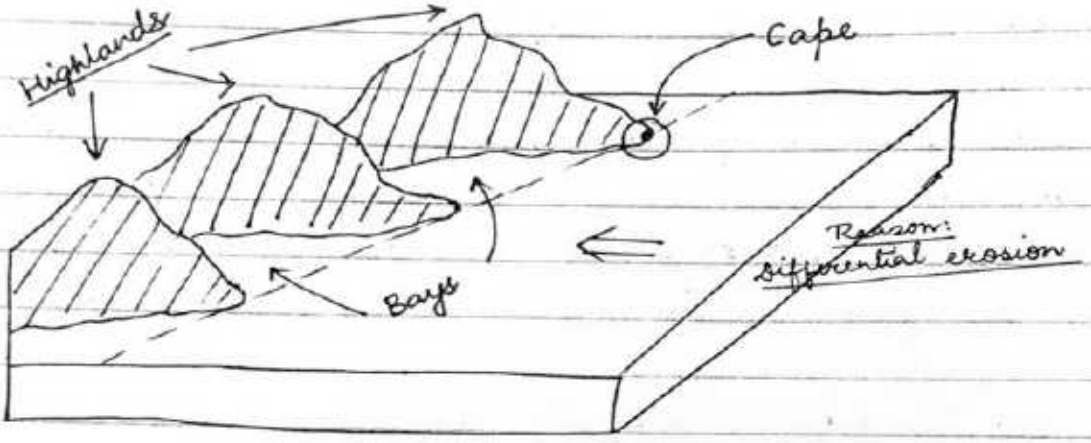
In the mature stage of Karst Cycle, apart from the effectivity of underground water as an active agent of gradation, surface solution continues. It is either due to solutional termination of the surface limestone strata or due to collapse of remaining limestone strata, when there is decrease in absolute height, the Karst Cycle enters old stage. In this stage, apart from the destruction of Karst Cave and all the ceiling deposits, there is the possibility of reappearance of disappeared surface stream. The Karst Cycle marks its termination with such developments. However as the undear baselevel of erosion, the end-product i.e. exposed cave ground largely reveal prevalence of dotting carbonate mounds called HUMS. With the exposure of humus, there can be the fresh beginning of the Karst Cycle.

Humus are remmanents but not resistant.

× × × × ×

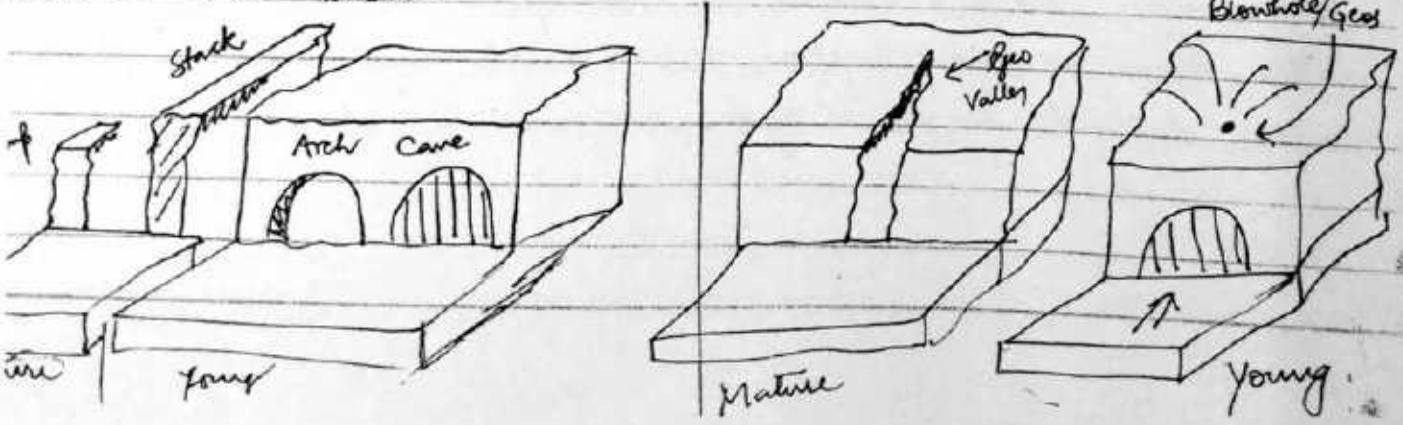
4. COASTAL CYCLE OF EROSION: Based on the identification of positive change in the Mean Sea Level, post pleistocene ice-age, Davis developed coastal cycle of erosion. He therefore emphasized that the shoreline on the map of world collectively represents shoreline of submergence either climatologically or tectonically. When the specific type of shoreline is developed, beginning of coastal cycle is outlined. Sea Waves as an active agent of gradation involves "surfing effect related hydraulic action" as the course of erosion and "backwash" as the course of transportation and deposition. Like other erosion cycles, the coastal cycles also includes, 3 well defined stages: Young, mature & old.

young stage of coastal cycle dominantly correlates with development coastal erosional features. The category of these features includes & Bay topography, coastal cave with either arch or s formation along with development of sea-cliff with notch. The Cape Bay topography forms the commonest topographical features oped by sea waves as an effective agent of gradation. It ates the example of differential erosion.

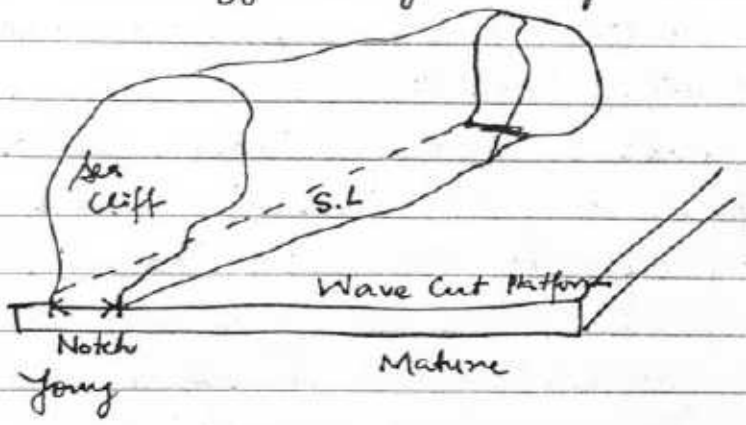


Features that develops in Young Stage: Cape & Bay.

In the coastal areas, where structurally variable resistant rock ade coastal mountains are specified, there is the projected development of the coastal features called coastal cave. It is this feature that in accordance to the subsequent evelopments either facilitates or formation of coastal bridge alled coastal arch or marks the development of blow-ole also called geos. These developments in the coastal areas parks their continuity in the mature stage of coastal rosion as well.



In the homogenous structural coastal mountains, young stage also marks the development of sea-cliff with notch, that is the point where maximum hydraulic action is experienced by the coastal mountains. Transition of coastal cycles from young to mature stage is primarily correlated to further development of erosional features as narrowing down of projecting headlands decrease of absolute height with arch paving way to stack that is coastal pillar or stump i.e. eroded stack or geos paving way to the geo-valleys terminating the existence of coastal cave. This transition is also correlated to the retreat of sea-cliff in generating wave-cut platform



Sea - Wave
Wind
Find out which are the locations that qualifies as desert where winds are active agents of gradation.

March 2014

Lecture-16

Coastal cycle contd.

d Stage : The old stage of coastal cycle marks its giroving theoritically with noticed decrease in absolute ight. This stage, therefore, is prominently recognised th development of depositional features. Both transportation deposition carried on by sea waves corresponds to its reating movement that is called backwash. The coastal bosits are broadly classified into 2 categories :

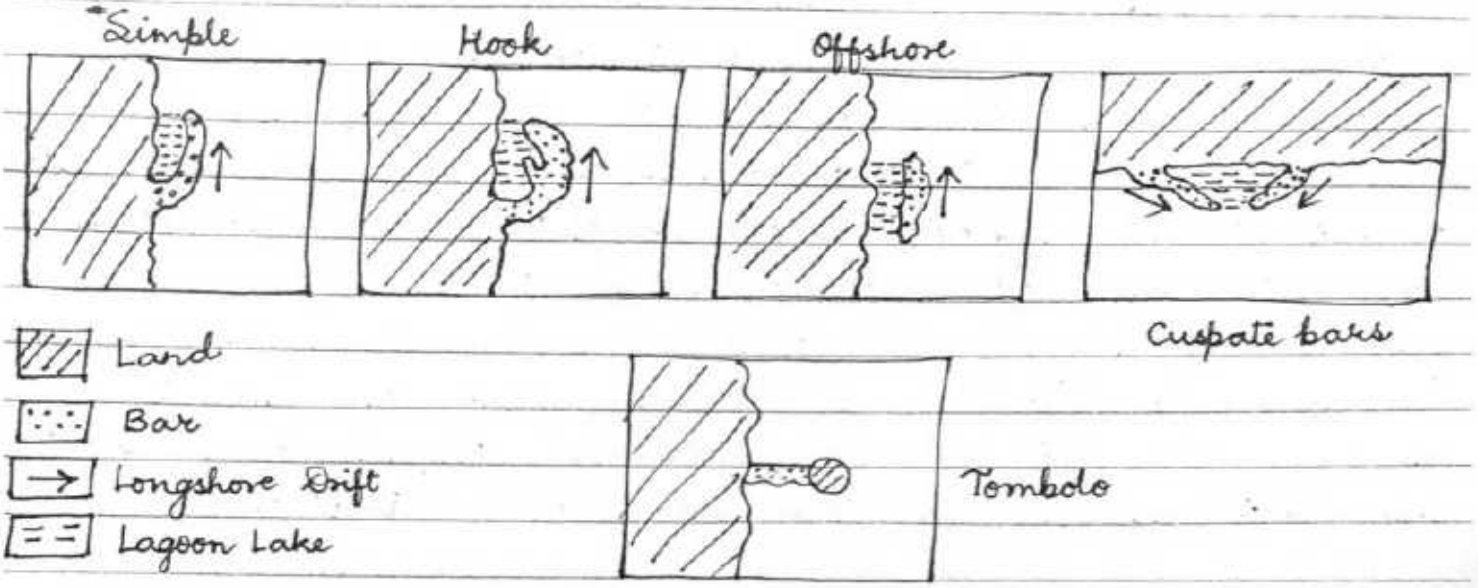
1. Onshore Deposits
2. Off-shore "

Inshore Deposits : The on-shore deposits are called beaches se are very well sorted, temporary, unconsolidated, rock posits deposited between coastline & shoreline. These are ssified as beach cusps & beach ridges. In accordance the specific platform of their deposition, beach cusps are mmon over the headlands generating "cape and swale iography" whereas beach ridges are common on wave-cut tforms generating wave-deposited platform.

ffshore Deposits : The off-shore deposits are called bar or t deposits. These represent elaborate effectivity of long ore drift that is recognised to be movement of sub-rface water parallel to the shoreline. After the surfing ect, the bar deposits form much more consolidated, thus, rmanent feature. When compared to the beach deposits, these posits are classified in accordance to their shape, size alignment. Among the prominently recognised types of deposits simple bar, hook, offshore bar, tombolo nd cuspate bars are included. The coastal cycle of osion marks its conclusion with the development of rofile of equilibrium i.e. erosion and deposition balancing ch other. The generated featureless lowland is called

coastal peneplain with dotting resistant, thus, remnant coastal monadnox.

Lagoon : Salt water lake which is carved out of sea/bay/gulf



Types of Bars

① Adam's Bridge (India-Sri Lanka) → Tombolo

② Vembanad Lake (Kerala) → Simple → Hook

③ Chilika

④ Long Island → Offshore bar near New York

EOLIAN CYCLE : Eolian cycle of erosion or arid cle of erosion is credited to the strongest disciple of Davis, King. He identified the cycle to be applicable in all deserts of the world where evaporational loss is more an precipitational gain. He identified deserts to be locationally pical west margin deserts as Sahara Desert or sub-tropical ntiental desert as Gobi Desert. Compositionally, deserts are assified as Erg Desert i.e. sandy desert, Reg Desert i.e. bles/ stony deserts and Hamada Desert i.e. big barren land- pe. In the deserts, absence of surface moisture, thus, rennial vegetation justifies wind to be the most effective ent. However, as the wind movement cannot be standardized generating sequential landforms, Eolian Cycle have been loped by utilising the effectivity of Secondary agents fluvial action generated by sudden torrential rains.

Eolian cycle is specified in the areas called BOLSON se are arid valleys surrounded by mountains. The serts are the location that commonly experiences non- pendable, unpredictable sudden torrential precipitation caused e to sudden heating of sand. It is this sudden rential rains that generates flash-floods setting in the vial action, thus, Eolian cycle of erosion. In the young ge, flashflood water is projected to be prominently gaged in vertical abrasion, descending down the slope of r arid mountains. This vertical abrasional feature lled wadis are recognised to be a variant of V-shaped ley which remains deprived of running water for major rt of year. The transported material by the flash flood ter generating the wadis is largely deposited at EDMONT location as fan-shaped deposits called y Delta.

These dominating activities in the young stage of arid cycles correlates to continuous ↑ in the relative height. When in the process absolute height marks its reduction, mature stage with dominating development of Bolson floor features is demarcated. In the Bolson floor, the flashflood water marks the development of horizontal abrasional plain called pediments. This marks simultaneous formations of depositional plains called Bajadas. Among the other features identified with Bolson floor, temporary lakes called playa lakes leading to the formation of hard salt layer called salinas and dry river valley called arroy are developed. In this stage, development of pediment & wadis continuously results in retreat of the mountain wall. Identified by LC King, this transition paves way to old stage of Eolian cycle where enlarged pediment paves way to the formation of pediplains, that is featureless erosional surface which incorporates resistant thus remnants, isolated mounds called inselbergs.

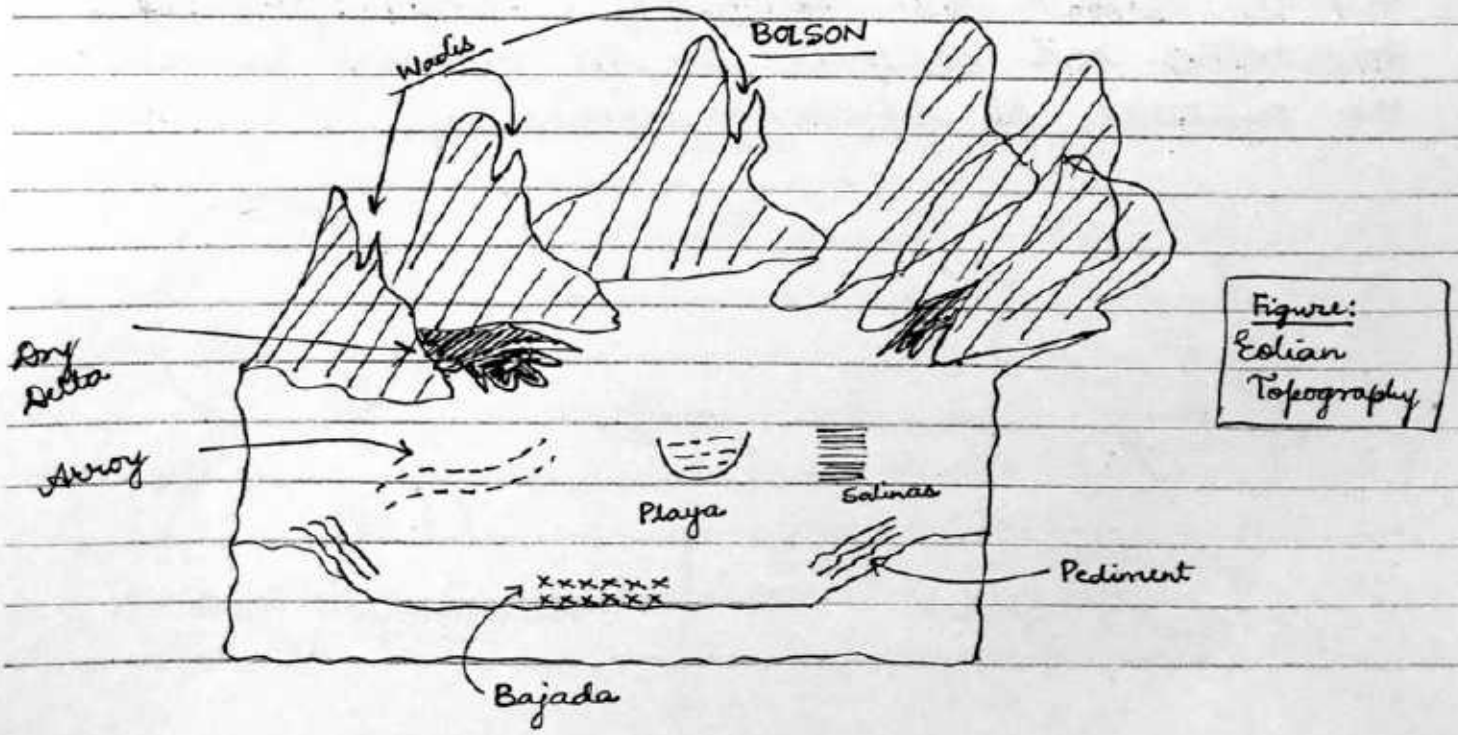
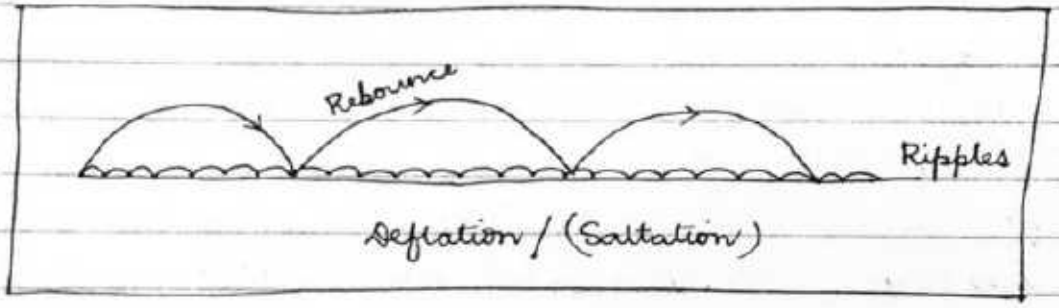


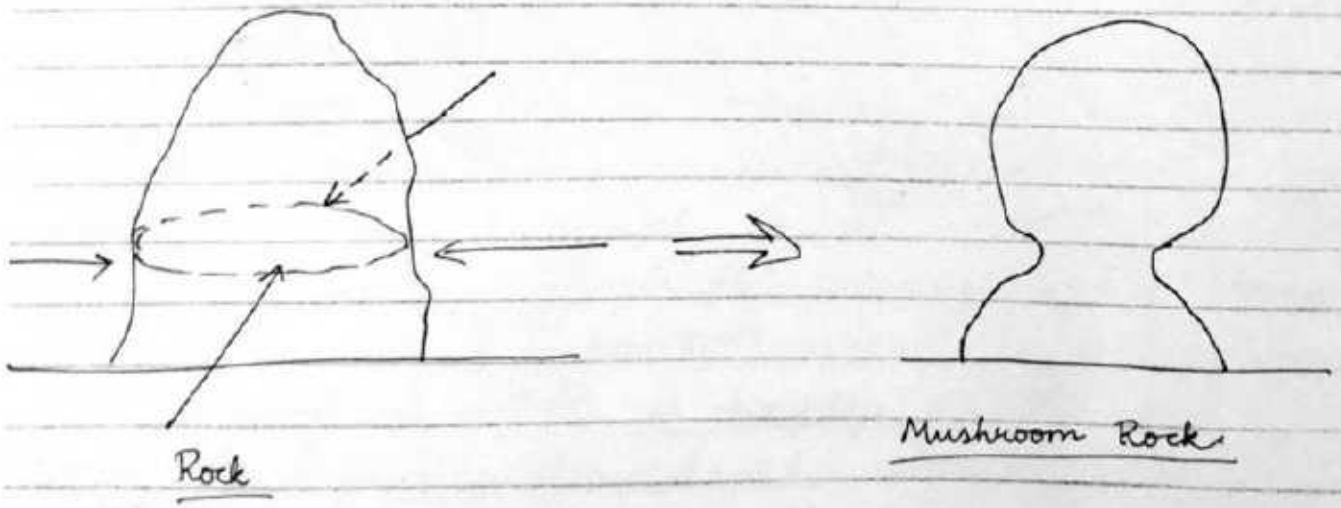
Figure: Eolian Topography

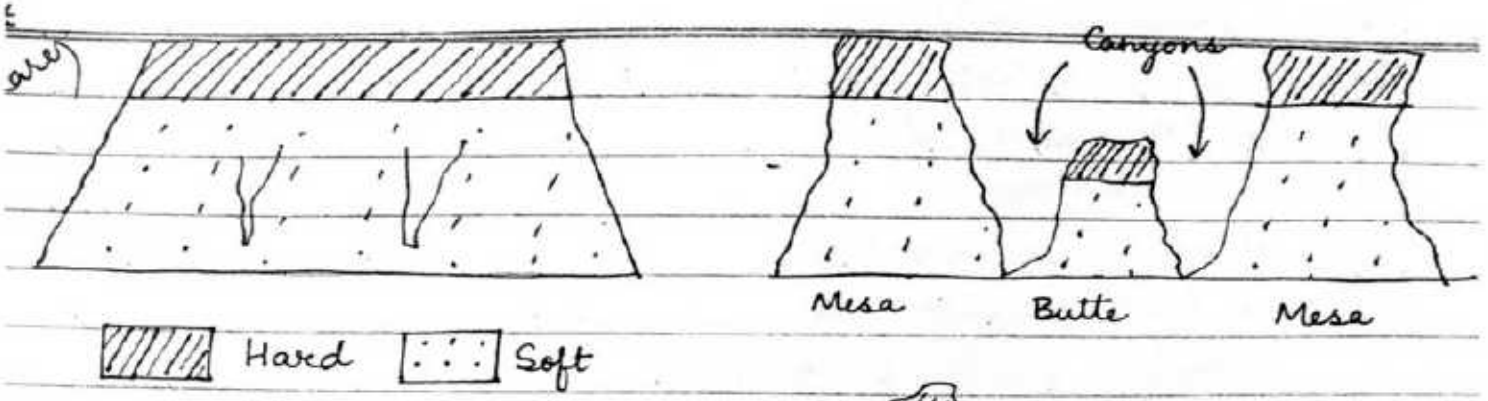
IND AS AN ACTIVE AGENT OF EROSION :

Horizontal movement of air is called wind which acts as most important agent of gradation in the deserts where moistureless sand and absence of vegetation facilitates uninterrupted deflation i.e. blowing effect. The deflative movement prominently includes saltation i.e. long leaps which creates sand blasting effect. On the absolute ground, however, rolling sand particles generates ripples.



The sand blasting effect generates erosional features which commonly includes mushroom rock as an absolute example of saltation effectivity and it also includes Mesa & Butte topography, Zueger-Yardang, tone lattice and Ventifacts as the erosional features that are outcomes of differential erosion.

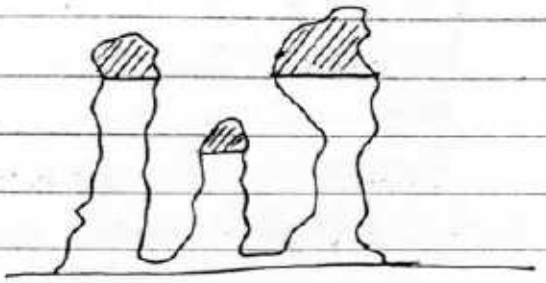




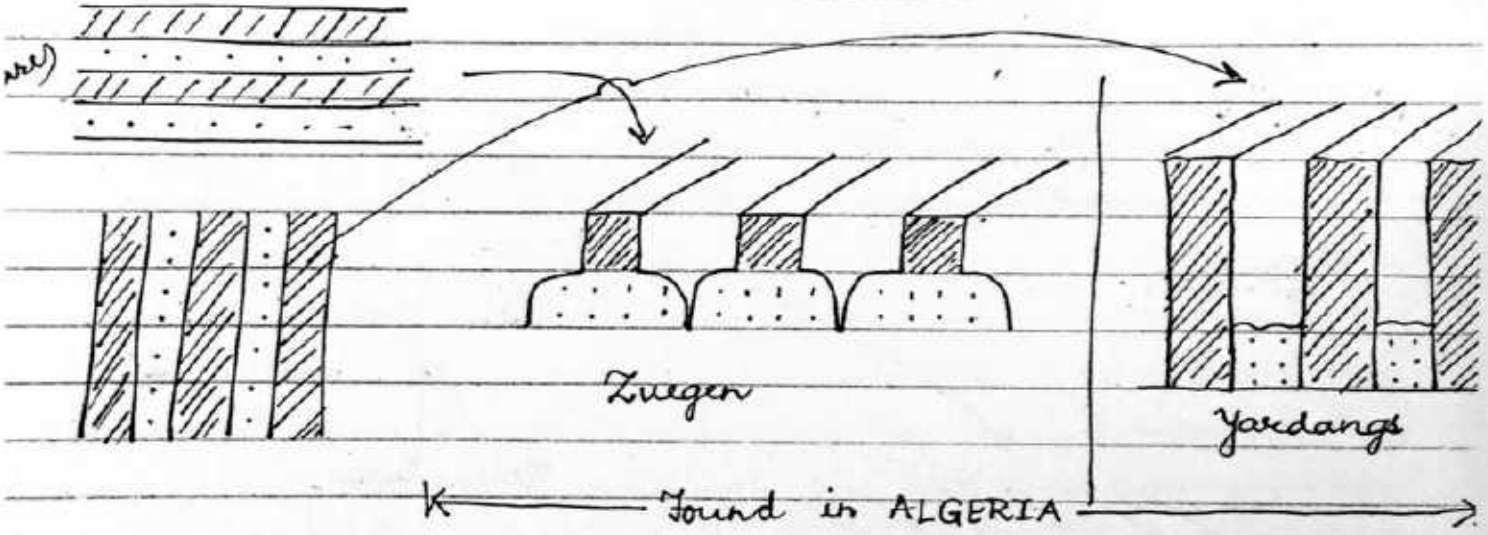
US Deserts.

→ Colorado Desert

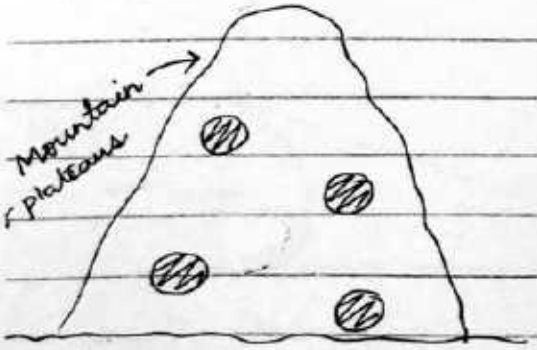
→ Mhave Desert



Hoodoos (Zig-Zag)

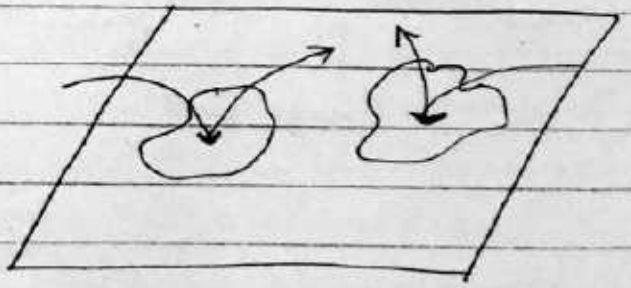


← Found in ALGERIA →



Stone Lattice or Stone Net (Common)

#



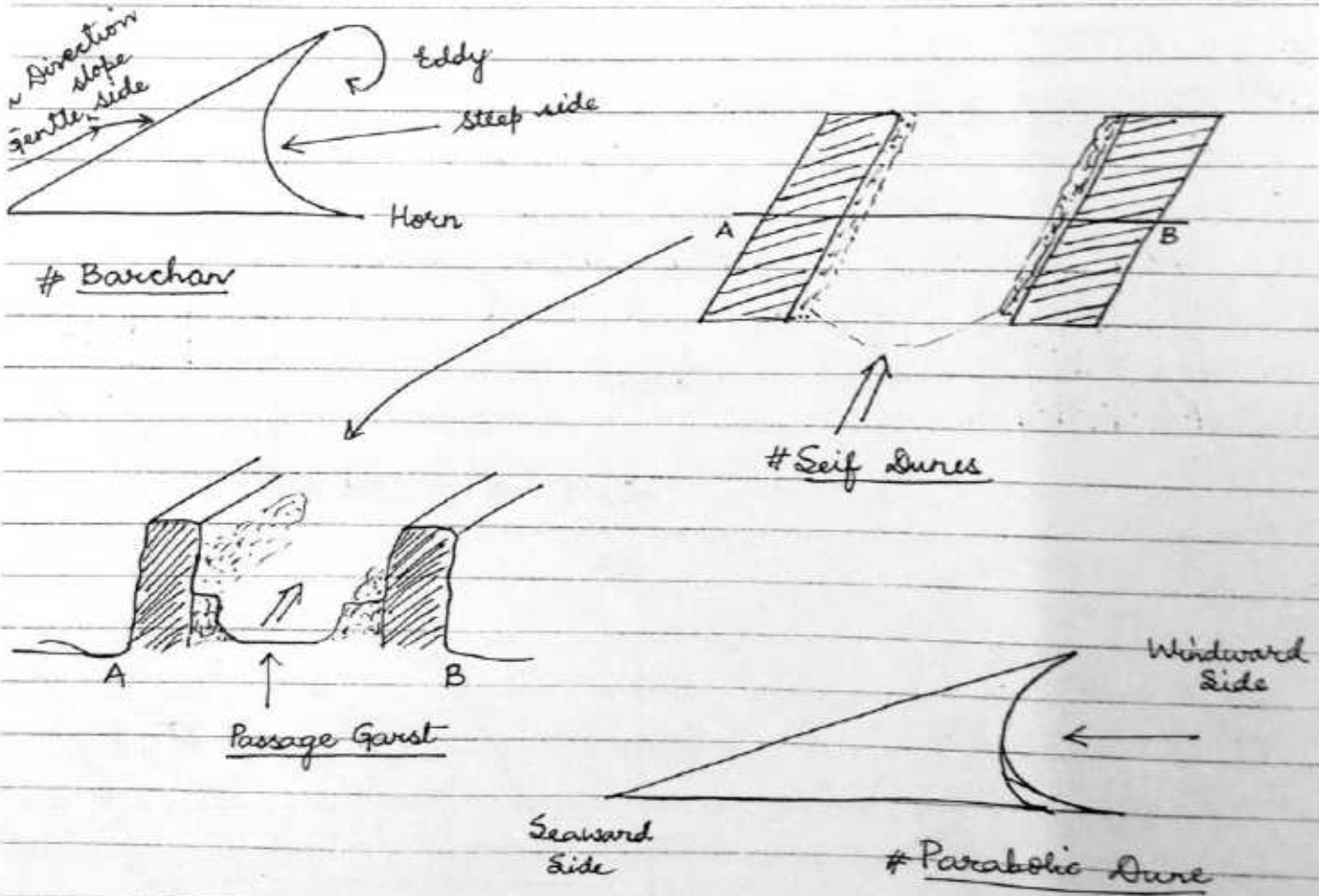
Ventifacts (Common)

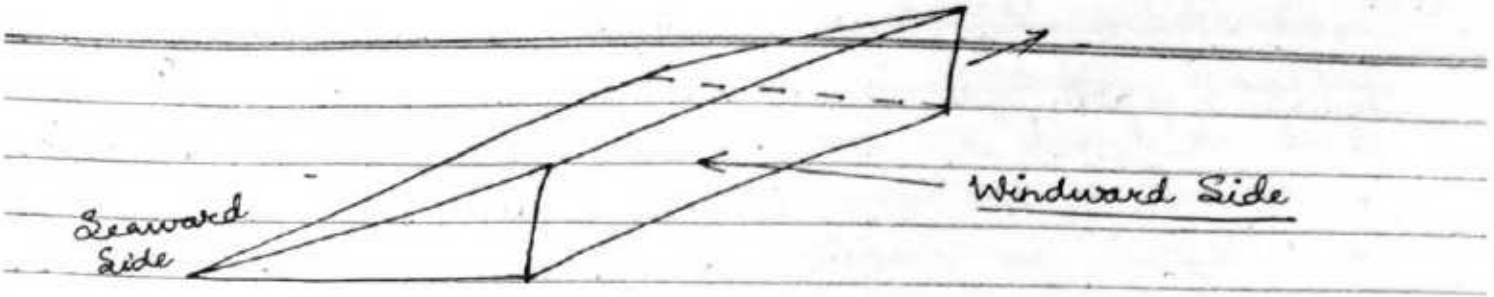
Wind as an active agent of deposition correlates to the formation of sand deposits. These deposits are classified to two well-defined categories : ① Sand Dunes & ② Loess. Sand dunes are classified in accordance to their shape as well as their location. The distinguished categories are :

(a) Crescent : Barchans (in desert)
Parabolic (in coastal areas)

(b) Linear : Seif (in desert)
Transverse (in coastal areas)

Loess is wind deposited soil outside the desert.





"Transverse Dune" on wave cut platform

Transverse dunes are ⊥r to wind flow while seif dunes are in the direction of wind flow.

Eolian Cycle → "Wind" + "Fluvial action by WC King"

Wind: Erosion — Deflation — Sand blasting

- Mushroom Rock
- Mesa & Butte
- Zuegen & Yardang
- Stone Lattice
- Ventifacts

↳ Due to differential erosion

Deposition — Sand dunes
Crescent: Barchans & Parabolic
Linear: Seif & Transverse
— Loess

Lecture # 17

/03/2014

Denudation Chronology & Erosional Surfaces:

istorical approach to the study of landform features hich is based on fundamental recognition of sequential evelopment of landform is denudation chronology.
e approach is credited to RJ Small, one of the most rominent modern geomorphologist.
e recognised that the primary objective of the study of enudation chronology is to reconstruct the development of opography which marks its evolution, denudation and -development.
e based denudation chronology onto fundamental eomorphic principles: (A) Davisian Cycle of Erosion and Hutton's uniformitarianism { Laws of Nature do not change with time }.
enudation chronology is not applied to the relief that ere developed in tertiary period.
ature and modifications of the topography is generalised be dependent on:

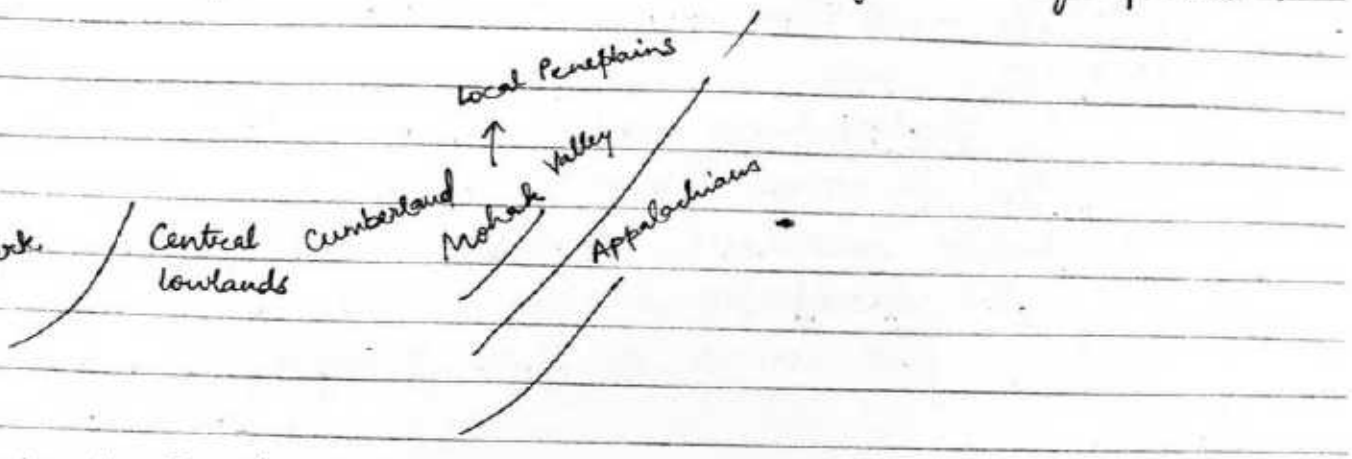
(a) Age of the surface
(b) Relative hardness of rocks
(c) Stream frequency or drainage density

e denudation chronology is strongly criticised by another odern geomorphologist Bloom. He recognised the approach be highly deductive and can be applied for the analysis ; general sequence of landform study.
or the specific cases, following this approach can result to misleading conclusions.
For example: Notre Dame mountain, Allegeney and Blue Ridge mountain of Appallachians are significantly taller than rest of the ranges not because they are made up of resistant structure but that they have

Facebook Group: Indian Administrative Service (Raz Kr)

Ozark
Cumberland } Earlier Part of Appalachians

experienced negative rejuvenation during tertiary period.



Cumberland & Appalachians : Mohak Valley (Local peneplains)
Ozark & Cumberland : Central lowland.

) The erosional surfaces are defined to be end product of erosional cycles which was given the expression peneplain by Davis.
) R.J. Small identified erosional surfaces to be the features cutting across all the geological structure, distinguished from erosional features which are structurally controlled.
) Erosional surfaces are identified to be cryoplain for glacial cycle, pediplain for Eolian cycle, peneplain for fluvial cycle and "Etch" plain (Thomas and Pugh) for Savannah Cycle.
) Erosional surfaces are classified into 3 prominent categories:
(i) Local peneplain
(ii) Regional "
(iii) Reuplifted "
The local peneplain forms first stage of peneplain development where in the influence of most effective agent or least resistant strata, part of relief is eroded to the base level. Mohak Valley and Central lowlands (USA) along the

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Karst Cycle does not have erosional surface.

Tropical continental climate – Savannah

ppalachians forms most recognised example.

he regional peneplains form the ultimate stage of development f peneplain where near complete relief is eroded to the ase level. Yenisey river valley relating to western Siberian lain forms the most projected example.

he re-uplifted peneplain denoting negative rejuvenation r polycyclic landform involves valley-in-valley topography eveloped in mid-course of River Yenisey as an excellent xample.

MAPS

Vest Siberian Plain : Peneplain

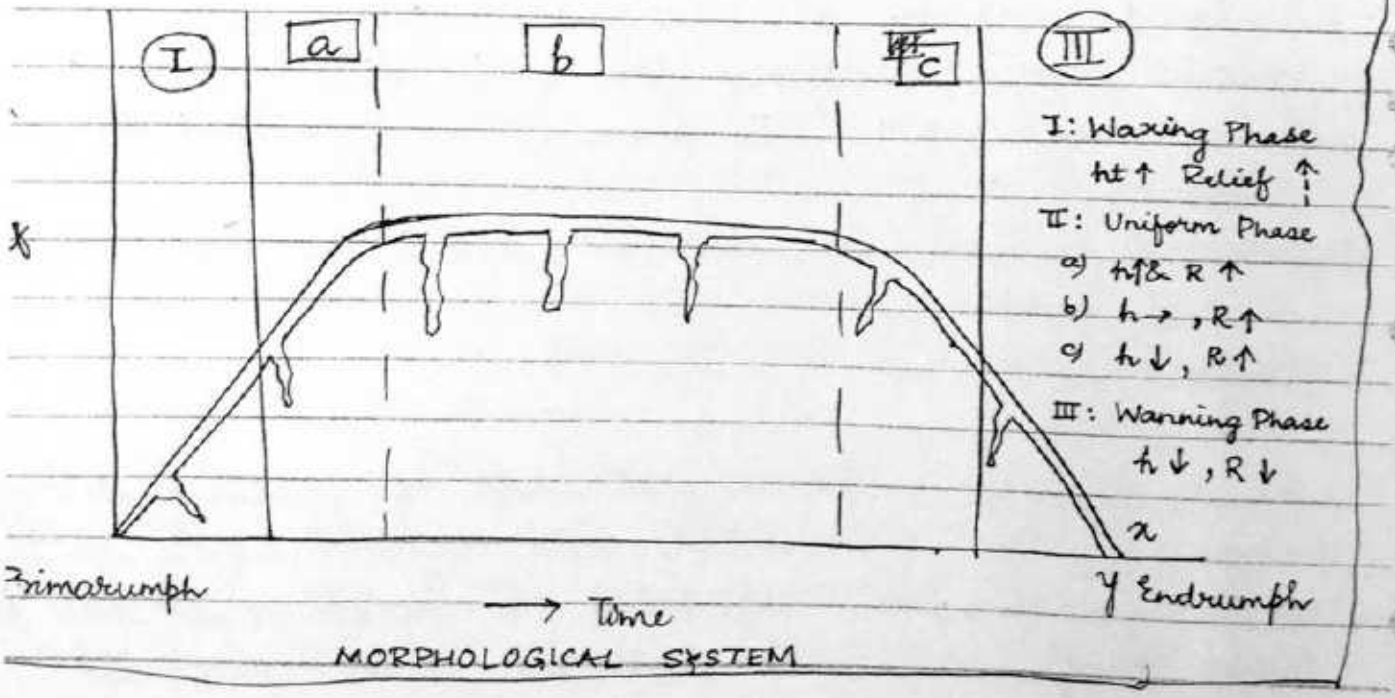
Vid-course of River Yenisey : Re-uplifted peneplain

ycle of erosion – Don't go beyond Young, Old, Mature

rosional surface → Must talk about regional & reuplifted

SODIC EROSION By PENCK

x = Water Divide Summit
y = Valley Floor

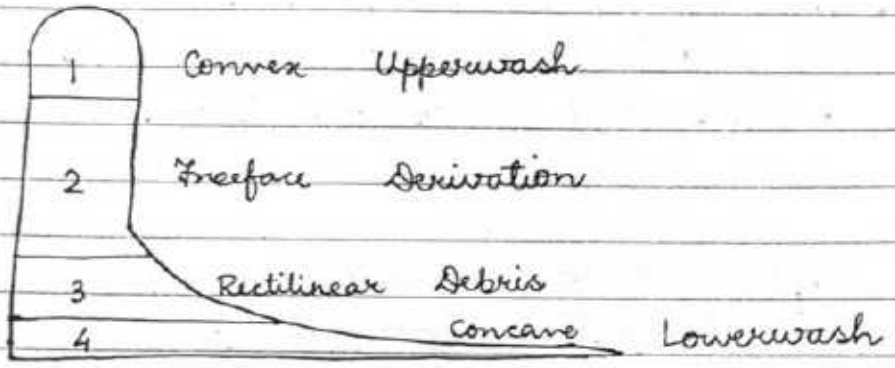


MORPHOLOGICAL SYSTEM

SLOPE DEVELOPMENT : CONCEPTS & APPROACHES

Concepts of Hill Slopes :

These concepts in geomorphology involves the definition of hill slopes to be angular indination of the topography b/w the hill-tops & valley-bottoms. These angular indinations are precisely caused by structure, age & process. The idealised hill slope is correlated with 4 different elements or segments. These includes :



Elements

The convex elements is the hill crest elements which corresponds to waxing phase of development of landform. It generates its profile with simultaneous effectivity of upliftment & downcutting and is therefore also recognised as upperwash slope.

The freeface represents the hill slope segment that is near absolute vertical slope of bare rocks. It is due to its gradient that it is completely deprived of weathered rock materials. This slope element is in accordance also referred to be slope of Derivation which correlates to the parallel retreat or back-wasting of the slope.

Rectilinear: Transitional element b/w freeface & concave element t represents constant slope. Lesser gradient in this hillslope

nent makes it prominently incorporating deposition of oken rock materials. It is in accordance that it is also lled debris slope.

concave element is the foot element of the hillslope ich may or may not have deposited materials. It is related primarily to the waning phase making it ar completely deprived of recognisable gradient. It is e to its location that it is recognised also as lowerwash ope.

These hillslope elements represent variable combination in real ground perspective so much so that every hillslope m revealing the combination of one or more than one llslope element making every hillslope 'complex compound rm'. It is therefor that to generate distinction, dominating lslope element is taken into account. To name hillslope ms, young reliefs are largely convex hillslope forms, l reliefs are largely rectilinear hillslope forms and cient reliefs are largely concave hillslope form. It is s combination of hillslope forms that are analysed in fferent geomorphic approaches of the hillslope development ese approaches are classified into 2 broad categories: uantitative & qualitative.

IANTITATIVE METHOD: The quantitative methods are the ontical methods which involves lab-based calculation of gradient of hillslopes. These methods are more applied the field techniques of geography, that is, cartography. ese methods are primarily contributed by Wentworth, iimz & Henry.

B) QUALITATIVE METHODS :

The qualitative methods are the empirical methods that are based on actual slope analysis. These methods take into account tectonic processes, erosional processes & depositional processes to be generating hillslope forms. Within this category, monoprocess and & polyprocess slopes are distinguished. The most followed ideologies:

(A) Slope Decline theory of Davis } Downcutting as process (erosional)
(B) Slope Replacement theory of Penck }

belongs to monoprocess category that is emphasizing exogenic processes in the development & modification of hillslope forms (with prime focus on erosional processes)

1) Slope Decline Theory of Davis :

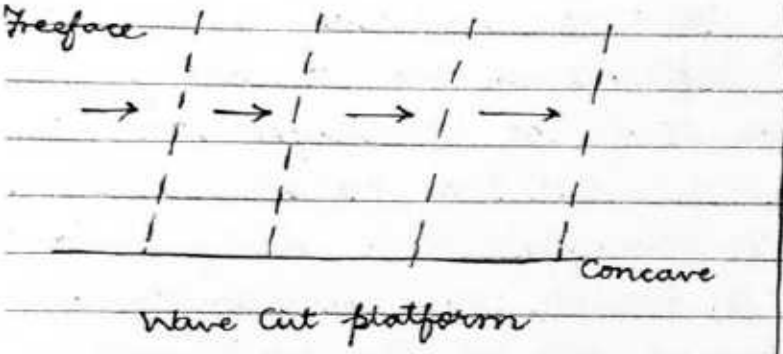
As part of his normal cycle of erosion, Davis emphasized that like cyclic landforms, hill and valley slopes undergoes cyclic process of development. In this cycle, there is progressive decline in the slope angle and sequential change in the hillslope form from convex to rectilinear and concave. In the young stage, steep convex slope evolves due to active downcutting or valley deepening. In this stage, there is limited slope retreat & continuous increase in the slope angle. The beginning of mature stage corresponds to the erosion of water divide summits that marks the decline in both slope & slope angle. In this stage, dominating horizontal abrasion continuously marks decrease in the slope generating much smoother slope profile corresponding to rectilinear hill slope form. With such development, the graded slope when only sustains transportation & deposition of the weathered rock materials, old stage of slope decline is demarcated.

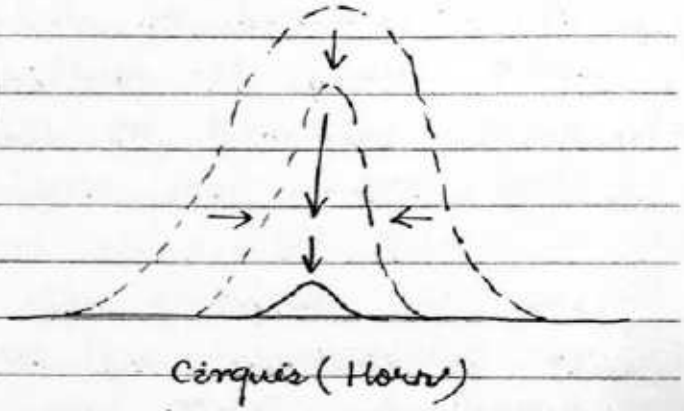
this stage, continuation of weak horizontal abrasion
ñ dominating deposition creates the general hillslope
ms to be concave with maximum of elevation eroded
d maximum of depression filled. This concave hillslope
m, however, marks the potentiality of presence of remnants
wex elements depicting the role of geological structure.
s approach of slope decline at par with normal cycle
erosion due to its simplicity makes it the most
owed ideological school with all the major geomorphologist
lowing this school of thought.

ope Replacement Theory The slight deviative approach which is
ɔ significantly followed is the slope replacement theory
Penck. Unlike his morphological system, Penck could
erate slope replacement as "well defined theory" counter-
ancing Davisian ideological school. He developed his
structural
broach in presuming the presence of homogenous cliffs
h non-eroding, non-depositing river at its foot. With
ɔ pre-assumed condition, he idealised the approach of
be replacement; where homogenous weathering of the
jected cliff, mass wasting of the material will be
ntinuously transported by the effective river at the foot;
and
using slope replacement converting the landscape from
eface slope form to concave form.

s theoritical orientation of Penck, wher, is practically
pplied, it depicts its validity in the functioning of sea waves
s an active agent of gradation as well as Alpine
aciers. In the case of sea waves, the retreat of
a cliff excellently depicts slope replacement applicable
ely on one side which marks the conversion of

hillslope form either from convex or freface to concave form as wave-cut platform. Similarly when we this approach is applied to more than one side of the slope the basal sapping effect carried on by Alpine glaciers in generating horn topography. In this topography, parallel retreat of the hillslope can convert the hillslope form from convex to concave form depicting collapse of horn in the old stage of glacial cycle.





Channel Morphology → (Running Water as active agent) + (Drainage Pattern)

Will be covered later.

Lecture # 18

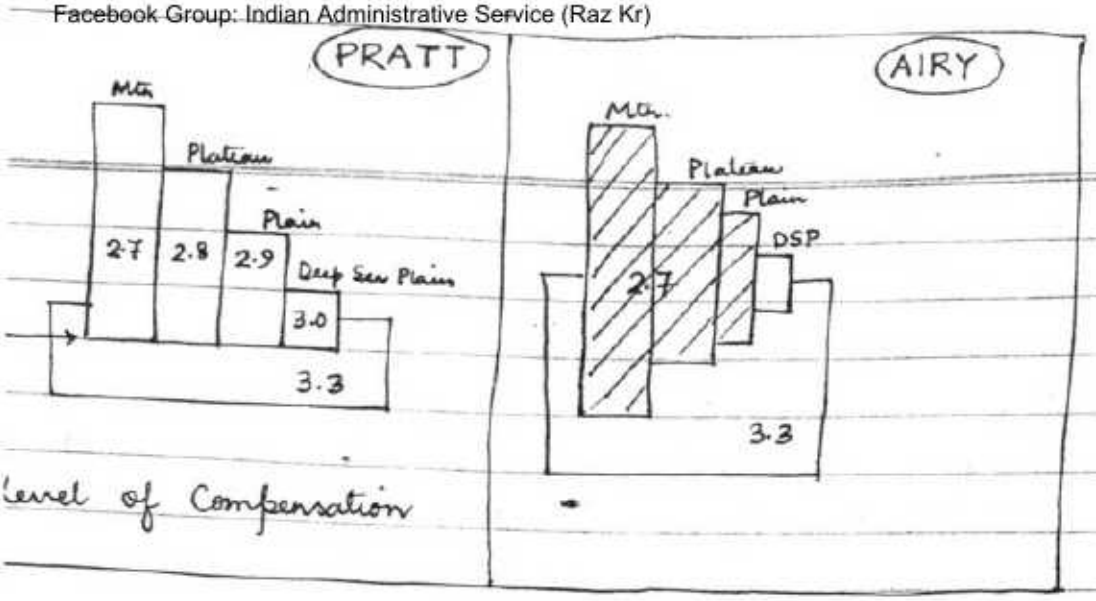
03/2014

ISOSTASY

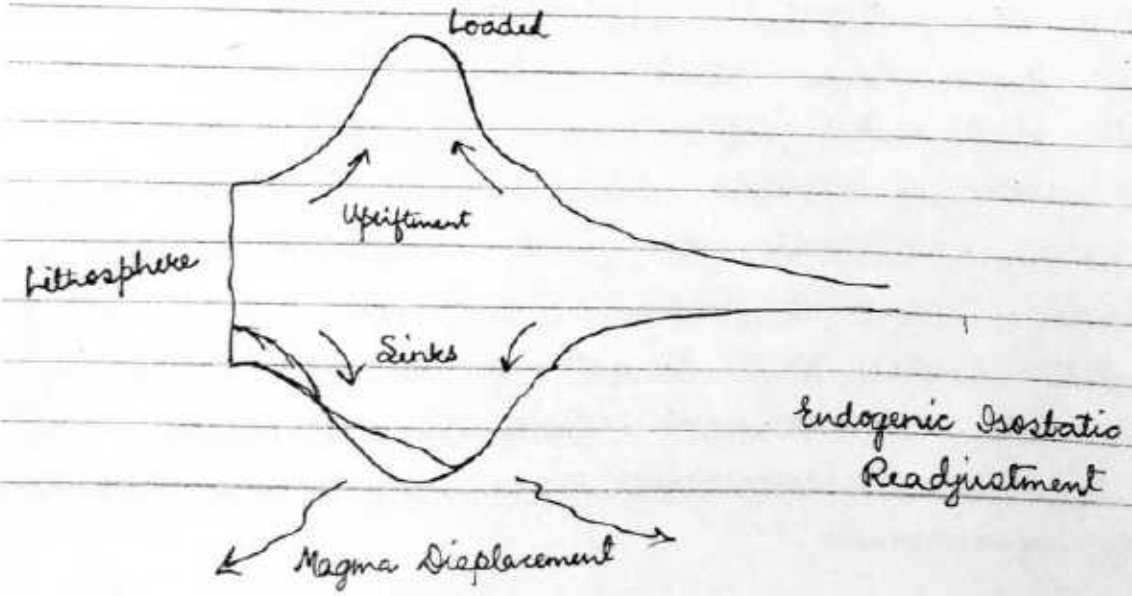
stasy is the concept of equilibrium that is prevalent b/w
structure lithospheric features irrespective of their height &
th on the fast spinning Earth. The genesis of this
proach in the former lines is traced back to first half
19th century when Sir George Everest attempted the
graphical surveys in order to determine structure of
malayan mountains. Utilising 2 different methods i.e.
round surveying and remote platform survey, he concluded
fferent record of distance b/w the same 2 centres
cated along the southern slopes of Himalayas. Greater
tance calculated on the reference of ground survey
thod made him conclude that as Himalayas are pulling
ir mass, there has been a deflection caused in the
mp bob resulting into the development of greater distance-
is establishement by Sir G. Everest was further carried
ward by Pratt who became one of the significant
pounder of approach in isostasy. In his study, Pratt
ncluded that if Himalayas would have been actually
ting the mass projected for its size & height, the differenc
the distance b/w the 2 places by Sir G. Everest would
e been more than 3 times than actually calculated. This
alysis made him conclude that "Himalayas are not
ting its projected mass" as it is made up of lighter
sity rocks. It is this conclusion that made him propound
ological school of isostasy that is called "Different
rities and Same Depth". This ideological school popularly
own as "Level of Compensation Concept"; projects that
ller features are made up of lighter density rocks
d deep seated features - denser rocks. It is, therefore,
at they mark the equilibrium or balance on the

fast spinning Earth. In accordance to his approach all the differences of density gets compensated at the specific level of aesthenosphere called "Level of Compensation". His analysis established one of the most referred ideological school in the geomorphic studies. It is followed on by big range of contributors to the study of isostasy specifically Hayford & Bowie. It is his analysis that led to the establishment of equilibrium study that was given the name isostasy by Dutton, who explained it to be "The concept of floating crust in gravitational balance".

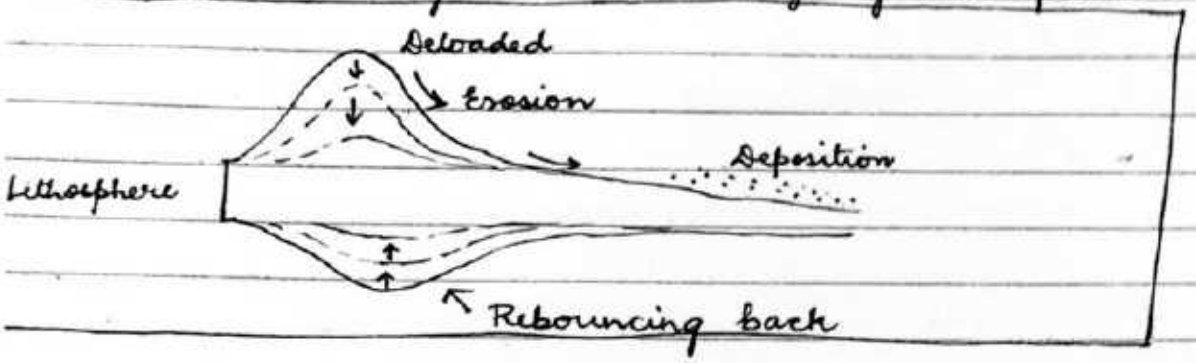
Pratt's level of compensation excellently explains the cause of equilibrium b/w the features of continent crust (granitic lighter density) and oceanic crust (basaltic denser). However, when the approach is applied to the individual crustal features, it avails the faulty description of variable density. It is this aspect of limitation that led to the development of mountain root concept of Airy, that makes the 2nd ideological school in the study of isostasy. Airy's approach emphasizes on "similar density" and variable depths" i.e. all the lithospheric features are projected to be made up of same density rocks. They mark equilibrium because they have different sizes in accordance to which taller features have deeper roots which makes them in balance with deep seated features with Shallower roots. This established school of thought best explains: "Intra-crustal features", However, avails the faulty description in identifying that continental crust & oceanic crust are made up of same density rocks. This school of thought is strongly contributed by HEISKANEN. Both these ideologies are taken to be "COMPLIMENTING EACH OTHER" in describing Isostasy or equilibrium.



PRACTICALLY, isostasy or the floating crust in gravitational balance continuously experiences interruptions by the geomorphic and climatic factors. Geomorphologically, both endogenic forces and exogenic forces induces modifications to persisting eqmb. the effect of endogenic forces, development of variation it is increase in the absolute height results in: "Loading the Crust" i.e isostasy is interrupted and in order to maintain the floatage, every upliftment resulting in the sinking of the crust not just generate "mountain roots" it also causes "displacement of magma" in asthenosphere with the mountain root to sustain the floatage. This upliftment called isostatic rebouncing is applicable to all the locations where endogenic upliftment dominates.



by the effect of exogenic forces, the induced levelling effect deloads the crust. This makes isostasy interrupted with simultaneous readjustment caused by rebouncing of deloaded crust and thus the termination of displacement of magma from underneath mountain root. This type of readjustment are applied to be more practical with ancient relief where the levelling task is majorly completed.



Climatologically isostasy marks its interruptions and readjustments in terms of ice age and its termination. During ice age loaded continental crust registers sink and thus, magma displacement in aesthenosphere in order to sustain isostatic balance. However, after ice age deloaded continental crust registers rebouncing to regenerate isostatic balance simultaneous to the loaded oceanic crust registering sink and thus magma displacement in aesthenosphere underneath it. It is post ice-age isostatic readjustment that is making Scandinivian shoreline continuously becoming shallower restricting commerciall potentiality of the developed sea-ports.

